

CEN/TC 278

Date: 2025-XX

prCEN/TR XXXXX:2025

Secretariat: NEN

**Public transport – Guidelines for building multimodal travel purchase
APIs based on Transmodel**

Einführendes Element — Haupt-Element — Ergänzendes Element

Élément introductif — Élément central — Élément complémentaire

CCMC will prepare and attach the official title page.

Contents	Page
European foreword	3
Introduction	4
1 Scope.....	5
2 Normative references.....	6
3 Terms and definitions	6
4 Abbreviations.....	8
5 Context.....	9
5.1 Purchase and reservation as the enabler of cross-European travel	9
5.2 Laying the foundation for future automated redress handling and independent trip-repair agencies	10
5.3 Learning from other industries opening their purchase & reservation systems.....	10
5.4 The definition of "booking" or lack thereof	11
5.5 Methodology	12
5.6 Exclusions	14
5.7 Preliminary conclusions	14
6 Multimodal travel purchase APIs basics, roles and use-cases.....	15
6.1 Basics.....	15
6.2 Roles	15
6.3 Additional recommendations.....	26
6.4 Travel Guarantees	26
6.5 Use-cases.....	28
7 Existing Transmodel Elements relevant for a multimodal travel purchase API specification.....	70
7.1 Introduction.....	70
7.2 Using Transmodel to model these use cases.....	72
7.3 Overview of existing Transmodel Model for trips and fares.....	84
7.4 Overview of existing Transmodel queries for Trip Planning and Fare Finding.....	103
8 New Transmodel concepts to support Multimodal Travel Purchase/Reservation APIs	107
8.1 Extensions to the Transmodel Trip Model	107
8.2 Additional Transmodel multimodal travel purchase API.....	122
8.3 Example of multi-supplier travel packages and the travel basket concept.....	123
Annex A (informative) BoB.....	128
Annex B (informative) OSDM	131
Annex C (informative) TOMP-API.....	136
Annex D (informative) Entur API	142
Annex E (informative) FerryGateway	143
Annex F (informative) TAP TSI.....	147
Annex G (informative) Comparison between OSDM & Transmodel elements	149
Bibliography	157

European foreword

This document (prCEN/TR XXXX:2025) has been prepared by Technical Committee CEN/TC 278 “ITS for Public Transport”, the secretariat of which is held by NEN.

This document is currently submitted to the CEN Vote on TR.

This document has been prepared in response to a formal request addressed to CEN by the European Commission. The request led to the opening of the CoRoM (Coordination & standardisation for Rail & Mobility) project for which this TR is one of the main deliverables.

The CoRoM project aims to provide support to the work to be carried out by the European Standardisation Organisations (ESOs) for the coordination and standardisation of purchase (including basic after-sale) interfaces (also called Booking APIs) for multimodal travel (including long-distance rail).

Introduction

For the past decade or so, Mobility-as-a-Service (MaaS) has been developed and implemented in Europe, mostly in an urban and peri-urban context. The original idea behind such a concept is very simple: allow travellers to go seamlessly from A to B using one single digital interface and based on their personal preferences. To roll-out this concept, the travellers' journey was divided into four main stages: plan, purchase (with or without a reservation), pay, and travel. These stages are not necessarily linear nor unique, and most certainly encompass other operational stages for mobility providers.

The planning stage can itself be split into searching through passenger information and planning based on the desired travel characteristics (e.g., date and time, number of passengers, etc.). This fundamental activity has been the main focus of the development of Transmodel-based data exchange formats such as the Network and Timetable Exchange (NeTEx) (CEN/TS 1664 series) for planned information and the Service Interface for Real time Information (SIRI) (CEN/EN & TS 15531 series) for real-time information. It also led to the development of the Transmodel-based API specification for trip planning with the Open API for distributed journey planning (CEN/TS 17118 series).

The purchase stage is one for which further standardisation is needed to support the development of MaaS solutions. Ideally, it would lead to a common API specification that facilitates the integration of different types of transport modes into one seamless application, including long-distance travel. It would then be a standard supporting the European Commission effort to open the distribution of tickets across Europe, especially the upcoming Multimodal Digital Mobility Services (MDMS) regulation. To support this standardisation work, a review of existing state-of-the-art existing APIs has been published by another CEN project team under the name of "Public transport - Distribution APIs for MaaS" (CEN/TR 00278582:2022). The current technical report (TR) takes into consideration the above-mentioned TR but is more focused on the inclusion of a deeper analysis, providing guidelines for formulating distribution standards in better alignment with Transmodel in order to facilitate the purchase for travellers (including other modes). The goal is to define principles to make existing travel purchase APIs compatible. It is also based on the latest revision of both Transmodel and NeTEx, which are key to understanding parts of the present TR.

Attention is drawn to the fact that authors of this document have chosen to use the word "purchase" rather than "booking". As further explained later in the document, the term "booking" bears several different meanings in colloquial use, which can generate misunderstanding.

Attention is also drawn to the fact that this document has been written before the European Commission has released a working draft of the upcoming MDMS regulation. Terms used in the latter might not match some used in the present document.

1 Scope

This document being one of the main deliverables of the CoRoM project, it details:

- The context of the CoRoM project and its methodology,
- The data information flow and steps required for purchasing any multimodal travel (including basic after-sale processes), from the perspective of travellers,
- The identification of the relevant concepts and data structure from Transmodel (EN 12896 series) to represent the above-mentioned data flow and steps,
- An identification and study of existing APIs that are used in multimodal travel for reservation and purchasing, which mostly include OSDM, TOMP-API, BoB, Entur API, FerryGateway, and (EU) 454/2011 (TAP TSI),
- The identification of concepts and data structures that should be added to Transmodel to cover the scope of the studied APIs,
- The guidelines for a multimodal travel purchase API specification based on Transmodel concepts.

The latter two parts of this TR will inform the revision of Transmodel Parts 5 and 6 (EN 12896-5 and EN 12896-6) and NeTEx Part 3 (TS 16614-3).

This TR covers the following functional scope:

- Provision of a catalogue of fares
- Request for specific fare offers (including specific search criteria),
- Selection of one or several formal offers by the customer (pre-purchase),
- Initiation of the purchase,
- Commitment to buy,
- Reservation of a place on a specific journey,
- Reception of evidence of the purchase,
- Management after purchase (basic after-sale actions)
- All interactions that can be related to a persistent identity.

These functions can be seen as the minimum required for the data flow of reservation from the travellers' perspective.

Though some of the studied APIs have functions beyond just reservation and involving other parties, the following activities are not in scope of this TR nor the CoRoM project:

- Creation and distribution of the travel document,
- Travel planning,
- Payment interfaces with banking information systems,

- Validation of purchased access rights during travel,
- Control of purchased access rights during travel

2 Normative references

The following documents are referred to in the text in such a way that some or all their content constitutes requirements for this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

[1] EN 12896-1, Public transport - Reference data model - Part 1: Common concepts

[2] EN 12896-5, Public transport - Reference data model - Part 5: Fare Management

[3] EN 12896-6, Public transport - Reference data model - Part 6: Passenger information

[4] TS 16614-3, Network and Timetable Exchange (NeTEx) - Part 3: Public transport fares exchange format

[5] TS 17118, Intelligent transport systems - Public transport - Open API for distributed journey planning

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardisation at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp/>
- IEC Electropedia: available at <https://www.electropedia.org/>

In the below section, whenever the definition has been taken from the EN 12896 series, their exact part has been referred to.

3.1

booking

placing of a reservation to use a specific transport service, but not necessarily entailing a purchase of a ticket (EN 12896-5).

3.2

consumption

using up of access rights by a transport customer as a result of travel (EN 12896-5).

3.3

consumption of a trip

consumption related to a trip.

3.4

customer purchase package

purchase of a SALES OFFER PACKAGE by a TRANSPORT CUSTOMER, giving access rights corresponding to those of one or several FARE PRODUCTS materialised as one or several TRAVEL DOCUMENTS. (EN 12896-5).

3.5

end-to-end trip

a TRIP from an origin to a final destination.

3.6**fare**

from the customer's perspective, the amount that a customer has to pay for a journey or for acquiring a specific fare product (EN 12896-5).

3.7**fare product**

an immaterial marketable element (access rights, discount rights, etc.), specific to a CHARGING MOMENT (EN 12896-5).

3.8**individual traveller**

individual travelling person (EN 12896-10).

3.9**inspection**

control, i.e., checking of a passenger's entitlement to consume access rights, involving identification of the traveller and/or the travel rights (EN 12896-5).

3.10**parameter assignment**

access right parameter assignment: the assignment of a fare parameter (referring to geography, time, quality or usage) to an element of a fare system (access right, validated access, control mean, etc.) (EN12896-5).

3.11**price**

value of fare or tariff (EN 12896-5).

3.12**query**

a function consisting in a request for and a delivery of information (EN 12896).

3.13**regional trip**

a TRIP limited to a region.

3.14**ticket**

unitary travel document allowing a single ride.

3.15**transport customer**

a specific person or organisation involved in a fare process.

NOTE: There may be a FARE CONTRACT between the TRANSPORT CUSTOMER and the OPERATOR or the AUTHORITY ruling the consumption of services (EN12896-5).

3.16**travel document**

a particular physical evidence to be held by a passenger, (ticket, card, etc.) allowing the determination of the right to travel or to consume joint-services. May comprise just a token associated with an online account, or some form of representation of the access rights (EN 12896-5).

3.17

travel guarantee

a promise that a travel service will be provided to a certain quality level and that a TRAVEL REDRESS will be offered if the guarantee is not met.

3.18

travel package¹

aggregated information selected from any SALES OFFER PACKAGE (satisfying the trip requirements of the TRANSPORT CUSTOMER) presented to the TRANSPORT CUSTOMER and enriched by specific information (i) as to the travel guarantee, (ii) as to the organisation providing the travel guarantee.

3.19

travel party

a group composed of at least two (02) INDIVIDUAL TRAVELLERS.

3.20

travel redress

a remedy or compensation that will be offered for failing to meet a TRAVEL GUARANTEE.

3.21

trip

a part of a TRIP PATTERN describing the movement of a passenger from one PLACE of any sort to another.

NOTE: A TRIP may consist of one or more consecutive LEGs having some common characteristics (EN 12896-6).

3.22

trip planning

delivery of information related to a request for trip proposals.

3.23

validation

verification that an access right is valid, has been or is being consumed, and that this consumption is allowed, achieved by checking the validity of the passenger's travel rights and comparing them to the parameters collected by the previous inspections and/or inspection context

4 Abbreviations

API	Application Programming Interface
BoB	Biljettdistribution och Biljettvisering
CEN	European Committee for Standardisation
CoRoM	Coordination & standardisation for Rail & Mobility
DG MOVE	Directorate-General for Mobility and Transport
ERA	European Union Agency for Railways
EU	European Union
GDS	Global Distribution System
GIS	Geographical Information System

¹ The new Transmodel TRAVEL PACKAGE element is described in §8.1.1

GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
IATA	International Air Transport Association
IFM	Interoperable Fare Management
IFOPT	Identification of Fixed Objects in Public Transport
ISO	International Standards Organisation
IS	Information System
IT	Information Technology
MaaS	Mobility-as-a-Service
MDMS	Multimodal Digital Mobility Services
NeTEx	Network and Timetable Exchange
PT	Public Transport
OJP	Open API for Distributed Journey Planning
OpRa	Operating Raw Data and statistics exchange
OSDM	Open Sales and Distribution Model
SIRI	Service Interface for Real-time Information
TAP TSI	Telematics application for passengers
TM	Transmodel
TOMP	Transport Operators – MaaS Providers
TR	Technical Report
UIC	International Union of Railways
UML	Unified Modelling Language
URI	Uniform Resource Identifier
URL	Universal Resource Locator

5 Context

5.1 Purchase and reservation as the enabler of cross-European travel

In 2021, President of the European Commission Ursula von der Leyen sent a Letter of Intent to David Sassoli, the President of the European Parliament, and Prime Minister Janez Janša, as the Presidency of the Council, in which she detailed the actions the Commission intends to take in the following year by means of legislation and other initiatives. Amongst them, within the pillar named “A Europe fit for the digital age”, was the mention of a “legislative proposal on multimodal digital mobility services”, which refer mostly to route-planners and ticket vendors that help to compare travel options.

The main objective of such a legislative proposal was to serve travellers of the European Union in their long-distance travels and to ease the integration of public transport and rail services to achieve seamless multimodal passenger transport, delivering the EU “Green Deal”.

Considering that the provision of EU-wide multimodal travel information to support the development of route planners is already addressed in the Delegated Regulation (EU) 2017/1926 amended by the Delegated Regulation (EU) 2024/490 of 29 November 2023, it is logical that the facilitation of purchase,

reservation and payment of mobility services would be covered by the MDMS legislative proposal. In a sense, it can be considered that “purchase and reservation” is the next piece required to enable cross-European travel for all.

Also considering that sustainable long-distance travel has a high impact on the Green Deal; and that MDMS services in the urban and peri-urban areas lead to a stronger integration of several modes of transport, a better integration of rail services with public transport can be considered as a necessary next step to enable cross-European travel for all. Additionally, implementations have to comply with GDPR framework.

The CoRoM project addresses both “purchase & reservation” and a better integration of rail services with public transport. It builds on pre-existing work as follows:

- The Fares and Timetable, Use-Cases description and mapping for PT and Rail, co-authored by experts of the CEN and UIC [11],
- CEN/TR 00278582:2022, Public transport – Distribution APIs for MaaS [10].

Finally, considering that the provision of EU-wide multimodal travel information is mandated variously in NeTEx CEN/TS 16614, SIRI CEN/TS 15531 and Transmodel EN 12896, in none of which there is a reference exchange protocol for “purchase & reservation”, it seems quite natural that the results of the CoRoM project should lead to a guidelines for API specification that facilitates multimodal travel purchase based on Transmodel.

5.2 Laying the foundation for future automated redress handling and independent trip-repair agencies

One of the hurdles encountered when trying to achieve and promote long-distance travel using combinations of rail services and other public transport is to solve the problem of broken connections due to delays and cancellations. It is not acceptable that a potential traveller be stranded half-way.

Today the passenger must often take action to initiate and get trip repair, or obtain other redress for a failure by the service provider. Instead, there should be an automated redress process, and the responsibility of initiating trip repair should be moved from the traveller to the carrier.

The redress process should be such that travellers should not have to delve into a complicated legal process to be reimbursed when their passenger rights are violated.

Instead, it would be better if the traveller could, at least as a second option after the carrier has failed to provide a solution, contact an EU-backed 24/7 reachable trip-repair / redress agency that is not linked to the carrier. The agency should judge the situation and provide the necessary trip-repair or redress without any charge to the traveller. The cost of such redresses could intermittently be covered by funds based on bank guarantees from the carriers and finally be settled in processes with the carrier without involving the traveller directly.

Such a handling should be based on real-time information as to the disruption (such as a cancellation or severe delay) and an electronic ticket that describes all necessary details such as who is travelling (i.e., a passenger or a group of passengers, their luggage and their passenger vehicles), comfort class, rebooking flexibility, accommodation and ancillary products, as well as travel guarantees and travel redress conditions. The API should thus support the exchange of all information needed to support an automated handling of travel redress, including any identifiers needed to link purchased tickets to real-time information from the carriers.

5.3 Learning from other industries opening their purchase & reservation systems

The airline industry and air travel, under the umbrella of International Air Transport Association (IATA), is currently going through a major transformation. IATA recognised that today’s customers’ expectations

are shaped by the digital experience they have of interacting with modern technology companies in other sectors and that modernisation is necessary.

Their goal was to develop a framework that would support providing the customer with a straightforward shopping cart experience, having a wide set of choices, towards a single ticket with seamless transfer and disruption protection also when the customer travels on multiple airlines.

A new framework for partnerships based on Offers and Orders has thus been developed. From a business perspective, it relies on a Retailer-Supplier model; similarly to the retailing industry, partners engage in a direct, bilateral relationship

The basis is to use direct API workflows between Retailers and Suppliers, as opposed to yesterday's complex, legacy flows using legacy technology (e.g., EDIFACT) and involving multiple third-party systems.

The idea is to provide the customer (air) transportation products and related services across Retailer and one or multiple Suppliers based on the following workflow:

- a) A Retailer requests offers from Suppliers
- b) Suppliers provide priced offers to the Retailer
- c) The Retailer, incorporating Suppliers' content, provides priced offers to the customer.

Accordingly, the Retailer is responsible for both the Customer and Supplier relationships, and all partners gain control of their offer. IATA believes that once implemented at scale, this framework will take travelling on multiple airlines and intermodal partners to the next level, addressing the challenges outlined above.

Within the ferry industry the *FerryGateway* initiative has brought forward an open source and free to use worldwide standard for communication between ferry operators and agent reservation systems. It provides a standardised set of messages that enables reservations, amendments and cancellations of ferry trips including all products and services on-board. It is stateless, SOAP-based and is defined using XML-schemas.

Learnings from the above in our context are:

- There should be a direct and simple way for Retailers and Suppliers to exchange offer details, including their travel guarantees and travel conditions.
- There should be enough room for Retailers to bear the responsibility of direct relationships with both Customers and Suppliers.
- All relevant stakeholders should have access to the full set of applicable products and fares available.

5.4 The definition of "booking" or lack thereof

Transmodel (EN 12896) has defined "booking" in its part 5 as the placing of a reservation to use a specific transport service — but not necessarily entailing a purchase of a ticket.

However, the term 'booking' is mostly avoided in this document as it is used differently across the stakeholders of the CoRoM project in different contexts that vary according of their diverse backgrounds. For example, depending on the types of mobility services they work with and the local context they experience, different meanings for 'booking' include:

- A long-term reservation of some resource, A purchase of a train ride,
- A written agreement between two parties, with two deliverables, one on each side,

- A seat reservation,
- The intended consumption of a trip,
- An act of reserving accommodation, a ticket, etc. in advance,
- An agreement that has been written down ('hit the books'),
- The entire process of reserving or securing a service, resource, or accommodation for a particular date or time.

The term "booking" is sometimes used as a noun, in other cases is related to the process of booking.

TSI Telematics uses the term "reservation" as an authorisation to assess as a service and "booking" rather as synonymous with "selling".

In **MMTIS** "booking" is used in the context of information provided by a fare query (i.e., by the delivery) and related to

- WHEN to book (purchase window, validity periods),
- WHAT choice of parameters is possible (space-related parameters: e.g., routing restrictions, zonal sequence or time-related parameters: e.g., minimum stay).

In **TOMP**, 'booking' is used in v1.x. as a container term for offers. In TOMP v2.0, it will closely be related to Transmodel and the term 'booking' will be replaced by terms related to the Transmodel SALES OFFER PACKAGE.

The **OSDM** specification says: "The booking represents the offers that have been selected and turned into a booking on request to the provider of the offers".

For Transmodel: "booking" is synonymous with "a process of **confirmed reservation** of an asset", an asset being an access right to a service, a service or specific equipment like a seat, a bike, etc. "Confirmed" has to be understood as: providing a record. Thus "booking" has to be understood as "**process of sales/purchase** excluding payment" (payment is a separate process).

If mentioned in this document, booking should be understood as "purchase", which is also the admitted term of the Transmodel concept.

5.5 Methodology

An API (Application Programming Interface) is a set of rules and tools that allows different software applications to communicate and exchange information in a standardised way. It defines the methods and data formats that applications can use to request and exchange information.

Designing an effective and robust API — whether for web services, frameworks, libraries, or other applications — equires a structured approach entailing the following steps:

- Gather Use Cases and Functional Requirements
- Define the Data Model
- Design the API Interfaces
- Determine Resource Representations

- Assign Operations
- Implement Error Handling & Validation
- Document the API
- Test and Iterate

This CEN/TR focuses on steps 1 (Gather Use Cases and Functional Requirements) and 2 (Define the Data Model).

Before drafting the guidelines for multimodal travel purchase APIs based on Transmodel, the CoRoM project team decided to analyse existing APIs. A detailed technical mapping table of the APIs queries has been elaborated to drive this CEN/TR. The choice of which API to select for analysis was based on three main criteria:

- The API is used in a multimodal context, which covers any modes of transportation from conventional public transport to long-distance rail, but exclude air travel,
- The API has already been adopted by several stakeholders (i.e., it must be implemented at least once) and is different enough from the others (i.e., it has functions not covered by other APIs),
- The API is European.

The selected APIs were:

- BoB,
- Entur API,
- FerryGateway,
- OSDM,
- TAP TSI,
- TOMP-API,

These APIs are detailed in Annexes A to F.

Attention is drawn to the fact that other APIs exist in European context (e.g., VDV used in Germany, Poland, Czech Republic, etc.), but they did not meet all the criteria above-mentioned. And some ongoing initiatives aims to specified national distribution APIs like in France, where a standardisation working group is dedicated to working on interfaces between transport ticketing systems and especially on "fare collection", how to sell and issue a ticket to a customer, aligned with CoRoM recommendations.

For all the selected APIs, the same methodology was applied:

- The mapping of their concepts to Transmodel, as the reference model,
- The identification of the commonalities between the API and Transmodel as to both:
 - Queries needed to provide a coherent set of online processes for a multimodal travel purchase API,
 - Data used in the queries

- An analysis of the commonalities and differences in the above was then used to identify additional concepts needed in Transmodel to allow the building of a multimodal travel purchase API.

The outcome of this CEN/TR aims to support interoperability of APIs provided they are aligned around Transmodel concepts. The evolution of each API will show the usefulness of different modes.

5.6 Exclusions

This document does not cover the following functions, that are variously associated with purchase and reservation:

- Journey Planning, considering it is a prerequisite action and is covered by the Open API for distributed Journey Planning (OJP) (TS 17118),
- Trip execution, i.e., travel by the passenger, considering it is an activity conducted after purchase,
- Ancillary activities such as manual assistance (customer service) for purchase or reservation activities,
- Payment, considering it is a well understood generic activity,
- Validation and control of travel documents, considering it is an activity conducted after purchase (though we note it will need data passed from the purchase transaction),
- Registration of a transport user (e.g., creation of an online account, against which travel purchases can be recorded), considering it is a well understood generic activity,
- Security management, considering it is a prerequisite action,
- Rights management (mostly related to Intellectual Property Rights in the data and contractual agreements to use it), considering it is a separate concern – the allocation of specific responsibilities to specific actors is however addressed.

5.7 Preliminary conclusions

Based on the analysis of the above-mentioned APIs and the work done within the CoRoM project, some preliminary conclusions have been made as to what to expect from a purchase & reservation API specification supporting multi-modal cross-Europe travel.

Such purchase and reservation API specifications should:

- Be “basic” enough to support all forms of public transport in that it should prescribe an easily achievable minimum level of implementation, while at the same time permit a more advanced use for more complex products and interactions where needed,
- Support combining offers from different distributors and aggregators. This requires support for a two-stage commit process in order to avoid customers ending up with having paid for “half a ticket”, e.g., for just one leg of an incomplete trip, without being able to secure onward travel,
- Be used as basis for interaction on all levels of interaction between the various actors:
 - Traveller to Retailer, Distributor, Aggregator, or Carrier,
 - Retailer to Distributor, Aggregator, or Carrier,
 - Distributor to Carrier,

- Give access to the full set of applicable products and fares available, needed to support the inclusion of local public transport offers within a multimodal travel offer (e.g., a pass for local public transport as an add-on to a long distance rail fare), whenever relevant for the travelling party,
- Support the automated handling of travel guarantees and travel redresses, which implies clarity on:
 - Who is travelling together (i.e., who is the travel party), so as not to break them up for a given travel redress,
 - Contact information for whenever a travel guarantee fails.

Obviously, the API specifications should use unambiguous and coherent terminology. The suggestion is to use Transmodel EN 12896, Public transport - Reference data model as the basis, and propose extensions to the related CEN group if needed.

6 Multimodal travel purchase APIs basics, roles and use-cases

6.1 Basics

From the purchaser's perspective, there are three main stages involved in buying the most optimal solution to travel from an origin to a destination in a multimodal/multi-distributor scenario:

Trip planning: Asking what possible options are available to travel, considering only combined timetable-data from different operators — available from National Access Points in accordance with MMTIS. At this stage price and reservation-availability information is not mandatory.

Asking for Offers, from one or several different Distributors, for selected Trips (or subsets of Trips) returned in the above Trip planning response. In this stage a price (e.g., € 195) and a preliminary availability (e.g., only 5 seats remaining/fully sold out) of the requested Trip-option or a part of should be provided. At this stage it is not required that the offered product is exclusively locked to the requestor. Note that separate Offers covering only certain parts of a Trip may be requested separately from different Distributors without any connection between them. The Offers may be managed by a Fare calculator, using various optimisation criteria: for example; most complete solution from start to end, least expensive, best touristic value, best using my travel pass, best using my commercial profile, etc. Note that all relevant Fare options must be provided (and any conditions for purchasing or using them).

Purchasing the preferred offer(s): The purchaser, i.e., the end customer (potentially through a retailer or an intermediate distributor) can decide that it is best to use a combination of offers, from different distributors, covering different sub-parts of the preferred Trip option. The separate purchases from all the involved distributors must then be synchronised, such that overall, it is an all-or-nothing purchase. Thus, the purchase must be preceded by first locking all necessary offers in parallel. Each of the involved distributors/attributors must lock the respective sub-Trip offer in their respective inventory for a short period. The successful locking should be reported back allowing the end customer to finalise the necessary purchases with all the different distributors in parallel. This should take place as soon as a positive response has been received from all parties. Alternatively, if one of the distributors does not respond within a given timespan, all purchases should be aborted.

6.2 Roles

Presentation of Transmodel functional roles with definitions and a list of IFM, TAP-TSI and OSDM roles playing the Transmodel role (partial role between brackets).

Table 1 – Roles – List and definitions

Functional role – Type and sub-types		Definition	IFM, TAP-TSI, TOMP-API and OSDM roles
FARE ORGANISATION ROLE		Any corporate role in providing or managing transport services.	
	FARE PRODUCT DISTRIBUTOR ROLE	The role, in large networks, of organising and accounting for the wholesale distribution of FARE PRODUCTS, as delegated by the FARE PRODUCT OWNER ROLE and to supply products to the FARE PRODUCT RETAILER ROLE.	IFM: N/A TAP-TSI: N/A OSDM: Fare Provider (Allocator)
	FARE PRODUCT OWNER ROLE	The role of the organisation which has commercial ownership of a FARE PRODUCT and is responsible for specifying pricing, usage conditions and commercial rules for the product, for ensuring there are processes for marketing and retailing the product, customer payment, control and validation, back-office accounting and settlement and for customer service.	IFM: Product Owner TAP-TSI: Fare Provider OSDM: Fare Provider (Allocator)
	FARE PRODUCT ISSUER ROLE	A role of an organisation introduced when there are a large number of fare product owners and a large number of distributors. The FARE PRODUCT ISSUER ROLE makes a portfolio of products available to a product distributor to be sold via product retailers to TRANSPORT CUSTOMERS.	IFM: N/A TAP-TSI: Issuer OSDM: Allocator
	FARE PRODUCT ATTRIBUTOR ROLE	The role of being an agent of a FARE PRODUCT OWNER, i.e., the role of an intermediary organisation to whom certain responsibilities have been delegated by the FARE PRODUCT OWNER, for instance to provide the technical implementation of the products defined by the FARE PRODUCT OWNER. NOTE: in TAP-TSI, corresponds to the distributor role , in charge of all product combinations (e.g., fares without yield management and yield managed fares).	IFM: N/A TAP-TSI: Distributor OSDM: Allocator (Fare Provider)
	FARE PRODUCT RETAILER ROLE	The role of an organisation to sell products to TRANSPORT CUSTOMERS, as authorised by a FARE PRODUCT OWNER ROLE. The FARE PRODUCT RETAILER	IFM: Product Retailer TAP-TSI: Issuer, Retailer

		ROLE may be undertaken by the same party as the FARE PRODUCT OWNER ROLE, the ACCOUNT PROVIDER ROLE, or a secondary outlet such as a TRAVEL AGENT or, newsagent, or other third party, etc. NOTE: In TAP/TSI, Issuer and Retailer roles are authorised by Railway undertakings (Fare Providers).	OSDM: Retailer TOMP: MP (MaaS Service Provider)
	FARE DATA COLLECTOR ROLE	The role of collecting and forwarding fare data between the various parties concerned with product delivery, such as PRODUCT RETAILER, PRODUCT OWNER, etc. May be undertaken by a separate intermediary. NOTE: in TAP-TSI, the distributor role includes collecting and providing data about issued tickets (including related information like spot allocation).	IFM: Collection and Forwarding (set of functions) TAP-TSI: Distributor, Ticket Controlling Organisation OSDM: Allocator, Ticket Controlling Organisation TOMP: Transport operator
	FARE REGISTRAR ROLE	The ADMINISTRATIVE ORGANISATION ROLE of coordinating the issue of unique registration codes (for example, by determining rules for generating unique codes) for data linked to fare management such as FARE CONTRACTs, TRAVEL DOCUMENTs, etc.	IFM: Registrar TAP-TSI: Fare Provider, Registrar Body OSDM: Fare Provider, Registrar Body
	FARE REGISTRAR ROLE (FARE CONTRACT & TRAVEL DOCUMENT)	The ADMINISTRATIVE ORGANISATION ROLE of coordinating the issue of unique registration codes (for example, by determining rules for generating unique codes) for data linked to fare management such as FARE CONTRACTs, TRAVEL DOCUMENTs, etc.	IFM: Registrar TAP-TSI: Fare Provider OSDM: Allocator, Certification Body
	FARE SECURITY MANAGER ROLE	The ADMINISTRATIVE ORGANISATION ROLE of establishing and coordinating a security policy for fare data across the provider organisations, and of the certifying and auditing of organisations, applications, etc., for their conformance to the security policy.	IFM: Security Manager TAP-TSI: N/A OSDM: Certification Body

TRAVEL ORGANISATION ROLE		The role of an ORGANISATION in providing any kind of travel related service.	TOMP: Transport operator
	SERVICE OPERATOR ROLE	The role of an ORGANISATION that provides a service to the customer who uses a product - in particular the carrier role for public transport, but also possibly ancillary services such as catering, baggage handling, parking, etc.	IFM: Service Operator TAP-TSI: Carrier OSDM: Carrier
	CUSTOMER SERVICE PROVIDER ROLE	The organisation role of providing "help line" and similar retail and fulfilment support facilities for CUSTOMERS, including replacement of stolen or damaged customer medium and assistance with the installation of applications.	IFM: Customer Service TAP-TSI: Carrier, ticket vendor OSDM: (Fare Provider)
	TRAVEL DOCUMENT CONTROLLING ORGANISATION ROLE	The role of an ORGANISATION providing the services of checking the validity of passenger's travel documents to check they have sufficient access rights for the journey that they are making	IFM: N/A TAP-TSI: Ticketing Controlling Organisation OSDM: Ticketing Controlling Organisation
TRANSPORT USER ROLE (specific to Fares Management)		General user ROLE for consuming travel services, including information services.	
	PURCHASER ROLE	The role of the party who pays for a travel service for one or more TRANSPORT CUSTOMERs.	IFM: Customer ? TAP-TSI: Customer OSDM: Customer TOMP: Customer
	INDIVIDUAL PURCHASER ROLE	The role of a person in purchasing a FARE PRODUCT as a private individual. Often, but not necessarily the same as the PASSENGER ROLE.	IFM: Customer ? TAP-TSI: Customer OSDM: Customer
	CORPORATE PURCHASER ROLE	The role of purchasing FARE PRODUCTs on behalf of a corporate body such as a company, school, government department, etc.	IFM: Customer ? TAP-TSI: Customer OSDM: Customer
	PASSENGER ROLE	The general role of consuming a travel service.	IFM: Passenger TAP-TSI: Passenger OSDM: Passenger TOMP: Traveller

		INDIVIDUAL PASSENGER ROLE	The role of a passenger consuming a travel service as a single individual.	IFM: Passenger TAP-TSI: Passenger OSDM: Passenger
		OFFENDER ROLE	The role enacted by someone attempting to travel with no or insufficient access rights.	IFM: Passenger TAP-TSI: Passenger OSDM: Passenger
		COMPANION ROLE	The role of accompanying another passenger under the access rights granted by the other passenger's ticket.	IFM: Passenger TAP-TSI: Passenger OSDM: Passenger
		GROUP MEMBER ROLE	The role of a passenger travelling under the access rights granted by a GROUP TICKET.	IFM: Passenger TAP-TSI: Passenger OSDM: Passenger
	TRANSPORT CUSTOMER ROLE		A specific person or organisation involved in a fare process. There may be a FARE CONTRACT between the TRANSPORT CUSTOMER and the OPERATOR or the AUTHORITY ruling the consumption of services.	IFM: Customer ? TAP-TSI: Customer OSDM: Passenger
ADMINISTRATIVE ORGANISATION ROLE			Any organisational role involving some aspect of the management of data or the use of data.	
	REGISTRAR ROLE		The ADMINISTRATIVE ORGANISATION ROLE of coordinating the issue of unique registration codes for Organisations, Products, Travel Documents, etc. The Registrar. function may issue unique identifiers directly or devise rules for the distributed generation of unique identifiers.	IFM: N/A TAP-TSI: Fare Provider, Issuer, Registrar Body OSDM: Distributor, fare Provider, Allocator, Registrar Body
	SECURITY MANAGER ROLE		The ADMINISTRATIVE ORGANISATION ROLE of establishing and coordinating a security policy across the provider organisations, and of the certifying and auditing of Organisations, Applications, etc., for their conformance to the security policy.	IFM: N/A TAP-TSI: N/A OSDM: N/A

	DATA COLLECTOR ROLE	The ADMINISTRATIVE ORGANISATION ROLE of collecting and forwarding data between the various parties concerned with product delivery, such as PRODUCT RETAILER, PRODUCT OWNER, etc. May be undertaken by a separate intermediary.	IFM: N/A TAP-TSI: Fare Provider, Attributor, Issuer, Ticket Controlling Organisation, Registrar Body OSDM: Distributor, fare Provider, Allocator, Ticket Controlling Organisation, Passenger, Registrar Body, Certification Body
	TECHNOLOGY ORGANISATION ROLE	The role of an ORGANISATION that provides some aspect of electronic fare or information services and products.	
	ACCOUNT PROVIDER ROLE	In Account Based Ticketing, the ACCOUNT PROVIDER provides the online account by means of which the product is delivered. Normally this involves a direct relationship with the CUSTOMER, though it may also be done as a white-labelled service for another party	IFM: Account Provider TAP-TSI: N/A OSDM: Allocator TOMP: Account provider
	MEDIUM APPLICATION PROVIDER ROLE	For media-centric products, the APPLICATION PROVIDER supplies the software application that hosts a TRAVEL DOCUMENT and presents it to CONTROL DEVICES for automatic control and validation.	IFM: Application Retailer TAP-TSI: N/A OSDM: Distributor TOMP: Ticket stock
	MEDIUM APPLICATION OWNER ROLE	For media-centric products, the APPLICATION OWNER ROLE has commercial ownership of the software MEDIA APPLICATION hosting a product and is contracted by the PRODUCT OWNER for the use of the application with the customer.	IFM: Application Owner TAP-TSI: N/A OSDM: Distributor TOMP: Ticket stock
	PAYMENT PROVIDER ROLE	The role of channelling funds from the CUSTOMER to the FARE PRODUCT RETAILER for travel services, and particularly for usage-based travel	IFM: Payment Provider TAP-TSI: N/A OSDM: N/A

		products. For example, EMV provider, ApplePay, PayPal, Android Pay, etc.	TOMP: Payment Service Provider
	IDENTITY PROVIDER ROLE	The role of supplying a trusted identity token that can be associated with a CUSTOMER ACCOUNT, TRAVEL DOCUMENT, etc. The IDENTITY PROVIDER ROLE may be undertaken by transport AUTHORITIES and OPERATORS, but also non-transport organisations such as governments central and local, mobile network operators, or online service providers (e.g., Google, Facebook, etc.) and banks.	IFM: Identity Provider TAP-TSI: N/A OSDM: N/A TOMP: Identity provider
	MEDIA PROVIDER ROLE	The role of supplying the media used for the TRAVEL DOCUMENT, that is a physical support containing machine readable/writable data for a processor application. This can include transport industry smartcards, payment industry contactless cards, public sector issued cards, mobile phones or paper (for barcodes), as well as new formats such as watches and key fobs, and NFC enabled devices.	IFM: Media Retailer TAP-TSI: N/A OSDM: N/A

In this document, some Transmodel roles are included in a parent role because there is no need to detail them in the purchase process.

In this document, some Transmodel roles are on the periphery of the project scope and so auxiliary to the purchase process (marked here as “Secondary”).

Table 2 – Functional roles in the CoRoM context

Functional role – Type and sub-types		CoRoM project context (Purchase process)
FARE ORGANISATION ROLE		
	FARE PRODUCT DISTRIBUTOR ROLE	In scope.
	FARE PRODUCT OWNER ROLE	In scope. Fare rule definition is typically done before the purchase process but can be accessed during the purchase process.
	FARE PRODUCT ISSUER ROLE	In scope.
	FARE PRODUCT ATTRIBUTOR ROLE	In scope.
	FARE PRODUCT RETAILER ROLE	In scope. May manage personal details of customer (and other persons travelling) relevant for travel purchases.
	FARE DATA COLLECTOR ROLE	Secondary. The data collection is often done after the purchase process but can be needed for after-sales action.

	FARE REGISTRAR ROLE		In scop
	FARE REGISTRAR ROLE (FARE CONTRACT & TRAVEL DOCUMENT)		In the scope. Subsumed within FARE REGISTRAR ROLE.
	FARE SECURITY MANAGER ROLE		In scope.
TRAVEL ORGANISATION ROLE			
	SERVICE OPERATOR ROLE		In scope. The timetable definition is theoretically given before the purchase process except for perturbations (travel guarantees).
	CUSTOMER SERVICE PROVIDER ROLE		In scope.
	TRAVEL DOCUMENT CONTROLLING ORGANISATION ROLE		Secondary. The travel document can be, but rarely, controlled by retailer or customer service provider during the purchase process.
TRANSPORT USER ROLE			
	PURCHASER ROLE		In scope..
		INDIVIDUAL PURCHASER ROLE	Subsumed within PURCHASER ROLE.
		CORPORATE PURCHASER ROLE	Subsumed within PURCHASER ROLE.
	PASSENGER ROLE		Secondary. Access rights consumption is rarely involved in purchase process except for some after-sale actions.
		INDIVIDUAL PASSENGER ROLE	Secondary. Subsumed within PASSENGER ROLE.
		OFFENDER ROLE	Secondary. Subsumed within PASSENGER ROLE.
		COMPANION ROLE	Secondary. Subsumed within PASSENGER ROLE.
		GROUP MEMBER ROLE	Secondary. Subsumed within PASSENGER ROLE.
	TRANSPORT CUSTOMER ROLE		In scope.
	ADMINISTRATIVE ORGANISATION ROLE		
	REGISTRAR ROLE		In scope. Subsumed within FARE REGISTRAR ROLE.
	SECURITY MANAGER ROLE		In scope. Subsumed within FARE SECURITY ROLE.
	DATA COLLECTOR ROLE		Secondary. Subsumed within FARE DATA COLLECTOR ROLE.
TECHNOLOGY ORGANISATION ROLE			
	ACCOUNT PROVIDER ROLE		In scope.
	MEDIUM APPLICATION PROVIDER ROLE		Secondary. Application is typically pre-coded for the media before the purchase process.
	MEDIUM APPLICATION OWNER ROLE		Secondary. Application is typically pre-coded on the media before the purchase process.
	PAYMENT PROVIDER ROLE		Secondary. Out of the document scope.

	IDENTITY PROVIDER ROLE	In scope.
	MEDIA PROVIDER ROLE	Secondary. Media are typically pre-coded and supplied before the purchase process.

Roles may be undertaken by an actor, that is a person, organisation or other system that interacts with a system; some roles are restricted only to organisations (e.g., PRODUCT OWNER, ACCOUNT PROVIDER), or only to individuals (e.g., PASSENGER ROLE, DRIVER ROLE). An actor may perform more than one role.

Schema of the interactions between actors with Transmodel roles.

To cover all real-world variations, the diagram in figure 1 is drawn as if each actor is playing only one role, and each role is played by only one actor. Having this approach helps identify where APIs should be defined. This does of course not contradict the possibility that one actor *may* play multiple roles.

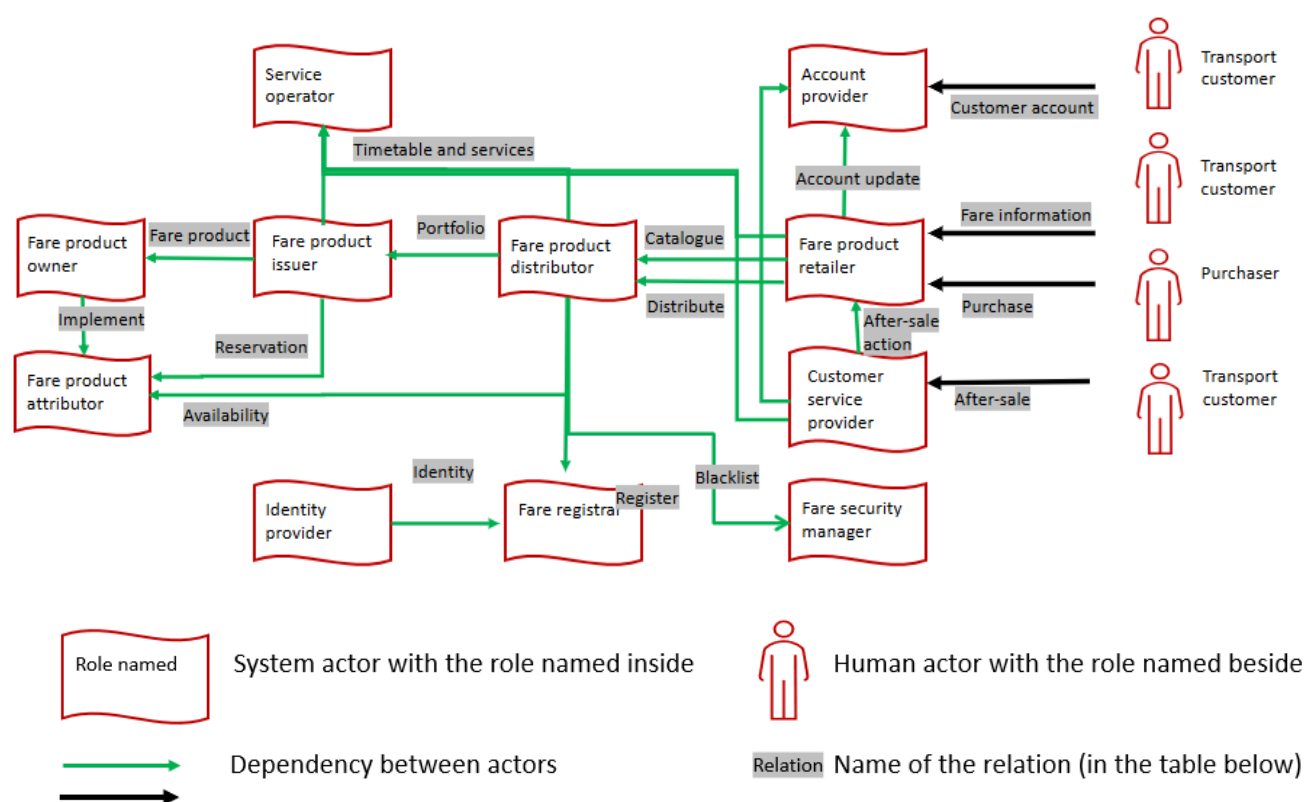


Figure 1 – Schema of the interactions between actors under Transmodel roles

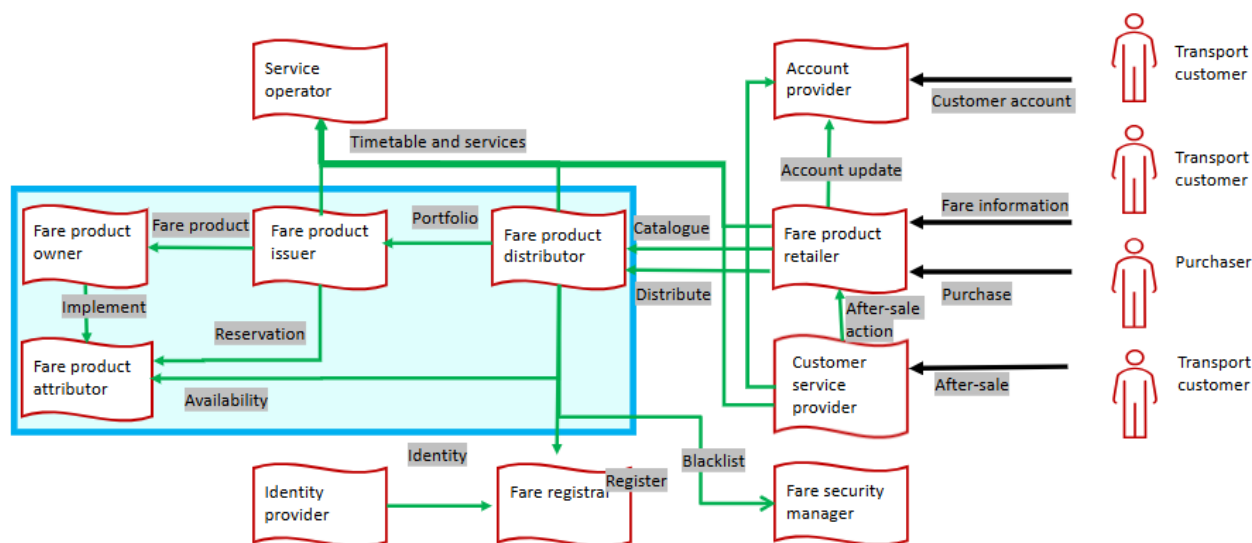


Figure 2 – Example when one actor plays multiple roles (all roles within blue rectangle)

The following table lists the dependencies between the actor-roles of the preceding schema

Table 3 – Dependencies between actor-roles in the traveller's perspective use cases (black arrows)

Flow identifier	Requestor	Service provider	Function
Customer account	Transport customer	Account provider	Request for management of CUSTOMER ACCOUNTs and CUSTOMER PURCHASE PACKAGES.
Fare information	Passenger	Fare product retailer	Request for FARE PRODUCT information (including TRAVEL GUARANTEEs).
Purchase	Purchaser	Fare product retailer	Request for a purchase of a TRAVEL PACKAGE ² (including payment).
After-sale	Transport customer	Customer service provider	Request for an after-sale action on a CUSTOMER PURCHASE PACKAGE (refund, change, cancellation).

Table 4 – Dependencies between actor roles in the retailer's and distributor's perspective use cases (green arrows)

Flow identifier	Requestor	Service provider	Function
-----------------	-----------	------------------	----------

² The new Transmodel TRAVEL PACKAGE element is described in §8.1.1

Fare product	Fare product issuer	Fare Product Owner	Get the FARE PRODUCTS and SALES OFFER PACKAGES definition.
Availability	Fare product distributor	Fare product attributor	Request availability for FARE PRODUCTS and services (reservation of a seat).
Portfolio	Fare product distributor	Fare product issuer	Request for FARE PRODUCT definition that the requestor can distribute.
Catalogue	Fare product retailer	Fare product distributor	Request for FARE PRODUCT definition that the requestor can sell.
Distribute	Fare product retailer	Fare product distributor	Request for distribution of TRAVEL PACKAGE ³ chosen by the Purchaser.
Implement	Fare product owner	Fare product attributor	Request for a delegated function on the FARE PRODUCT (technical implementation).
Reservation	Fare product issuer	Fare product attributor	Request for reservation management.
After-sale action	Customer service provider	Fare product retailer	Request for an after-sale action on a CUSTOMER PURCHASE PACKAGE that must be done by the retailer.
Timetable and services	Fare product issuer	Service operator	Request for timetable to organise the distribution.
	Fare product distributor		Request for timetable with services to organise the distribution.
	Fare product retailer		Request for timetable with services and real-time operational changes.
	Customer service provider		Request for timetable with services, real-time changes and historical data.
Register	Fare product distributor	Fare registrar	Request for generation and registration of unique identifier for TRAVEL DOCUMENT and FARE CONTRACT.
Blacklist	Fare product distributor	Fare security manager	Request about security on TRAVEL DOCUMENTs and FARE CONTRACTs. Can be with blacklist management.
Account update	Fare product retailer	Account provider	Request for consultation and update on the CUSTOMER ACCOUNT following the assigned rights.
	Customer service provider		Request for consultation and update on the CUSTOMER ACCOUNT following the assigned rights.
Identity	Fare registrar	Identity provider	Request for identity for FARE PRODUCTS.

³ The new Transmodel TRAVEL PACKAGE element is described in §8.1.1

6.3 Additional recommendations

Additional recommendations (They are not detailed in the use cases).

Recommendations for technical aspects of API implementation can be based on SIRI documentation part 2 (EN 15531-2:2015). An efficient API can be developed using the principles in this documentation:

- The data exchange patterns to ensure transaction completion.
- The discovery service mechanism that documents the interface and where it is available.
- The subscription mechanisms (can be used on FARE PRODUCT parameterisation) with a mediation layer implementation.
- The complementary monitoring services (Check status, Heartbeat and other utility services).
- The recovery and error management.
- The version management of the API through successive releases.
- The authentication and access permission management recommendations.

Other recommendations on security include:

- Implement fraud management mechanisms: media or contract blacklist management by the distributors provides a basic protection. The mechanism can be improved with an independent fare security manager servicing different actors (retailer, issuer, attributor, customer service provider, ...).
- Implement a protection against media falsification: use secured media (secure element embedded, ...) and enrolment mechanisms before sale transactions, use a secured machine-readable element (short time validity QR-code, ...).

6.4 Travel Guarantees

In the European Union, passengers are protected by a full set of consumer rights - whether they travel by air, rail, ship or bus and coach. The rights have been codified under European legislation, with different legislative acts to cover all modes of transport. This is to protect passengers' rights and ensure such rights are not lost in a myriad of national rules.

Passenger rights are based on three cornerstones:

- Non-discrimination.
- Accurate, timely and accessible information.
- Immediate and proportionate assistance.

Based on these principles, ten rights have been defined. In the context of this TR, the rights that were examined are:

- Right to information before purchase and at the various stages of travel, notably in case of disruption.
- Right to renounce travelling (reimbursement of the full cost of the ticket) when the trip is not carried out as planned.

- Right to the fulfilment of the transport contract in case of disruption (rerouting and re-reservation).
- Right to get assistance in case of long delays at departure or at connecting points.
- Right to compensation for failure to provide contracted services under certain circumstances.
- Right to liability of carrier to passengers and their baggage.

Our examination led to the definition various types of travel guarantees, which are the “promises” that a certain service will be provided to a certain quality level. We could classify them in the categories as follow:

- Travel quality, representing the delivery of some services to the passengers to an agreed quality (e.g., availability of promised facilities, punctuality, etc.),
- Trip repair, representing the delivery of some or all parts of a passenger’s trip even under disruption (e.g., making sure they can reach their final destination),
- Travel information, representing the delivery of passenger information (e.g., notice of disruption or change of time prior to the travel).
- Other, for all other types of travel guarantee.

Once the travel guarantees are detailed, it seems natural to articulate different types of travel redress for whenever the guarantees are not met by the transport operator. These redresses are to be understood as compensations for the travellers. The table below illustrates some of the travel guarantees and corresponding redresses that are commonly found and described from the travellers’ perspective. The last column identifies their categorisation.

Table 5 – Travel Guarantees & Redresses – Illustrations and categorisation

Travel Guarantee	Possible redress	Type of guarantee
30-minutes guarantee (the delay is under 30 minutes)	Digital voucher that equals a percentage of the ticket price.	Travel quality.
Reaching the final destination	Payment for the last leg to be made using a different transport mode (e.g., taxi instead of regional train).	Trip repair.
Information on the departure time	Text message to notify a change in departure time.	Travel information.
Possibility to have a vegetarian meal onboard	Other food options are offered (e.g., a second starter to compensate for the lack of a main course).	Other.

The information about the travel guarantees and their redresses for a given trip must be shared with the passenger before the purchase. This is especially necessary if more than one travel document is issued to

the passenger, where it must be stated if these tickets comprise a through ticket or if they represent separate transport contracts, where the whole passenger journey is not covered by the same passenger rights and travel guarantees.

The mandate to explain the travel guarantees and redresses for passenger information has led to their inclusion in Transmodel as new additional concepts (see subclause 8.1.2).

6.5 Use-cases

6.5.1 Introduction

The descriptions of the following use cases are inspired by the APIs in the scope of this report and the collective expertise of the contributors.

The order of the use cases follows the main MaaS use case, but the use cases have been reduced to address primarily what is inside the project scope: information (fare tables, prices and availability); the purchase process; and after-sale actions (before, during, and after the travel).

6.5.2 From the travellers' perspectives

The following use cases are described from the TRANSPORT CUSTOMER (undertaking a PURCHASER ROLE) point of view.

6.5.2.1 Choice of a solution by FARE PRODUCT catalogue consultation

The use case covers the choice of a SALES OFFER PACKAGE by a customer using a FARE PRODUCT catalogue of products set with an availability check (which may constitute a short-term pre-reservation). This use case does not involve a trip planner.

The green bubbles are indicative; they can be mandatory depending on each implementation.

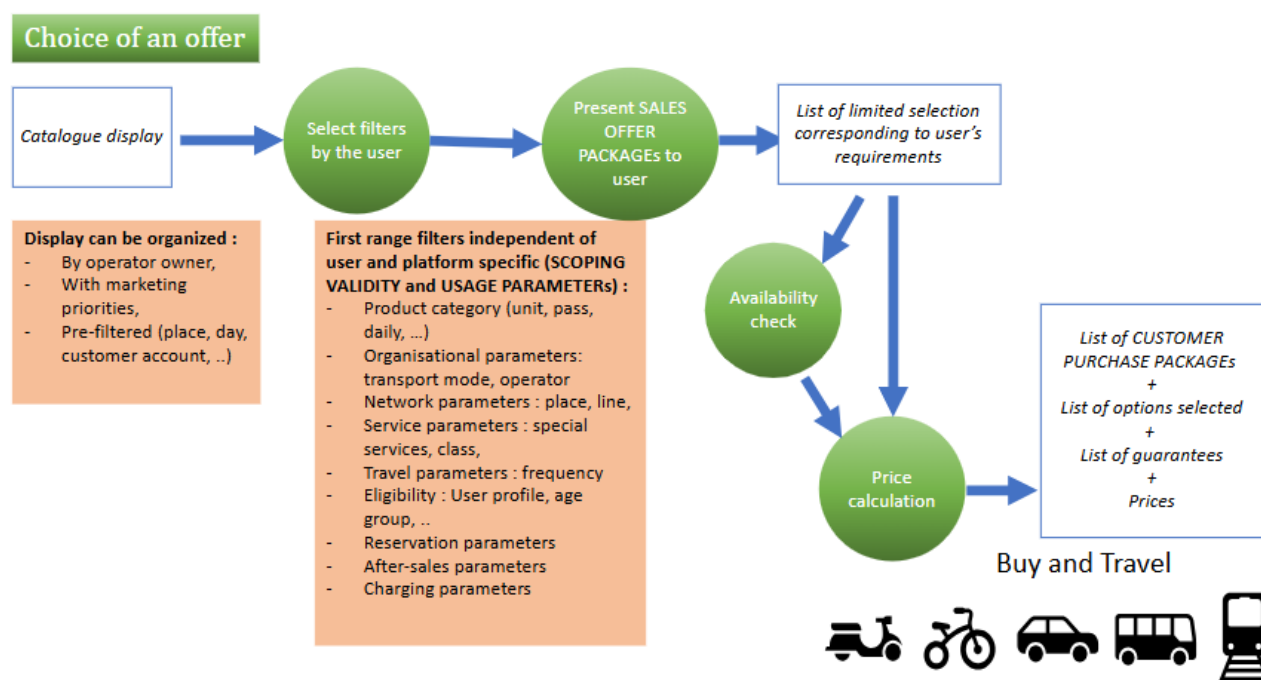


Figure 2 – Illustration of the use-case “Choice of a solution by FARE PRODUCT catalogue consultation”

Table 6 – Details of the use-case “Choice of a solution by FARE PRODUCT catalogue consultation”

Use Case 1: Choice of a solution by FARE PRODUCT catalogue consultation	
Summary	Using the FARE PRODUCT catalogue, the TRANSPORT CUSTOMER chooses one or more catalogue items and enters additional information to allow determining the best offer (FARE PRODUCT, SALES OFFER PACKAGE, the related PRICE and TRAVEL GUARANTEES).

Description	It is assumed that the TRANSPORT CUSTOMER knows where he/she wants to travel, and when or at which frequency; and also he/she may already have reduction cards, etc.. The TRANSPORT CUSTOMER opens the FARE PRODUCT catalogue with a mobile application, web site or vending machine, or it is done by desk agent. The catalogue is organised to be easy to use. It can have pages organised by TRANSPORT OPERATOR; or it can present the most requested products first, pre-filtered with locally usable products, etc.. The display can also be driven by platform marketing rules or other operator priorities. The FARE PRODUCT can be characterised in detail by USAGE PARAMETERS describing conditions of travel (e.g., are interchanges allowed, can the journey be broken, for how long is the ticket valid, etc.), as well as commercial conditions for EXCHANGING, REFUNDING, CANCELLING, RESERVING, etc.. Depending on the offer, these parameters may be mandatory, optional, or optional at an additional price (e.g., for differently priced LUGGAGE ALLOWANCES). If the TRANSPORT CUSTOMER is not satisfied by the presented products, he/she can navigate through the pages or use pre-defined filters to refine his/her search, until finding a SALES OFFER PACKAGE fitting his/her requirements. The TRANSPORT CUSTOMER can be helped by detailed information on each SALES OFFER PACKAGE (network and lines, stops, connections, etc..). Either the SALES OFFER PACKAGE is a “simple” product (Example: a Single ticket on a bus for immediate travel) or it needs additional data to be supplied by the TRANSPORT CUSTOMER to be completely defined (for example: time of travel, class of seating, possession of specific rail-card, etc.). The SALES OFFER PACKAGE may be sold with or without reservation following their definition (not needed, optional, mandatory). This process is described in the use case below. With reservation, on some platforms, the TRANSPORT CUSTOMER must start the reservation process to obtain availability and final price. The reservation can be a simple quota or a seat to select on a map for a scheduled journey. If there is no availability, the TRANSPORT CUSTOMER must change either an option (e.g., access right parameter such as comfort class) or FARE PRODUCT. Without a reservation or additional data, an indicative price can be given early, i.e., at the time of the product selection. In case of reservation, the TRANSPORT CUSTOMER obtains the price for a completely defined TRAVEL PACKAGE⁴. After these steps, the TRANSPORT CUSTOMER can choose to finalise the purchase process or not.
Main roles and actors	BoB: Ticket Vendor, Price & product service, Ticketing service, Customer OSDM 3.2: Fare Provider, Allocator, Customer TOMP-API: Reseller (MaaS service Provider) and/or Transport operator (including parking organisations and EV charging organisations), data aggregators, and indirect: customer, payment service provider, identity provider.

⁴ The new Transmodel TRAVEL PACKAGE element is described in §8.1.1

	TAP TSI: Carrier, Distributor, retailer. Transmodel: TRANSPORT USER ROLE (PURCHASER ROLE), TRANSPORT CUSTOMER, FARE ORGANISATION ROLES (in particular FARE PRODUCT ATTRIBUTOR ROLE but also FARE PRODUCT OWNER/ DISTRIBUTOR/RETAILER ROLES).
Main objects	BoB: Product, Manifest, Ticket Bundle OSDM: Products, offers, coach-layout, zone, reduction-card, bookings Transmodel: FARE PRODUCT, FARE PRODUCT PRICE, SALES OFFER PACKAGE, VALIDITY CONDITIONS, USAGE PARAMETERS, SALES OFFER PACKAGE PRICE, TYPE OF TRAVEL DOCUMENT, FULFILMENT METHOD, DISTRIBUTION CHANNEL, FULLFILMENT METHOD, TEMPORAL VALIDITY CONDITIONS, VEHICLE JOURNEY, TYPE OF SERVICE, TYPE OF PRODUCT CATEGORY, FACILITY SET, LINE TARIFF ZONE, DECK PLAN, LOCATABLE SPOT, SCHEDULED STOP POINT TOMP-API: Package (v1.x: Booking), Product (v1.x: Asset Type), Leg, Asset, Price, Ancillary, Travel Document., Cards, Card Types, License, License Type, User Profile. And it refers to external data sources like GBFS, DATEX-II, NeTEx, GTFS, APDS, OCPI, or Journey planner output (OJP, OTP). TAP TSI: Products, Fares, reduction cards.
Contributions	BoB: Product API: POST /product <i>[retrieve products based on filter]</i> Product API: POST /manifest <i>[retrieve manifest based on productIds]</i> Ticket API: POST /ticket <i>[request ticket based on manifest]</i> Ticket API: GET /ticketbundle <i>[retrieve ticket bundle]</i> OSDM 3.3.0 API: - GET /products, - GET /coach-layouts, - GET /coach-layouts/{layoutId}, - GET /zones, - GET /reduction-cards, - GET or POST /availabilities/place-map, - GET /availabilities/nearby, - GET /availabilities/preferences, - GET / POST /offers, - POST /bookings (in state prebooked) Transmodel: - FARE PRODUCT QUERY - STOP FARE QUERY NeTEx: - RequestTravelSpecification SIRI services: - Stop Monitoring,

	Vehicle Monitoring, Facility Monitoring OJP 2.0: FareRequest (MultiTripFareRequest, StaticFareRequest, StopFareRequest, TripFareRequest) TOMP-API 1.x: GET /operator/available-assets, POST /planning/inquiries TOMP-API 2.0: POST /processes/search-offers/execution (contains offer request, and optional availability check & fare calculation), POST /processes/purchase-offer/execution POST /processes/commit-purchase/execution TAP TSI: Provision of the fare catalogue by the retailer in a file format (e.g. NeTEx) ENTUR: POST /sales/v1/offers/search/graphql ?
--	---

6.5.2.2 Information on and choice of a solution for a specific trip

This use case presents the customer information on the SALES OFFER PACKAGE corresponding to a particular trip, at two levels of detail.

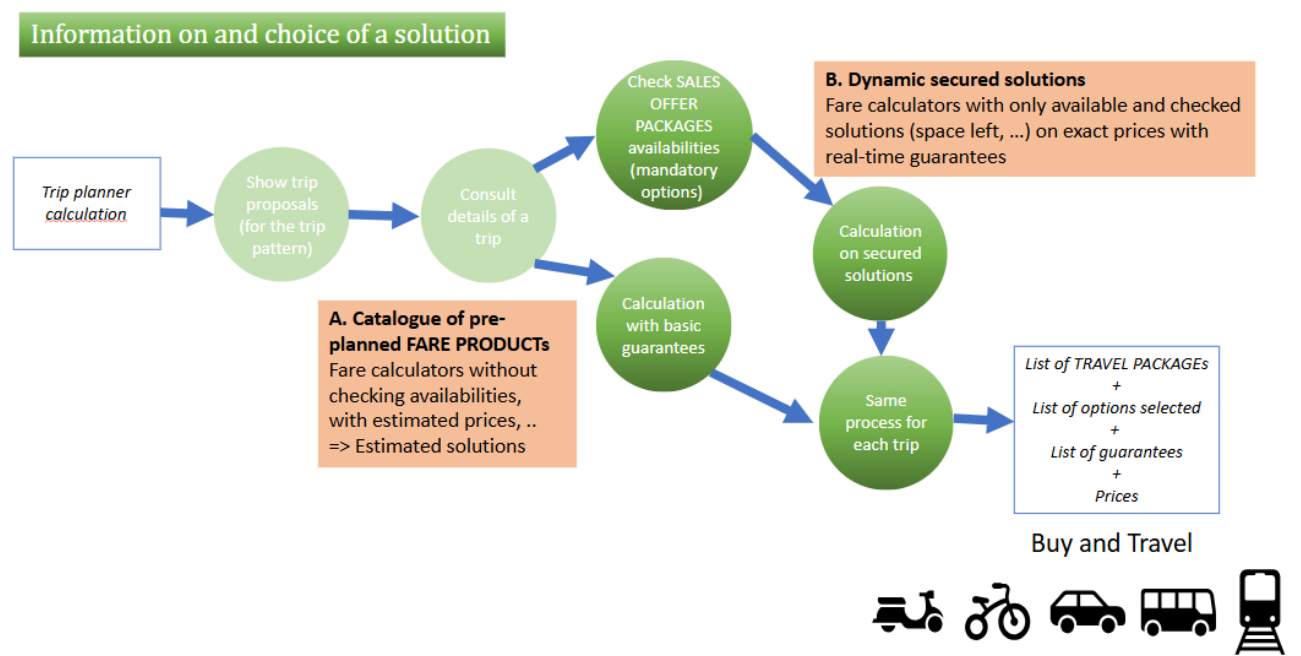


Figure 3 – Illustration of the use-case “Information on and choice of a solution for a specific trip”

Table 7 – Details of the use-case “Information on and choice of a solution for a specific trip”

Use Case 2: Information on and choice of solution for a specific trip	
Summary	After having received trip proposals from the trip planner, the TRANSPORT CUSTOMER chooses one of them and requests information on the corresponding TRAVEL PACKAGE (FARE PRODUCTS, SALES OFFER PACKAGES, PRICES and TRAVEL GUARANTEES).
Description	This use-case aims to prepare a purchasable package. It is assumed that possible trip options (TRIP PATTERNS) are provided by a trip planner which determines the mobility solutions for trips according to the required locations and time⁵. The trip planner answers with one or many solutions presented as TRIP PATTERNS which, in general, comprise a sequence of the different LEGs of the trip; the related transport modes; and the points where a connection is necessary together with the corresponding timing. For each of the TRIP PATTERNS⁶, the TRANSPORT CUSTOMER may request the fare corresponding to the different TRIP PATTERNS⁷ if the fare is not already provided in the delivery. The output he/she will get is a solution⁸ for the request. Depending on the operators and systems involved, a TRAVEL PACKAGE⁹ can be proposed in two ways: A1. A solution with SALES OFFER PACKAGES as it exists in a catalogue of pre-assigned FARE PRODUCTS, i.e., a basic proposal without checks for space availability, customer profile, It includes only basic, pre-planned TRAVEL GUARANTEES. It may comprise one or several simple SALES OFFER PACKAGES or several SALES OFFER PACKAGES in sequence. It provides pre-planned or estimated prices. 2. The solution can be more elaborated and based on a non-exhaustive list of strategies that adapts in real-time the resulting solution to a particular context and secures the travel. The TRAVEL PACKAGE is based on an exhaustive FARE PRODUCT catalogue. It involves several checks as regards seat availability, class options, transfer options, information on travel guarantees. Both cases (1) and (2) include a price calculation that can adapt according to the customer profile and/or an already existing contract (information stored in the customer account). The TRANSPORT CUSTOMER choses an offer among the TRAVEL PACKAGES before proceeding to the purchase stage.
Main roles and actors	BoB: Ticket Vendor, Price & product service, Ticketing service, Customer OSDM 3.2: Fare Provider, Allocator, Customer TOMP-API: Reseller (MaaS Service Provider) & Transport operator (including parking organisations and EV charging organisations), data aggregators, and indirect: customer, payment service provider, identity provider. Transmodel: TRANSPORT USER ROLE (in particular PURCHASER ROLE), TRANSPORT CUSTOMER, FARE ORGANISATION ROLE (in particular FARE

⁵ see Transmodel TRIP REQUEST

⁶ see Transmodel TRIP DELIVERY

⁷ see Transmodel TRIP FARE REQUEST

⁸ see also Transmodel TRIP FARE DELIVERY

⁹ The new Transmodel TRAVEL PACKAGE element is described in §8.1.1

	PRODUCT ATTRIBUTOR ROLE but also FARE PRODUCT OWNER/DISTRIBUTOR/RETAILER ROLES)
Main objects	BoB: Product, Manifest, Ticket Bundle OSDM: productCategory on timedLeg, zone, fare, Product, TripCoverage, ProductLegAssociation, Transmodel: TRIP PATTERN, LEG, TRAVEL SPECIFICATION, SERVICE JOURNEY, PASSENGER FARE OFFER, FARE PRODUCT, SALES OFFER PACKAGE, FARE PRICE, SCOPING PARAMETERS (TARIFF ZONE, TOPOGRAPHIC AREA), USAGE PARAMETERS, TYPE OF TRAVEL DOCUMENT, DISTRIBUTION CHANNEL, TRAVEL GUARANTEES, DECK PLAN, LOCABLE SPOT, AVAILABILITY CONDITION, TRAVEL SPECIFICATION, CLASS OF USE, FACILITY SET, ORGANISATION, TYPE OF PRODUCT CATEGORY, USER PROFILE, TYPE OF CONCESSION, TYPE OF ENTITLEMENT, ENTITLEMENT, TYPE OF SPOT AFFINITY, LUGGAGE ALLOWANCE, TYPE OF BAGGAGE, TYPE OF LUGGAGE USE, REPEATED TRIP FARE REQUIREMENTS, PASSENGER FARE OFFER, SALES OFFER PACKAGE, FARE OFFER PRICE, VALIDABLE ELEMENT, TRIP PATTERN, OFFERED TRAVEL SPECIFICATION TOMP-API (v2): Package (v1.x: Booking), Product (v1.x: Asset Type), Leg, Asset, Ancillary, Travel Document, Usage parameter. And it refers to external data sources like GBFS, DATEX-II, NeTEx, GTFS, APDS, OCPI, or Journey planner output (OJP, OTP)
Contributions	BoB: Product API: POST /product <i>[retrieve products based on trip plan]</i> Product API: POST /manifest <i>[retrieve manifest based on productIds]</i> Ticket API: POST /ticket <i>[request ticket based on manifest]</i> Ticket API: GET /ticketbundle <i>[retrieve ticket bundle]</i> OSDM 3.3.0 API: - GET /trips/{tripId} - GET /products - GET /coach-layouts - GET /coach-layouts/{layoutId} - GET /reduction-cards - GET /zones - GET or POST /availabilities/place-map - GET /availabilities/nearby - GET /availabilities/preferences - POST /offers - POST /bookings (in state prebooked) Transmodel: COMMON / SINGLE / REPEATED TRIP FARE QUERY NeTEx : FARE PRODUCT request ? OJP: TripInformationRequest

	SIRI services: - Stop Monitoring - Vehicle Monitoring - Facility Monitoring TOMP-API 1.x: - POST /offers, - POST /bookings/{id}, - POST /planning/inquiries, - GET /operator/available-assets, TOMP-API 2.0: - POST /processes/search-offers/execution, - POST /processes/select-offers/execution - POST /processes/purchase-package/execution - POST /processes/purchase-offers/execution - POST /processes/commit-package/execution ENTUR: - GET /sales/v1/offers/search/trip/trip-pattern/{id}, - POST /v1/orders
--	--

6.5.2.3 Getting availability with or without reservation

This use case covers the provision of information to a customer on a TRAVEL PACKAGE, along with the potential reservation of an available seat and schedule. This is a pre-purchase step during which the reservation process is started.

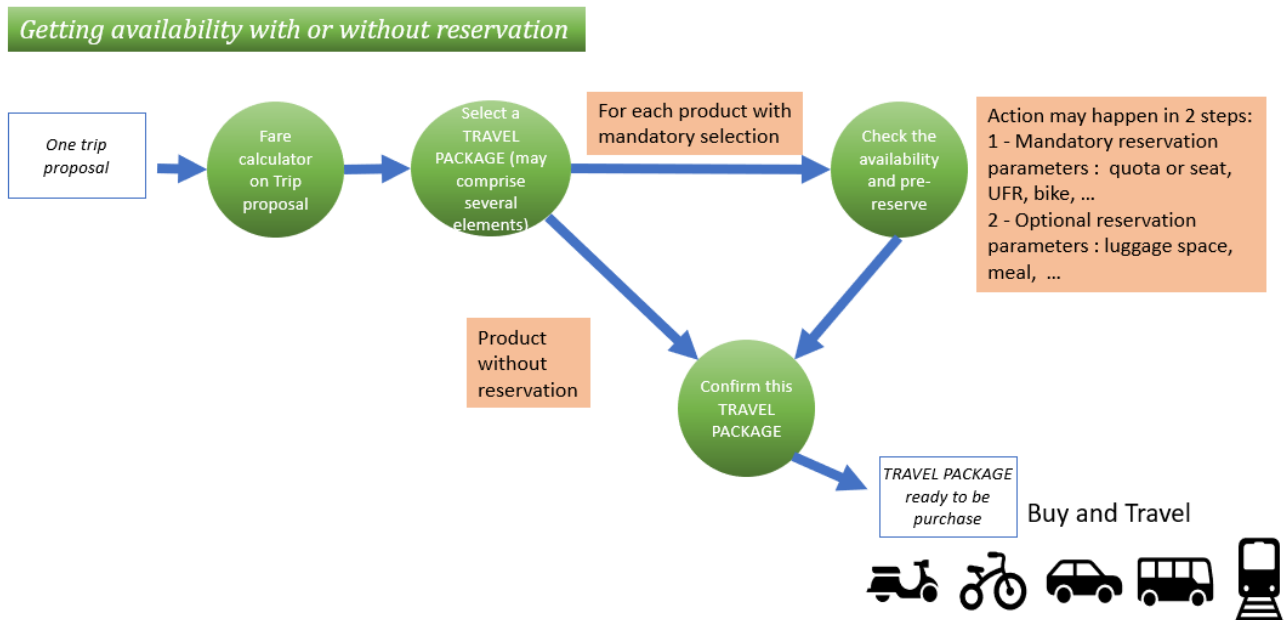


Figure 4 – Illustration of the use-case “Getting availability with or without reservation”

Table 8 – Details of the use-case “Getting availability with or without reservation”

Use Case 3: Getting availability with or without reservation	
Summary	When choosing a TRAVEL PACKAGE for a specific trip, with or without making a reservation, the TRANSPORT CUSTOMER wants to be informed about the availabilities of each solution and their options before making a purchase. <u>The customer may or may not place a reservation, temporarily or confirmed. An availability check may be done in advance to preplan (e.g., to check availability of product or seat) or also at the time of travel to check the current availability of assets.</u>
Description	It is assumed that possible trip options (trip patterns) are provided by a trip planner and that the TRANSPORT CUSTOMER wants to know about the different solutions composed with different SALES OFFER PACKAGES (in particular, prices). One trip proposal is sent to the fare calculators. Solutions are returned, each expressed by a TRAVEL PACKAGE (based on relevant SALES OFFER PACKAGES). They can be several on each trip proposal. A TRAVEL PACKAGE is composed with zero, one or more FARE PRODUCTS on each leg or is combining many legs. If the product has no mandatory reservation nor options that need separate pricing, the final price is given by the fare calculator, and the TRANSPORT CUSTOMER can directly select and buy this FARE PRODUCT. If a FARE PRODUCT has mandatory reservation and/or options that need pricing, the TRANSPORT CUSTOMER must select every needed data: class, deck, window or corridor, bike space, etc.. The platform starts the reservation process to block the selected options until the TRANSPORT CUSTOMER purchases it (or abandons it or after a limited period). It is the pre-purchase step. The TRANSPORT CUSTOMER can modify/add ancillaries/options at this stage. If there is an issue with the reservation (example: another TRANSPORT CUSTOMER has purchased the last seat in the meantime), the TRANSPORT CUSTOMER must be alerted in order to modify a parameter and thus the TRAVEL PACKAGE. Which parameters and options are mandatory depends on the product. For a point-to-point fare, they include the specific origin and destination of the trip (with scheduled timing). Further parameters may be needed to fully specify access rights, such as the selection of a seat reservation, travel class, luggage space and onboard services. For pass products allowing repeated travel, they might include a zone or an origin-destination pair, and a time period. For all the FARE PRODUCTS, the TRANSPORT CUSTOMER is informed about the travel conditions and related TRAVEL GUARANTEES. Once any mandatory reservation and other options are selected, the TRANSPORT CUSTOMER can proceed to the purchase and travel.
Main roles and actors	BoB: Ticket Vendor, Price & product service, Ticketing service, Customer OSDM 3.2: Fare Provider, Allocator, Customer, Distributor TOMP-API: Reseller (MaaS Service Provider) & Transport operator (including parking organisations and EV charging organisations), data aggregators, and indirect: customer, payment service provider, identity provider.

	TAP TSI: Availability request for a set of products from the product catalogue. (Only applicable for products subject to reservation). Transmodel: TRANSPORT USER ROLE (in particular PURCHASER ROLE,), FARE ORGANISATION ROLE (in particular FARE PRODUCT ATTRIBUTOR ROLE but also FARE PRODUCT OWNER/DISTRIBUTOR/RETAILER ROLES).
Main objects	BoB: Product, Manifest, Ticket Bundle, Booking OSDM: Product, offer, coach-layout, zone, reduction-card, bookings, fare, TripCoverage, ProductLegAssociation, seat Transmodel: TRIP, TRAVEL SPECIFICATION, SERVICE JOURNEY, PASSENGER FARE OFFER, FARE PRODUCT, SALES OFFER PACKAGE, FARE PRICE, SCOPING PARAMETERS (TARIFF ZONE, TOPOGRAPHIC AREA), USAGE PARAMETERS, LOCABLE SPOT, PASSENGER SPOT ALLOCATION, DECK PLAN, TRAVEL GUARANTEES, CUSOMER ACCOUNT, TRAVEL PACKAGE, CUSTOMER PURCHASE PACKAGE, MOBILITY SERVICE, TYPE OF MOBILITY SERVICE, TOPOGRAPHIC PLACE, MOBILITY SERVICE GEOMETRY RESTRICTION, TRANSPORT ORGANISATION, EQUIPMENT, AVAILABILITY CONDITION, SERVICE BOOKING ARRANGEMENT, MOBILITY SERVICE CONSTRAINT ZONE, TRANSPORT TYPE, MODE OF OPERATION, VEHICLE, TRAVEL PACKAGE, TRIP PATTERN, VEHICLE JOURNEY, CLASS OF USE, FACILITY SET, SPOT ALLOCATION METHOD, TRIP PROFILE REQUIREMENTS, VEHICLE SPOT ALLOCATION, LOCABLE SPOT, PASSENGER SPOT ALLOCATION TOMP-API (v2): Package (v1.x: Booking), Product (v1.x: Asset Type), Leg, Asset, Ancillary, Travel Document, Card, Card Types, License, License Types, User Profile. And it refers to external data sources like GBFS, DATEX-II, NeTEx, GTFS, APDS, OCPI, or Journey planner output (OJP, OTP) TAP TSI: products zones, reduction card, fares
Contributions	BoB: Product API: POST /product [<i>retrieve products based on trip plan</i>] Product API: POST /manifest [<i>retrieve manifest /w booking required</i>] Booking API: POST /booking [<i>check availability, create booking</i>] Ticket API: POST /ticket [<i>request ticket based on manifest</i>] Booking API: PATCH /booking/{bookingId} Ticket API: GET /ticketbundle [<i>retrieve ticket bundle</i>] OSDM 3.3.0 API: GET /products GET /coach-layouts GET /coach-layouts/{layoutId} GET /zones GET or POST /availabilities/place-map GET /availabilities/nearby GET /availabilities/preferences POST /offers

	POST /bookings (in state prebooked) POST /bookings/{bookingId}/booked-offers/ Transmodel: TRAVEL PACKAGE SPOT AVAILABILITY REQUEST (TRAVEL PACKAGE, CLASS OF USE, ACCESSIBILITY ASSESSMENT, TRIP PROFILE REQUIREMENTS, USER PROFILE, ENTITLEMENT, TYPE OF SPOT AFFINITY) & TRAVEL PACKAGE SPOT AVAILABILITY DELIVERY (VEHICLE JOURNEY SPOT ALLOCATION, PASSENGER SPOT ALLOCATION, FACILITY SET), OSDM 3.3.0 API: GET /products GET /coach-layouts GET /coach-layouts/{layoutId} GET /zones GET or POST /availabilities/place-map GET /availabilities/nearby GET /availabilities/preferences POST /offers POST /bookings (in state prebooked) POST /bookings/{bookingId}/booked-offers/ Transmodel: TRAVEL PACKAGE SPOT AVAILABILITY QUERY TRAVEL PACKAGE SPOT RESERVATION QUERY MOBILITY SERVICE QUERY OJP: TripInformationRequest SIRI services: Stop Monitoring Vehicle Monitoring Facility Monitoring TOMP-API 1.x: POST /offers POST /bookings/{id} POST /planning/inquiries GET /operator/available-assets TOMP-API 2.0: GET /collections/offers/items POST /collections/assets/items POST /processes/select-offers/execution POST /processes/assign-asset/execution POST /processes/purchase-package/execution POST /processes/purchase-offers/execution POST /processes/commit-package/execution ENTUR: GET /sales/v1/offers/search/trip/trip-pattern/{id} POST /sales/v1/reserve-offers POST /v1/orders PUT /v1/orders/{orderId} POST /v1/orders/{orderId}/reservations
--	--

	TAP-TSI: AvailabilityRequest ReservationRequest
--	---

6.5.2.4 Purchase process

This use case presents the main steps of the purchase process from the TRANSPORT CUSTOMER point of view.

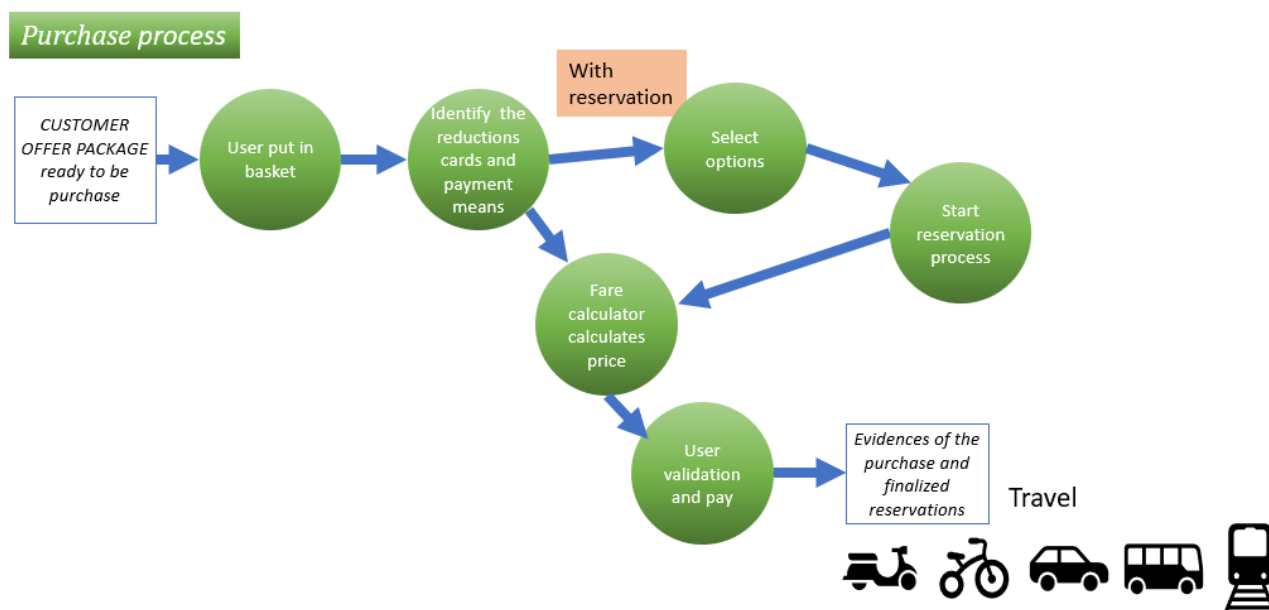


Figure 5 – Illustration of the use-case “Purchase process”

Table 9 – Details of the use-case “Purchase process”

Use Case 4: Purchase process	
Summary	When a TRANSPORT CUSTOMER has selected one or more PASSENGER FARE OFFERS and any required choices for a TRAVEL PACKAGE, he/she proceeds to the purchase.
Description	It is assumed that the TRANSPORT CUSTOMER has chosen and parameterised the FARE PRODUCTs and SALES OFFER PACKAGEs presented in a TRAVEL PACKAGE he/she wants to buy. He/she can select at the same time any reduction rights for which he/she is eligible, for example, from a rail card, as represented by a SALE DISCOUNT RIGHT or USAGE DISCOUNT RIGHT (i.e., other types of FARE PRODUCTs) that they already own. All the SALES OFFER PACKAGEs are added to his/her "basket". If on some SALES OFFER PACKAGEs there are mandatory (or optional) parameters to select (e.g., luggage constraints, seat class, seat location, etc.), the TRANSPORT CUSTOMER has to make his selection. For example, the seat reservation can be done by selecting a spot on a deck plan (with all information about class, window, ...) or just with a class selection (reservation with quotas), leaving the system to assign a place.

	Once a selection is done, the holding process is started. The holding process (locking the resources) lasts a limited time (as a parameter), enough to validate and pay for the purchase by the TRANSPORT CUSTOMER (some minutes). Once the payment is made, the reservation is finalised, and all options and seat are definitively reserved. In case of unexpected issues arising, a rollback process is possible. If successful, the traveller receives the evidence of the purchase, reservations and TRAVEL DOCUMENTs associated with the SALES OFFER PACKAGES he has bought.
Main roles and actors	BoB: Ticket Vendor, Price & product service, Ticketing service, Booking service, Customer OSDM 3.2: Fare Provider, Allocator, Customer, Distributor TOMP-API: Reseller (MaaS Service Provider) & Transport operator (including parking organisations and EV charging organisations), data aggregators, ticket stocks and (optionally) dataspace administrators, and indirect: customer, payment service provider, identity provider. Transmodel: TRANSPORT USER ROLE (in particular PURCHASER ROLE), FARE ORGANISATION ROLE (in particular, FARE PRODUCT ATTRIBUTOR ROLE but also FARE PRODUCT OWNER/DISTRIBUTOR/RETAILER ROLES)
Main objects	BoB: Product, Manifest, Ticket Bundle, Booking OSDM 3.3.0: booking, offer, passenger, fulfilment Transmodel: TRIP, SALES OFFER PACKAGE, FARE TABLE, SCOPING PARAMETERS, USAGE PARAMETERS, PASSENGER SPOT ALLOCATION, DECK PLAN, LOCABLE SPOT, TRAVEL GUARANTEES, CUSTOMER ACCOUNT, TARIFF ZONE, CUSTOMER PURCHASE PACKAGE, DISTRIBUTION, TRAVEL DOCUMENT, TRANSPORT CUSTOMER, FARE CONTRACT, SALES TRANSACTION, GROUP TICKET, TYPE OF REQUEST, TRAVEL PACKAGE, TRAVEL PREFERENCES, FARE CONTRACT, CUSTOMER PURCHASE STATUS, OFFERED TRAVEL SPECIFICATION, TRAVEL DOCUMENT, FULFILMENT METHOD, CUSTOMER ACCOUNT, TRAVEL PREFERENCES, LOCATABLE SPOT, SPOT RESERVATION, TRAVEL SPECIFICATION, TRAVEL PACKAGE DOCUMENT FULFILMENT REQUEST POLICY, MEDIUM ACCESS DEVICE, MEDIUM APPLICATION INSTANCE TOMP-API (v2): Package (v1.x: Booking), Product (v1.x: Asset Type), Leg, Asset, Ancillary, Travel Document, Usage parameter. And it refers to external data sources like GBFS, DATEX-II, NeTEx, GTFS, APDS, OCPI, or Journey planner output (OJP, OTP) TAP TSI: reservation, accommodation (seat, couchette, berth, bike place, car place)
Contributions	BoB: Product API: POST /product [retrieve product offerings] Product API: POST /manifest [retrieve manifest /w discounts etc.] Booking API: POST /booking [check availability, create booking] Ticket API: POST /ticket [request ticket based on manifest]

	Booking API: PATCH /booking/{bookingId} Ticket API: GET /ticketbundle [<i>retrieve ticket bundle</i>] OSDM 3.3.0: POST /bookings in state prebooked, POST /bookings/{bookingId}/fulfillments,¹⁰ PATCH /bookings/{bookingId}, POST /bookings DELETE /bookings/{bookingId}, GET /bookings/{bookingId}/booked-offers/{bookedOffersId} POST /bookings/{bookingId}/booked-offers/ GET /bookings/{bookingId}/passengers/{passengerId}, PATCH /bookings/{bookingId}/passengers/{passengerId}, GET /fulfillments/{fulfillmentId}, Transmodel: TRAVEL PACKAGE QUERY SPOT QUERY FULFILMENT QUERY ENTUR: POST /sales/v1/reserve-offers, PUT /v1/orders/{orderId}/travellers/{travellerId}, POST /v1/pdf-tickets/{orderId}/createOrGet TOMP-API 1.x: POST /bookings POST /bookings/{id}/events – commit PUT /bookings/{id} GET /bookings/{id} TOMP-API 2.0: GET /collections/offers/items POST /processes/select-offers/execution POST /processes/assign-asset/execution POST /processes/update-traveller/execution POST /processes/purchase-offers/execution POST /processes/purchase-package/execution POST /processes/confirm-purchase/execution GET /collections/packages/items TAP-TSI: ReservationRequest, GetStoredPersonalData, AddPersonalDataRequest
--	--

6.5.2.5 Change before the travel

This use case covers the possible modifications of the acquired access rights (i.e., after the completion of the purchase) and before the start using the acquired access rights.

¹⁰ “fulfilment” in Transmodel is the delivery of the ticket when in OSDM “fulfilment” is a confirmation of purchase

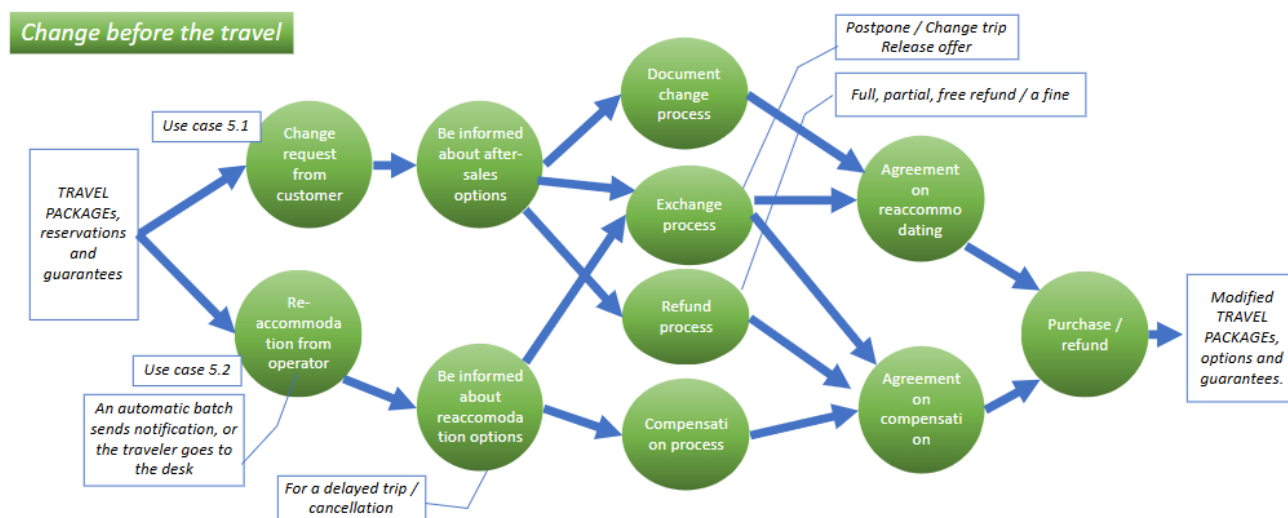


Figure 6 – Illustration of the use-case “Change before the travel”

Table 10 – Details of the use-case “Change before the travel”

Use Case 5: Change before the travel	
Summary	UC5.1 “After-sale request by the traveller” Once a purchase has been made, a TRANSPORT CUSTOMER can request an after-sale action at any moment during travel, but the focus here is on the process before the travel. UC5.2 “Change under the responsibility of any providers involved in the trip for a through ticket (not the traveller)” Change in the previously advertised services (re-accommodation)
Description	For both use cases, it is assumed that the TRANSPORT CUSTOMER has prepared his travel. He has purchased a TRAVEL PACKAGE that fits with his travel. It can be one or several unit ticket or a complex through ticket with guaranteed connections and more than one TRANSPORT MODE (bike and rail, rail and urban bus). It is also assumed that the requested action is authorised by the terms of sale. UC5.1 “After-sale request by the traveller” A TRANSPORT CUSTOMER can decide to change some parameters of his/her trip before travel according to the commercial conditions placed on the FARE PRODUCTS and SALES OFFER PACKAGES being sold, represented in Transmodel by USAGE PARAMETERS such as CANCELLING, REFUNDING, EXCHANGING, TRANSFERABILITY, MINIMUM STAY, etc. For all such requests, the TRANSPORT CUSTOMER must provide evidence of the TRAVEL PACKAGE he bought (with its associated USAGE PARAMETERS and TRAVEL GUARANTEES) by connecting to his online CUSTOMER ACCOUNT, going to a desk, or identifying with his TRAVEL DOCUMENT (anonymous) on a ticket vending machine.

	For each type of request, a process is starting between both parties in order to find an agreement. The after-sales proposals may be different depending on the TRAVEL PACKAGE and its specified after-sale conditions and TRAVEL GUARANTEES,, the passenger's specific choice of options, the moment of the request and the TRANSPORT CUSTOMER's particular wishes. Before the travel starts, when he/she cancels the travel, the TRANSPORT CUSTOMER can usually obtain a full or partial refund (depending on the REFUNDING parameter) For a change in reservation or to the TRAVEL DOCUMENTs, usually, the TRANSPORT CUSTOMER must pay extra-charges (depending on the RESERVING user parameter) <i>To be noted: During the travel or when it is very close to start, the TRANSPORT CUSTOMER can often request for a change in his reservation, take the next train or add a step. The after-sale usually proposes a FARE PRODUCT exchange with an extra charge.</i> <i>After the travel, usual after-sale requests are based on issue during the travel. The TRANSPORT CUSTOMER had a delayed train and missed his connection or received a fine for an issue with his TRAVEL DOCUMENT. In this case the TRANSPORT CUSTOMER must request the after-sale desk for a compensation or to pay/not pay the fine.</i> When the TRANSPORT CUSTOMER request is accepted by the DISTRIBUTOR, the after-sale action and payment are executed. At the end of the after-sales process, the TRANSPORT CUSTOMER receives the evidence, the new/updated TRAVEL PACKAGE and TRAVEL DOCUMENTs. UC5.2 "Change under the responsibility of any providers involved in the trip for a through ticket (not the traveller)" An OPERATOR can decide schedule changes or experience perturbations to the normal schedule (Re-accommodation). When such changes occur, the OPERATOR should inform the future travellers that their TRAVEL PACKAGEs must be modified (adapted or cancelled). Either the OPERATOR contacts the TRANSPORT CUSTOMERs, or the TRANSPORT CUSTOMERs take the initiative (if they have been monitoring their travel). For each type of request, a process is starting between both parties to find an agreement. Usually, the OPERATOR proposes an exchange on another JOURNEY or another TRANSPORT MODE for a part of the trip, in case of broken connection, or for the full trip in the case of a service cancellation. If the TRANSPORT CUSTOMER doesn't accept the exchange or prefers instead to have compensation because the proposal doesn't fit with his/her expectations (e.g., if the arrival is too late, or travel requires use of another transport solution not previously included in the OPERATOR proposal), both parties find an agreement as to a suitable compensation. It can be a refund or a monetary value to spend on future travel with the same OPERATOR.
--	--

Main roles and actors	BoB: Ticket Vendor, Price & product service, Ticketing service, Customer OSDM 3.2: Fare Provider, Allocator, Customer, Passenger, Distributor TOMP-API: Reseller (MaaS Service Provider) & Transport operator (including parking organisations and EV charging organisations), data aggregators, ticket stocks and indirect: customer, payment service provider, identity provider. Transmodel: TRANSPORT USER ROLE (in particular PURCHASER ROLE,), FARE ORGANISATION ROLE (in particular FARE PRODUCT ATTRIBUTOR ROLE but also FARE PRODUCT OWNER/DISTRIBUTOR/RETAILER ROLES), GROUP TICKET
Main objects	BoB: Ticket OSDM: booking, offer, refund, exchange, reimbursement, passenger, document, Transmodel: TRAVEL SPECIFICATION, TRAVEL PACKAGE, CUSTOMER PURCHASE PACKAGE, USAGE PARAMETER, CUSTOMER RESERVATION, TRAVEL DOCUMENT, TRANSPORT CUSTOMER, FARE CONTRACT, SALES TRANSACTION, TRAVEL GUARANTEE, SPOT ALLOCATION, LOCATABLE SPOT TOMP-API 1.x: Deposit, Fare structure, Journal-entry, Additional costs (covering REFUND) TOMP-API 2.0: Deposit, Fare structure, Journal-entry, Additional costs, Refund, Damage, Fine, Voucher, Replacements, Stolen, Extra_usage ENTUR: order, reservation, supplement product
Contributions	BoB: <i>Requesting a refund:</i> Ticket API: GET /ticket/{ticketId}/refundableStatus [<i>Check refundable amount of ticket</i>] Ticket API: PUT /ticket/{ticketId}/refundStatus [<i>Cancel and request a refund of ticket</i>] <i>In BoB, all tickets are immutable. Any changes are carried out by requesting a new ticket while recovering the remaining value of the previous:</i> Ticket API: GET /ticket/{ticketId}/recoverableStatus [<i>Check recoverable amount of ticket, if trading it in for a new</i>] OSDM 3.3.0: POST /bookings/{bookingId}/refund-offers, PATCH /bookings/{bookingId}/refund-offers/{refundOfferId}, DELETE /bookings/{bookingId}/refund-offers/{refundOfferId}, POST /bookings/{bookingId}/exchange-offers, POST /bookings/{bookingId}/exchange-operations, PATCH /bookings/{bookingId}/exchange-operations, DELETE /bookings/{bookingId}/exchange-operations, POST /bookings/{bookingId}/release-offers, GET /bookings/{bookingId}/release-offers/{releaseOfferId}, PATCH /bookings/{bookingId}/release-offers/{releaseOfferId}, DELETE /bookings/{bookingId}/release-offers/{releaseOfferId}, POST /bookings/{bookingId}/cancel-fulfillments-offers, GET /bookings/{bookingId}/passengers/{passengerId}, PATCH /bookings/{bookingId}/passengers/{passengerId}, POST /bookings/{bookingId}/documents,

	GET /bookings/{bookingId}/documents/{documentId}, DELETE /bookings/{bookingId}/documents/{documentId}, POST /bookings/{bookingId}/reimbursements, GET /bookings/{bookingId}/reimbursements, PATCH /bookings/{bookingId}/reimbursements Transmodel: TRAVEL PACKAGE ALTERATION / CANCELLATION / REFUND QUERY, TRAVEL PACKAGE REVOKE FULFILMENT REQUEST, TRAVEL PACKAGE SPOT CANCELLATION QUERY TOMP-API v2: GET /collections/packages/items GET /collections/payments/items POST /collections/redress-options/items POST /processes/claim-redress-option/execution POST /processes/confirm-redress-option/execution (for options & confirmation) POST /processes/cancel-package/execution (for options & confirmation) POST /processes/request-payments/execution POST /processes/assign-asset/execution POST /processes/assign-ancillary/execution BoB: It is possible to get the refundable amount, support requests: put / get / change, BoB.Ticket.API GET /ticket/{ticketId}/refundableStatus, PUT /ticket/{ticketId}/refundStatus, BoB.Traveller.API PUT /traveller/{travellerId} ENTUR: refundOptions request, cancelOrder request, updateOrder request, DELETE /v1/orders/{orderId}/reservations/{reservationId}, PUT /v1/orders/{orderId}/travellers/{travellerId} TAP-TSI: CancelRequest, PartialCancelRequest, GetStoredPersonalData, AddPersonalDataRequest
--	---

6.5.2.6 Change during travel

This use case presents some of the impacts that may happen when the travel is changed after it has begun. The change can be the result of decisions by wither the TRANSPORT CUSTOMER or the OPERATOR.

The changes permitted to travel will depend on the detailed specification of the TRAVEL PACKAGE, in particular on the characteristics of the selected FARE PRODUCTS and their associated after-sales conditions.

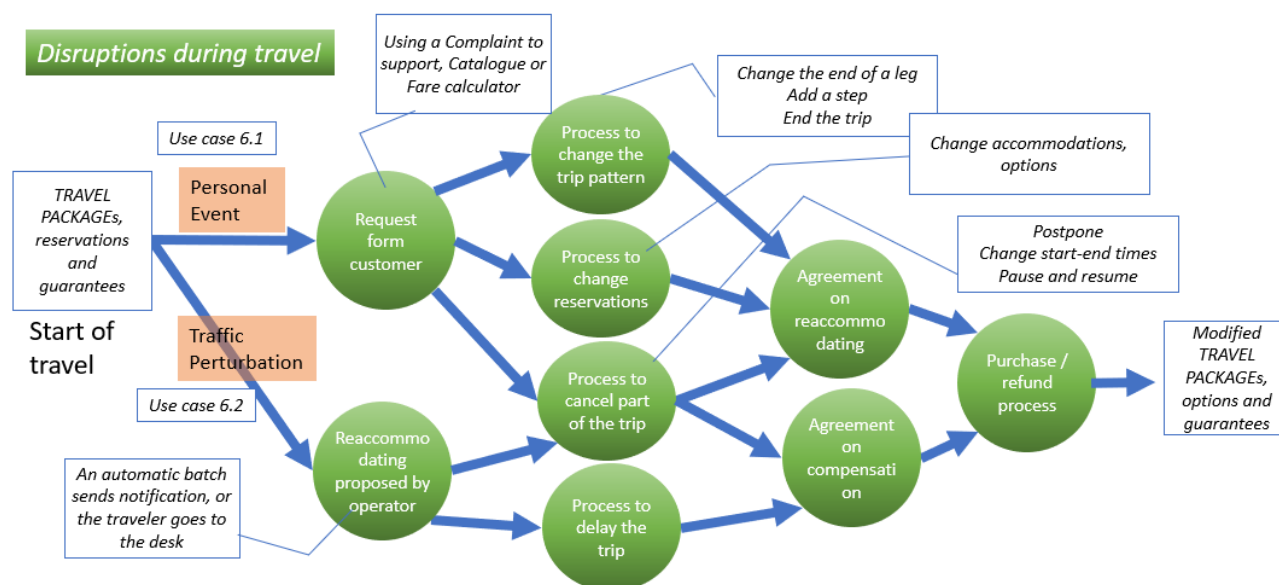


Figure 7 – Illustration of the use-case “Disruptions during travel”

Table 11 – Details of the use-case “Disruptions during travel”

Use Case 6: Change during travel	
Summary	UC6.1 “Change by the traveller” After-sale request by the traveller After the trip departure, a personal event (e.g., a change of plans, sickness, a forgotten meal, missed connection) can lead the traveller to modify his/her passenger accommodation or to cancel or extend the trip or a part of the trip. UC6.2 “Change under the responsibility of any providers involved in the trip (not the traveller)” Disruptions (e.g., accidents) and related trip modifications made by the OPERATOR may modify the trip impacting SALES OFFER PACKAGES, reservations, final prices, TRAVEL GUARANTEES and TRAVEL DOCUMENTS.
Description	For both use cases, it is assumed that the TRANSPORT CUSTOMER prior to travel has made a purchase and also possibly made reservations (through requests to a trip planner, purchase of SALES OFFER PACKAGES, seat reservation, etc.), and that the corresponding TRAVEL DOCUMENTS have been issued. (e.g., paper ticket, printed CB2D on a sheet, smart card, mobile

	application, etc). It is also assumed that the requested action to make a change is permitted by the terms of sale (as described by USAGE PARAMETERS). When a traveller starts his travel, he possesses TRAVEL DOCUMENTs which provide evidence of his purchased TRAVEL PACKAGES. UC6.1 “Change by the traveller” During travel, travellers may wish to initiate changes to their CUSTOMER PURCHASE PACKAGE due to a personal change of plan, missed connections, desire for additional services, etc. The TRANSPORT CUSTOMER may consequently request a change: such as modification of the trip pattern to cancel or add to parts of the trip (e.g., a destination change), or to alter the travel times (as say with a missed connection due to the traveller arriving late), or to change the trip options (e.g., with an additional meal). The traveller can review the new solutions using a mobile application or be informed / make arrangements at an information desk or ticket office. He can modify the trip pattern, change an option or postpone his travel. For each case, the traveller must check his CUSTOMER PURCHASE PACKAGE (and its USAGE PARAMETERS describing the commercial conditions of sale), TRAVEL DOCUMENTs, and TRAVEL GUARANTEEs. He may need to buy new FARE PRODUCTS and possibly be refunded for the old ones. In each case, the USAGE PARAMETERS (EXCHANGING, CANCELLING, REFUNDING, etc.) provide information about the conditions of sale governing a refund or other change initiated by the user, and whether any additional payment is required. The TRAVEL GUARANTEEs associated with the purchase may also be relevant for deciding the conditions permitting changes, in particular if it was some failure on the part of the operator that led to the travellers wish for a change. TRANSPORT CUSTOMER requests for alterations are processed as an after-sale action (purchase) with possible further payment and with evidence of the purchase. UC6.2 “Change under the responsibility of any providers of the trip (i.e., not the traveller)” Coming from a traffic perturbation, further disruptions can impact the CUSTOMER PURCHASE PACKAGES: operator delays (technical, weather, strikes, ...), un-guaranteed connections, modification of trip options by the operator. In some circumstances, the OPERATOR may propose a re-accommodation to the TRANSPORT CUSTOMER. The proposal can be a substitution of the original trip with another trip (new schedule, another day, alternative route, etc.) or/and a refund when the option is no longer available (e.g., a meal or Wi-Fi service). In some cases, the trip or a part of the trip may be cancelled and refunded. Then the TRANSPORT CUSTOMER decides to accept one proposal or not and to request for a compensation for an issue during the travel (for example, a broken guaranteed connection, or a very late arrival) according to the TRAVEL GUARANTEEs provided with his/her CUSTOMER PURCHASE PACKAGE. The chosen solution is processed as an after-sale action (purchase) with payment and with evidence of the purchase. The TRANSPORT CUSTOMER can receive new TRAVEL DOCUMENTs, new TRAVEL GUARANTEEs in a new or modified CUTOMER PURCHASE PACKAGE.
--	---

	Note that for some large-scale disruptions, no change is made to the individual TRAVEL PACKAGES and TRAVEL DOCUMENTs but rather a general 'easement' is announced and the checking of access rights relaxed. For example, in the case of a derailment, passengers holding a ticket valid only for a specific train might be permitted to travel on any train or bus on various alternative routes.
Main roles and actors	BoB: Ticket Vendor, Price & product service, Ticketing service, Customer OSDM 3.2: Fare Provider, Allocator, Customer, Passenger, Distributor TOMP-API: Reseller (MaaS Service Provider) & Transport operator (including parking organisations and EV charging organisations), data aggregators, ticket stocks and (optionally) dataspace administrators, and indirect: customer, payment service provider, identity provider. Transmodel: TRANSPORT USER ROLE (in particular PURCHASER ROLE), FARE ORGANISATION ROLE (in particular FARE PRODUCT ATTRIBUTOR ROLE but also FARE PRODUCT OWNER/DISTRIBUTOR/RETAILER ROLES)
Main objects	BoB: Ticket OSDM 3.3.0: Transmodel: LEG, TRAVEL SPECIFICATION, TRAVEL PACKAGE, CUSTOMER PURCHASE PACKAGE, USAGE PARAMETER, CUSTOMER RESERVATION, TRAVEL DOCUMENT, TRANSPORT CUSTOMER, FARE CONTRACT, SALE TRANSACTION, TRAVEL GUARANTEE, SPOT ALLOCATION, LOCATABLE SPOT, TYPE OF REQUEST, PASSENGER FARE OFFER, OFFERED TRAVEL SPECIFICATION, REQUESTED TRAVEL SPECIFICATION, FARE OFFER PRICE, CUSTOMER PURCHASE STATUS, TYPE OF RESALE, RESALING, REBATING, TRANSFERABILITY, TRIP PATTERN, PASSENGER SPOT ALLOCATION, SERVICE JOURNEY, SPOT AVAILABILITY STATUS TOMP-API (v2): Package (v1.x: Booking), Product (v1.x: Asset Type), Leg, Asset, Ancillary, Travel Document, Usage parameter, Traveller (incl. User needs & entitlements), Customer (incl. payment details). And it refers to external data sources like GBFS, DATEX-II, NeTEx, GTFS, APDS, OCPI, or Journey planner output (OJP, OTP)
Contributions	BoB: <i>In BoB, this use-case is identical to use-case 6.</i> OSDM 3.3.0: PATCH /bookings/{bookingId}, DELETE /bookings/{bookingId}, POST /bookings/{bookingId}/cancel-fulfillments-offers, GET availabilities/place-map, POST /availabilities-placemap, POST /bookings/{bookingId}/booked-offers/, POST /legs/{id}/ancillaries/{category}/{number}, POST /booking, POST /bookings/{bookingId}/refund-offers

	Transmodel: TRAVEL PACKAGE ALTERATION / CANCELLATION / REFUND QUERY TRAVEL PACKAGE SPOT CANCELLATION QUERY TOMP-API 1.x: POST /bookings/{id}/events - COMMIT , POST /bookings/{id}/events - CANCEL, GET /bookings/{id}, DELETE /bookings/{bookingId}, POST /bookings/{id}/notifications, GET /legs/{id}/available-assets, POST /legs/{id}/events - ASSIGN_ASSET, DELETE /legs/{id}/ancillaries/{category}/{number}, POST /legs/{id}/events - POSTPONE_TIME, POST /legs/{id}/events - PREPARE, POST /legs/{id}/events - CANCEL TOMP-API 2.0: GET /collections/packages/items POST /processes/update-travel-specification/execution POST /processes/start-leg/execution POST /processes/end-leg/execution POST /processes/postpone-leg/execution POST /processes/extend-leg/execution POST /processes/assign-asset/execution POST /processes/request-support/execution POST /processes/update-traveller/execution POST /processes/update-ancillary/execution POST /collections/redress-options/items POST /processes/claim-redress-option/execution POST /processes/confirm-redress-option/execution ENTUR: refundOptions request, cancelOrder request (hold time can be extended via updateOrder request), updateOrder request, getOrder request, GET /sales/v1/offers/search/trip/trip-pattern/{id}, delivers seatingCapacity, POST /v1/orders/{orderId}/reservations, DELETE /v1/orders/{orderId}/reservations/{reservationId}, POST /v1/update-order/supplement-products TAP-TSI: CancelRequest, PartialCancelRequest
--	---

6.5.2.7 Customer account consultation

This use case covers the actions of TRANSPORT CUSTOMERs when connected to their CUSTOMER ACCOUNTs (by web site, mobile application, on desk) to access or alter their purchases.

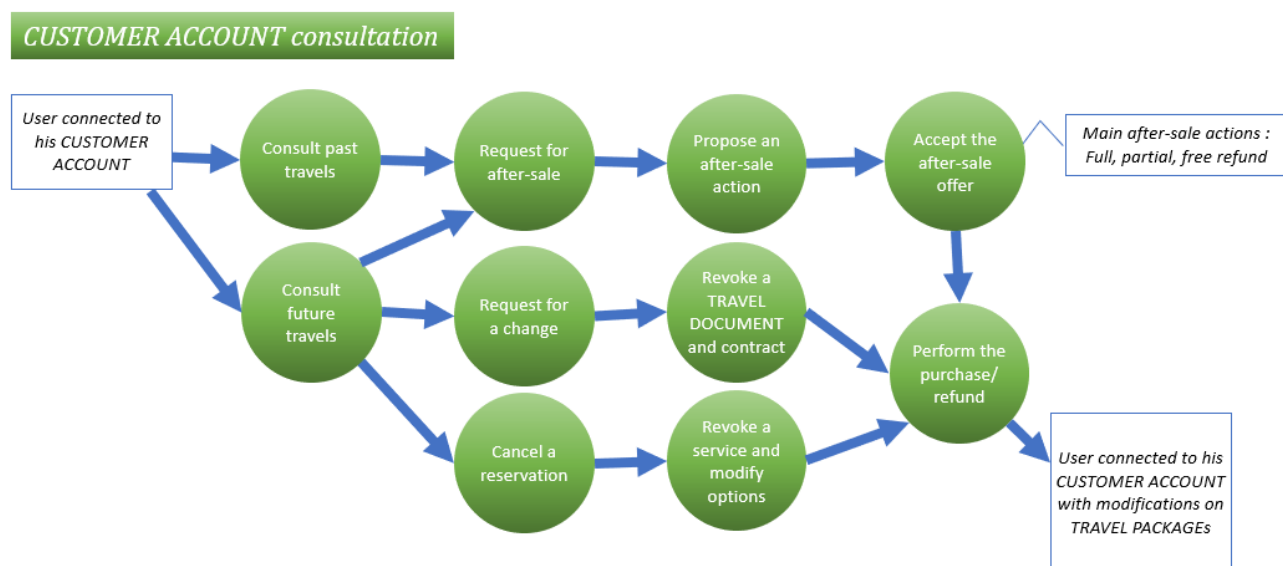


Figure 8 – Illustration of the use-case “Customer account consultation”

Table 12 – Details of the use-case “Customer account consultation”

Use Case 7: Customer account consultation	
Summary	A TRANSPORT CUSTOMER can consult his/her CUSTOMER ACCOUNT and perform some actions on FARE PRODUCTS-SALES OFFER PACKAGES he/she has purchased.
Description	It is assumed that TRANSPORT CUSTOMERs have a CUSTOMER ACCOUNT on a mobility platform with at least one contract (and associated CUSTOMER PURCHASE PACKAGES they have purchased). They can consult their CUSTOMER ACCOUNTs and depending on the platform, carry out some actions. For CUSTOMER PURCHASE PACKAGES associated with travel already completed (or not actually executed), the TRANSPORT CUSTOMER can request an after-sale action such as a refund (for example, in the case of problems having occurred during the travel). For CUSTOMER PURCHASE PACKAGES associated with future travel, the TRANSPORT CUSTOMER can, according to the sales conditions, modify one or more elements of the CUSTOMER PURCHASE PACKAGE, modify his travel documents, or change the service or certain options. In both cases, the customer may change his TRAVEL DOCUMENT and possibly pay an extra charge. Moreover, for future travel, the reservation (and the trip) can be cancelled according to the after-sales conditions as specified by USAGE PARAMETERS. The access rights are revoked, and the price can be refunded. At the end of the process, the TRANSPORT CUSTOMER can consult the modifications on TRAVEL PACKAGES associated with his CUSTOMER ACCOUNT.
Main roles and actors	BoB: Ticket Vendor, Customer OSDM 3.2: Allocator, Customer, Distributor

	TOMP-API: Reseller (MaaS Service Provider) & Transport operator (including parking organisations and EV charging organisations), data aggregators, ticket stocks and indirect: customer, payment service provider, identity provider. BoB Transmodel: TRANSPORT USER ROLE (in particular PURCHASER ROLE,), FARE ORGANISATION ROLE (in particular FARE PRODUCT ATTRIBUTOR ROLE but also FARE PRODUCT OWNER/DISTRIBUTOR/RETAILER ROLES), Entur: TAP TSI: Attributor, Customer, Distributor
Main objects	OSDM: booking, passenger, complaint, fulfilment Transmodel: CUSTOMER ACCOUNT, TRAVEL PACKAGE, CUSTOMER PURCHASE PACKAGE, CUSTOMER RESERVATION, TRAVEL DOCUMENT, TRANSPORT CUSTOMER, FARE CONTRACT, FARE CONTRACT ENTRY, SALES TRANSACTION, CUSTOMER PAYMENT MEANS, GROUP TICKET, TYPE OF REQUEST, PASSENGER FARE OFFER, OFFERED TRAVEL SPECIFICATION, REQUESTED TRAVEL SPECIFICATION, FARE OFFER PRICE, CUSTOMER PURCHASE STATUS, TYPE OF RESALE, USAGE PARAMETERS (EXCHANGING, REFUNDING, RESERVING, RESELLING, REBATING, TRANSFERABILITY), TYPE OF TRAVEL DOCUMENT, TRANSPORT CUSTOMER, PASSENGER FARE OFFER, TRIP PATTERN, OFFERED TRAVEL SPECIFICATION, PASSENGER SPOT ALLOCATION, SERVICE JOURNEY, LOCABLE SPOT, SPOT AVAILABILITY STATUS, MEDIUM APPLICATION INSTANCE, SECURITY LISTING, SECURITY LIST TOMP-API 1.x: TOMP-API 2.0: a new extension, customer registration
Contributions	OSDM: Find / consult / change / Cancel a booking, GET /bookings/{bookingId}/passengers/{passengerId}, PATCH /bookings/{bookingId}/passengers/{passengerId}, POST /complaints, GET /complaints/{complaintId}, PATCH /complaints/{complaintId}, PATCH /fulfillments/{fulfillmentId}, PATCH /bookings/{bookingId}/exchange-operations Transmodel: TRAVEL PACKAGE ALTERATION / CANCELLATION / REFUND QUERY TRAVEL PACKAGE SPOT CANCELLATION QUERY TRAVEL PACKAGE REVOKE FULFILMENT QUERY TOMP-API 1.x: GET /bookings, POST /bookings-search, DELETE /bookings/{bookingId}, PUT /bookings/{id}, POST /bookings/{id}/notifications, TOMP-API 2.0: \ GET /collections/redress-options/items

	POST /processes/claim-redress-option/execution POST /processes/confirm-redress-option/execution POST /processes/search-offers/execution with exchange package POST /processes/release-package/execution POST /processes/update-assets/executionBoB : BoB.Traveller.API PUT /traveller/{travellerId}, POST /traveller/{travellerId}/wallet/transaction, GET /traveller/{travellerId}, PUT /traveller/{travellerId}, BoB.Ticket.API PUT /ticket/{ticketId}/revoke, ENTUR: getOrder request, cancelOrder request, PUT /v1/orders/{orderId}/travellers/{travellerId} TAP-TSI: GetStoredPersonalData, AddPersonalDataRequest, CancelRequest, PartialCancelRequest
--	--

6.5.3 From the system's perspectives

The use cases detailed below follow the same order as the use cases above reflecting the traveller's perspective but instead are from the retailer's or distributor's perspective.

The schemas in the following use case descriptions are using theoretical dedicated actors as it is done in the role's schema in 6.2 [Roles](#) chapter (it is assumed that each role is undertaken by exactly one actor and that each actor plays only one role).

6.5.3.1 Retailer parameterisation

This use case covers how retailers can receive and update the catalogue they present to the TRANSPORT CUSTOMER at a desk, app, or on a vending-machine. The up-to-date data can be provided by actors in NeTEx.

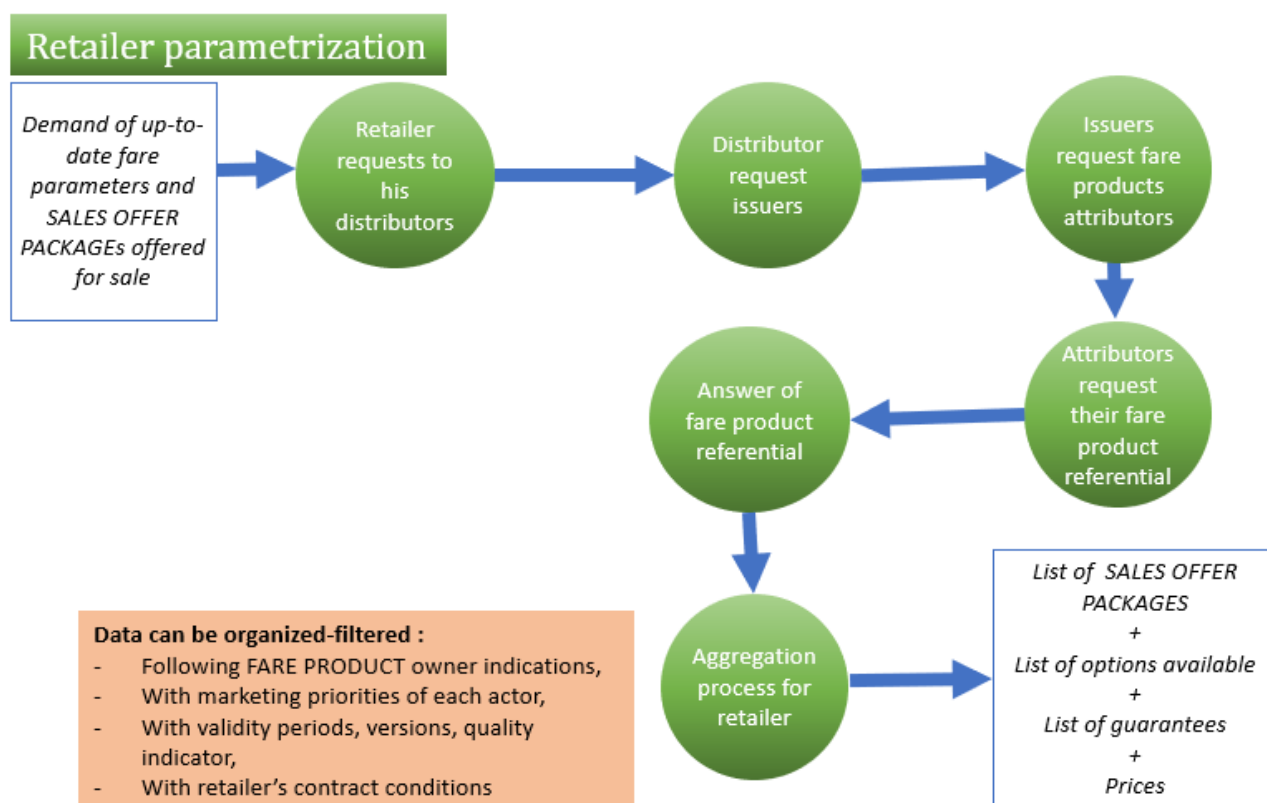


Figure 9 – Illustration of the use-case “Retailer parameterisation”

Table 13 – Details of the use-case “Retailer parameterisation”

Use Case 8 : Retailer parameterisation	
Summary	Periodically, a retailer must receive the up-to-date list of FARE PRODUCTs available to offer for sale to TRANSPORT CUSTOMERs. Mostly will be done in advance as a batch process. The data describes the fare products and the conditions pertaining to them. It may include static fare prices, or the prices may be fetched later using a dynamic pricing engine.
Description	It is assumed that all the FARE PRODUCT retailers are synchronised using an established process, with the distributors able to request the update of the data for fare parameterisation. Such actions can be carried out by a periodical batch process by a request on notification, or by other means. It is assumed that relations between retailers and distributors follow trade agreements and optimisation rules (performance). Any update activates the retailer's fare parameterisation module, requesting each distributor's parameterisation module to collect up-to-date FARE TABLEs and FARE PRODUCTs (and the related parameters) from the retailer. Each distributor can request in his turn the issuer's data, either before or during the retailer request for up-to-date FARE TABLEs. In the same way, issuers request attributors (or directly FARE PRODUCT OWNERS) for up-to-date FARE TABLE descriptions.

	When each actor has the up-to-date FARE TABLEs, on request, he can send the up-to-date data to the requestor. At each step of the delivery, the fare parameters are aggregated, consolidated, filtered and improved. The retailer receives one delivery by the distributor adapted to him and integrates the parameters on his sales platform. New parameters can be delivered by partial upgrades or fully each time. New parameters can be anticipated for a period (for next month for example) or can be applicable as soon as possible.
Main roles and actors	BoB: N/A OSDM 3.2: Fare Provider, Allocator, Distributor TOMP-API: The TOMP-API resides between a Transport Operator (or a façade hiding all the roles) towards a reseller (MaaS Provider) Transmodel: FARE PRODUCT RETAILER ROLE, FARE PRODUCT ISSUER ROLE, FARE PRODUCT DISTRIBUTOR ROLE, FARE PRODUCT ATTRIBUTOR ROLE, FARE PRODUCT OWNER ROLE TAP TSI: Carrier, Distributor
Main objects	BoB: N/A OSDM 3.3.0: Transmodel: FARE PRODUCT, SALES OFFER PACKAGE, TRAVEL DOCUMENT, USER PROFILE, FARE TABLE, PRICE, OPERATOR, VALIDITY PERIOD, VERSION, DISTRIBUTION CHANNEL, TOMP-API v2: Product, Package, Fare, Price TAP TSI: Products, Fares
Contributions	BoB: <i>This use-case is not applicable to BoB as the use case cannot support the types of complex business rules employed by most regional public transport providers. Also, as information on prices and products are provided in the moment they are needed, there is no need for periodical updating of offers.</i> OSDM 3.3.0: Transmodel: FARE PRODUCT QUERY NeTEx: Fare profile using Transmodel objects: with FARE PRODUCTs described with FARE PRODUCT ELEMENTs, SALES OFFER PACKAGES, TYPES OF TRAVEL DOCUMENT, FARE TABLEs with PRICES by DISTRIBUTION CHANNELs and USER PROFILES. The objects are organised in FRAMEs and in files. TOMP-API 2.0: N/A ENTUR: TAP-TSI: Fare profile using NeTEx

6.5.3.2 Collecting FARE PRODUCTs information

This use case covers how the distributors, attributors, issuers and the retailers collect the information to present the catalogue of the FARE PRODUCTs.

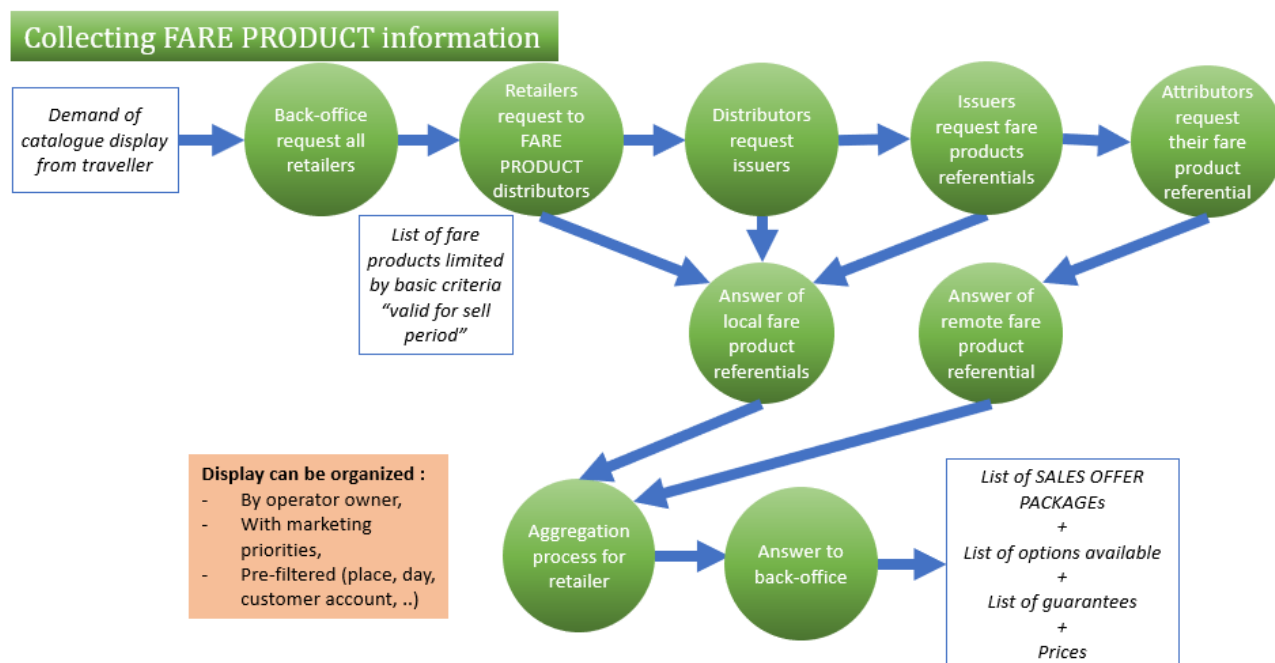


Figure 10 – Illustration of the use-case “Collecting fare product information”

Table 14 – Details of the use-case “Collecting fare product information”

Use Case 9 : Collecting FARE PRODUCT information	
Summary	When a retailer wants to present the catalogue required by the end-user for a specific trip, all the actors are required to provide their information.
Description	It is assumed that each actor of the schema (retailer, issuer, distributor, attributor) plays exactly one role. Each actor can organise the collect and management of the FARE PRODUCTs information on their own domain and with its own method. Each request/answer of the process can be done in real-time (for dynamic pricing, remote FARE PRODUCT referential) or with scheduled back-office process (locale FARE PRODUCT referential for performance management). When there is a request for FARE PRODUCT information from a system back-office, the retailer actors (can be the web site or mobile application back-office as well) is requested for information. Either the retailer can answer with locale data or/and can complete with a request to distributor actors. Each requested distributor actor (in large system) answers with local data or/and completes with a request to issuer actors. Each requested issuer actor answers with local data or/and completes (in large system) with a request to attributor actors. Each actor answers to his requestor with an aggregation of local and remote information.

	The retailers aggregate the information for the back-office (NeTEx ProductType, ConditionSummary, FareTable). The back-office formats the data and the presentation.
Main roles and actors	BoB: N/A OSDM 3.2: Fare Provider, Allocator, Distributor TOMP-API: N/A Transmodel: FARE PRODUCT RETAILER ROLE, FARE PRODUCT ISSUER ROLE, FARE PRODUCT DISTRIBUTOR ROLE, FARE PRODUCT ATTRIBUTOR ROLE. TAP TSI: Carrier, Attributor, Distributor
Main objects	BoB: N/A OSDM 3.3.0: Transmodel: FARE PRODUCT, SALES OFFER PACKAGE, FARE TABLE, PRICE, OPERATOR, AVAILABILITY CONDITION, TRAVEL SPECIFICATION, CLASS OF USE, FACILITY SET, ORGANISATION, TYPE OF PRODUCT CATEGORY, USAGE PARAMETER (USER PROFILE, COMMERCIAL PROFILE, GROUP TICKET, ENTITLEMENT, ROUTING, INTERCHANGING, STEP LIMIT, MINIMUM STAY, TRANSFEREABILITY, RESERVING, EXCHANGING, REFUNDING, SUSPENDING, REPLACING, REBATING, PURCHASE WINDOW, USAGE VALIDITY PERIOD, LUGGAGE ALLOWANCE, etc.), TYPE OF CONCESSION, TYPE OF PROOF, TYPE OF ENTITLEMENT, TYPE OF BAGGAGE, TYPE OF LUGGAGE USE, TYPE OF SPOT AFFINITY, REPEATED TRIP FARE REQUIREMENTS, PASSENGER FARE OFFER, FARE OFFER PRICE, VALIDABLE ELEMENT, TRIP PATTERN, OFFERED TRAVEL SPECIFICATION TOMP-API: Product, Package, Fare, Price, User profile, Operator, Usage parameter TAP TSI: FARE PRODUCT, SALES OFFER PACKAGE, FARE TABLE, PRICE, USER PROFILE, OPERATOR, VALIDITY PERIOD,
Contributions	BoB: <i>This use-case is not applicable to BoB as it cannot support the types of complex business rules employed by most regional public transport providers.</i> OSDM 3.3.0: Transmodel: COMMON / SINGLE / REPEATED TRIP FARE QUERYNeTEx: Fare profile using Transmodel objects: with FARE PRODUCTs described with FARE PRODUCT ELEMENTs, SALES OFFER PACKAGEs, TYPEs OF TRAVEL DOCUMENT, FARE TABLEs with PRICEs by DISTRIBUTION CHANNELs and USER PROFILEs. TOMP-API 2.0: POST /processes/search-offers/execution ENTUR:

	TAP-TSI: Fare profile using FARE PRODUCTs described with FARE PRODUCT ELEMENTs, SALES OFFER PACKAGEs, TYPEs OF TRAVEL DOCUMENT, FARE TABLEs with PRICEs by DISTRBUTION CHANNELs and USER PROFILEs.
--	--

6.5.3.3 Detailed information for a reliable offer (reservations checked)

This use case covers how the system reacts when a TRANSPORT CUSTOMER requests the availability of an offer for a long trip with many separate reservations for different legs (i.e., many TRAVEL PACKAGE ELEMENTs with reservations).

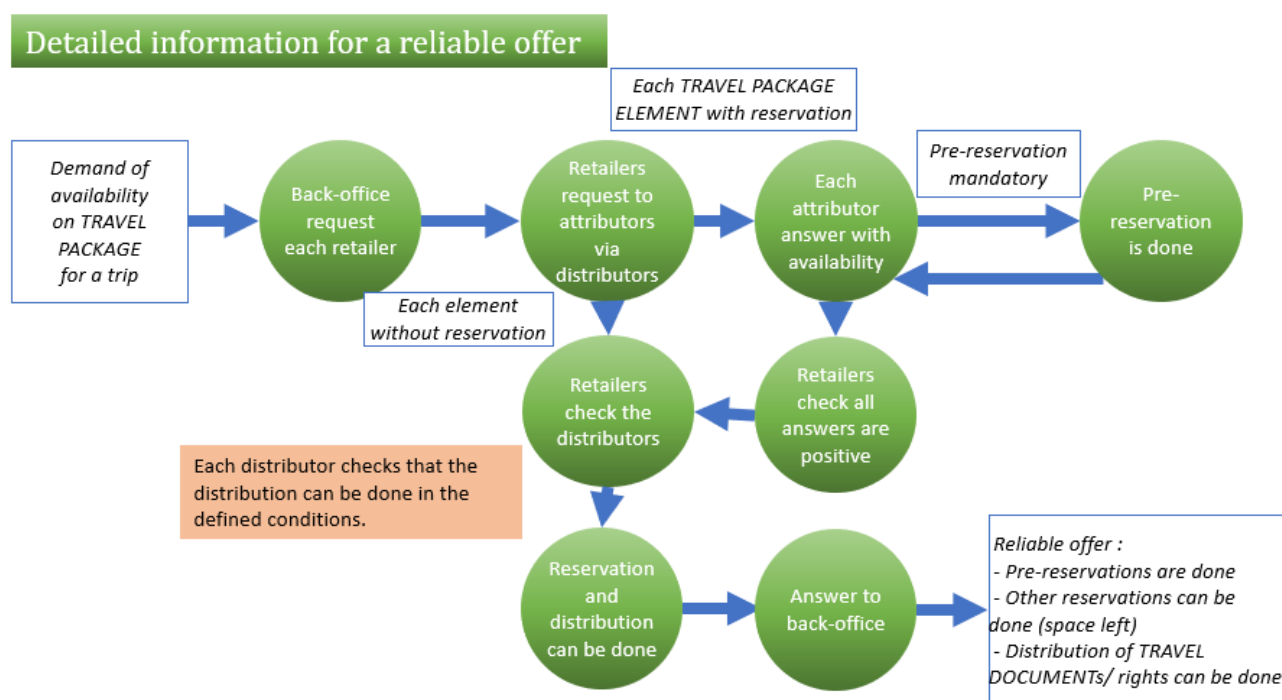


Figure 11 – Illustration of the use-case “Detailed information for a reliable offer”

Table 15 – Details of the use-case “Detailed information for a reliable offer”

Use Case 10: Detailed information for a reliable offer	
Summary	A request for the availability of an offer composed of multiple SALES OFFER PACKAGEs requested by a trip planner or by a traveller.
Description	It is assumed in this use-case, that to be reliable, an offer must be checked on two points: that the reservations can be made (and are held as soon as possible) and the TRANSPORT CUSTOMER can get all the TRAVEL DOCUMENTs needed to travel (i.e., that distribution can be done). It is assumed that each actor has only one role. It is assumed in this use case that the demand of reliability comes from a web site, a mobile application or any other system back-office.

	Before the sale (or at least at the beginning), the availability of the TRAVEL PACKAGE is checked. The back-office receives the request and requests each retailer involved as regards the reservation availabilities. A retailer can request the attributor for reservation availability possibly via the distributor. The attributor either start a pre-reservation or reply with actual availability depending on the FARE PRODUCT sales conditions. Some attributors involved in an availability check start a 2-stage reservation process (the check is followed by a time-limited pre-reservation). Then each retailer collects the answers of the concerned attributors . If all answers are positive, i.e., each leg can be reserved, the retailer starts additional checks. If one of the reservations can't be done (no seat left, ...), the offer can't be sold in that state. The answer to the demand of availability is negative on that offer as a whole. Then follow requests to the distributors for each FARE PRODUCT of the offer checking that the access right to travel can be delivered: The TRAVEL DOCUMENT is ready (examples: for contactless card, there is enough space on the support to add the new contract; for mobile application with CB2D, the version of the application is the last and the mobile is enlisted on the back-office), The TYPE OF TRAVEL DOCUMENT (media) is secured (examples: the contactless card is not in blacklist, the CB2D generator is ready), the e-mail address has been checked, the printer is ready, etc. Then when the distribution checks are positive, each retailer answers to the back-office requestor a positive answer. The back-office collects the retailer's answers and its turn, is able to inform that the offer is reliable.
Main roles and actors	BoB: Ticket Vendor, Price & product service, Ticketing service, Booking Service, Customer OSDM 3.2: Fare Provider, Allocator, Customer, Passenger, Distributor TOMP-API: N/A Transmodel: FARE PRODUCT RETAILER ROLE, FARE PRODUCT DISTRIBUTOR ROLE, FARE PRODUCT ATTRIBUTOR ROLE. TAP TSI: Fare provider, Attributor, Customer, Passenger, Distributor
Main objects	BoB: Product, Manifest, Ticket Bundle, Booking OSDM 3.3.0: Transmodel: TRAVEL PACKAGE, CLASS OF USE, ACCESSIBILITY ASSESSMENT, TRIP PROFILE REQUIREMENTS, USER PROFILE, ENTITLEMENT, TYPE OF SPOT AFFINITY, VEHICLE JOURNEY SPOT ALLOCATION, PASSENGER SPOT ALLOCATION, FACILITY SET, TRIP PATTERN, SPOT ALLOCATION METHOD, VEHICLE SPOT ALLOCATION, LOCABLE SPOT, MOBILITY SERVICE, TYPE OF MOBILITY SERVICE, TOPOGRAPHIC PLACE, MOBILITY SERVICE GEOMETRY RESTRICTION, TRANSPORT ORGANISATION, MOBILITY SERVICE, EQUIPMENT, AVAILABILITY CONDITION, SERVICE BOOKING ARRANGMENT, MOBILITY SERVICE CONSTRAINT ZONE, TRANSPORT TYPE, MODE OF OPERATION, VEHICLE

	TOMP-API: Product, Leg, Asset, Class, Traveller, Customer
Contributions	BoB: <i>There are no separate steps required to retrieve a reliable offer. Reliable offers are always provided in the manifest responses and (optionally) in the booking process. See the previous use-cases.</i> OSDM 3.3.0: Transmodel: TRAVEL PACKAGE SPOT AVAILABILITY QUERY TRAVEL PACKAGE SPOT RESERVATION QUERY MOBILITY SERVICE QUERY TOMP-API 2.0: POST /processes/search-offers/execution (availability & reservation) ENTUR: TAP-TSI: FARE PRODUCTS FerryGateway: GetSailingsRequest(anonymous Passengers, Pets, Vehicles, SearchCriterion(list of requested Sailings (i.e., legs/trips) with detail of their respective DeparturePlace, DestinationPlace and DepartureDateTime-window)) The customer adjusts the returned information-set by removing all but the preferred Sailing-option for each required departure-destination-pair. The next step is to investigate what possible optional services (i.e., ancillaries) are available for each selected Sailing (i.e., trip/leg) option. This is done by posting a GetServicesRequest The customer then removes unwanted optional Services (ancillaries) from the returned information-set. Now the information-set should reflect the complete package to be bought.

6.5.3.4 Calculate the price of a reliable offer (reservations checked)

This use case covers the pricing of an offer (the availability having been checked in the previous use case). This process is done for a TRAVEL PACKAGE composed of TRAVEL PACKAGE ELEMENTs (corresponding to SALES OFFER PACKAGES) in a context of multi-modal trip with distributed fare price calculators.

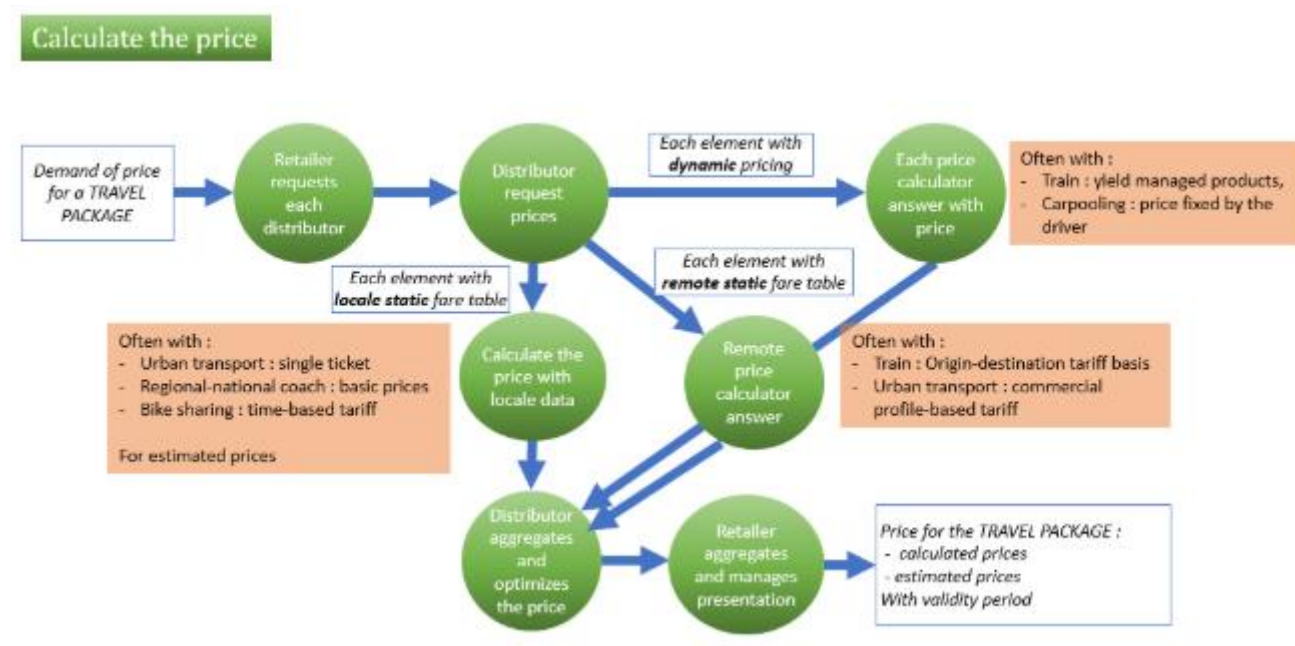


Figure 12 – Illustration of the use-case “Calculate the price of a reliable offer”

Table 16 – Details of the use-case “Calculate the price of a reliable offer”

Use Case 11 : Calculate the price of a reliable offer	
Summary	When there is a demand for the real price of a complex TRAVEL PACKAGE, each element of the offer must be priced. The final answer must be also managed.
Description	It is assumed that the TRAVEL PACKAGE is composed of one or more elements to be sold. It is assumed that some elements can be sold grouped together into one single package and for that, each price calculator answers with all the solutions (single price, packages with the element, combined offers, ...) It is assumed that every price calculator can answer with estimated price and the information that it is an estimation. A retailer may act as an aggregator of information supplied by various other retailers. The distributor represented in the diagram could in fact be a cascade of distributors, each requesting partial or complete information from one another. A request for the price of an offer implies the evaluation of the price of each element separately, thereby evaluating the price of the full offer. When a back-office requests distributors for prices of each offer element they retail, each distributor can calculate it in different ways depending on the FARE PRODUCT parameters. Some FARE PRODUCTS have static prices which may be distributed in advance as FARE PRICES with actual amounts in a specified currency.

	For some SALES OFFER PACKAGES, the price must be calculated in ‘real-time’ as it is yield managed or can be changed in real-time. This price has a short validity. In Transmodel such prices are indicated by a FARE PRICE that specifies a PRICING SERVICE to go to at run time, rather than an amount. For other SALES OFFER PACKAGES, the price must be requested from another system (which itself might again pass the request to yet another system) because, for example, the computation is complex, or the data are not shared by the back-office, or the distributor has no local data for the product. For some SALES OFFER PACKAGES, the price can be calculated by the distributor itself. In this case, the retailer has a local FARE PRODUCT specification, the tariffs are stable and/or simple, there is exchanges between systems, the distributor has its own calculator, ... Once every element has at least one price (or many, if the element can be bought in a package), distributor may apply further commercial rules to the results and their presentation (special offer, promotion of one product, legals, etc.). Then, at its turn, the back-office manages the final answer to the demand with its rules (legal, commercial, platform parameters, etc.) and sends the answer. <i>Note:</i> Sometimes there is a requirement to show a price breakdown, such as the tax component of a fare. Transmodel can represent a detailed justification of a FARE PRICE as one or more RULE STEP RESULTS, each showing the application of a PRICING RULE used to compute a price from another, along with any relevant pricing parameters such as ROUNDING factors. For example, there might be a base adult fare of 11 Euros; a PRICING RULE that a child fare is 50% and another PRICING RULE that VAT of 12.5 % is applied on top of any price; and a ROUNDING rule that final prices are rounded down to the nearest 0.50 Euros. A child fare could thus be derived to be 5.0 Euros. However, it normally suffices in a purchase API to give any justification as a text note, e.g., “50% adult fare including 12.5% added VAT” so the full representation does not necessarily have to be exposed. PRICING RULES can include minimum and maximum amounts for variable prices.
Main roles and actors	BoB: Ticket Issuer, Retailer, Customer OSDM 3.2: Fare Provider, Allocator, Customer, Passenger, Distributor TOMP-API: The TOMP-API hides this functionality; it is the responsibility of the ‘Transport Operator’ Transmodel: FARE PRODUCT RETAILER ROLE, FARE PRODUCT DISTRIBUTOR ROLE
Main objects	BoB: Product, Manifest, Ticket Bundle, Booking OSDM 3.3.0: Transmodel: TYPE OF REQUEST, TRAVEL PACKAGE, CUSTOMER ACCOUNT, TRAVEL PREFERENCES, FARE CONTRACT, CUSTOMER PURCHASE PACKAGE, CUSTOMER PURCHASE STATUS, OFFERED TRAVEL SPECIFICATION, TRAVEL DOCUMENT, FULFILMENT METHOD, TYPE OF REQUEST, LOCATABLE SPOT, SPOT RESERVATION, PRICING SERVICE, FARE PRICE, PRICING RULE, RULE STEP RESULT, ROUNDING.

	TOMP-API: Package, Fare, Price, Traveller, Customer
Contributions	BoB: <i>There are no separate steps required to retrieve a reliable offer. Reliable offers are always provided in the manifest responses and (optionally) in the booking process. see the previous use-cases.</i> OSDM 3.3.0: Transmodel: TRAVEL PACKAGE QUERY SPOT QUERY TOMP-API 2.0: POST /processes/search-offers/execution GET /collections/assets/items ENTUR: TAP-TSI: FerryGateway: GetPriceRequest resulting in a GetPriceResponse containing the complete information-set including total cost-details and possibly also a unique Token with an expiryTime showing for how long the price-info is valid.

6.5.3.5 Distributed purchase process

This use case context is the sale of a TRAVEL PACKAGE composed by several elements in a distributed sales platform. For the end-user, it is a “Market place” where the offer (composed of many SALES OFFER PACKAGES of different actors) can be bought in one sale with one payment.

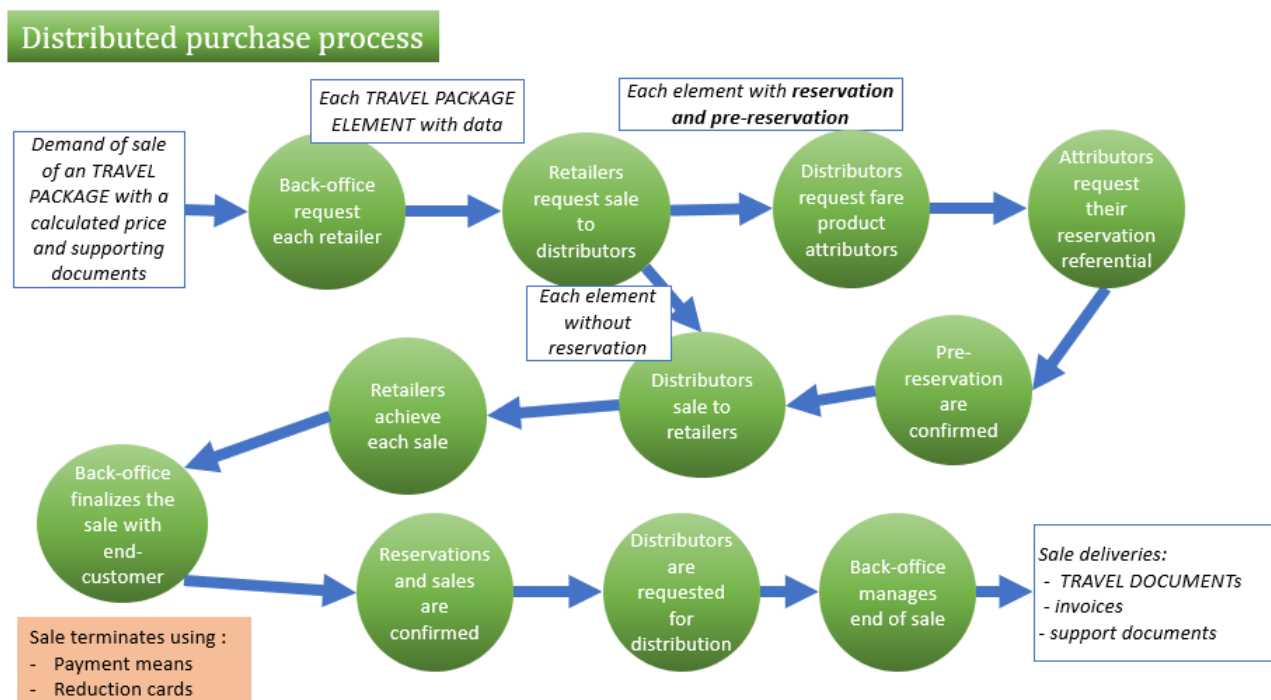


Figure 13 – Illustration of the use-case “Distributed purchase process”

Table 17 – Details of the use-case “Distributed purchase process”

Use Case 12 : Distributed purchase process	
Summary	When there is a request for the global sale for a complex offer, each element of the TRAVEL PACKAGE must be “sold internally” with each actor. Each single reservation and sale must be synchronised: The final sale must be managed in all-or-nothing mode.
Description	It is assumed that the offer has been checked for availability and the price and some elements have been pre-reserved. When the back-office receives a sale demand of a TRAVEL PACKAGE, each TRAVEL PACKAGE ELEMENT is requested for sale from the associated retailer. Each retailer starts an internal sale transaction with each distributor. Each distributor requests the attributors for the reservation process when needed by the FARE PRODUCT. For a FARE PRODUCT with a pre-reservation already made, the attributor confirms the pre-reservation (and it becomes a reservation). For a FARE PRODUCT with reservation yet to be made (and no pre-reservation process started), the attributor starts a reservation. If one reservation fails, the purchase process is cancelled (and other pre-reservations also). Then, for all TRAVEL PACKAGE ELEMENTs, each distributor’s sale to the retailer is made (without a final commit). Each retailer’s sales to the back-office is made (without a final commit).

	The back-office finalises the sale with the end-customer (payment, update of customer account, etc.) and then confirms the sales and reservations to all actors. If one action fails, the purchase process is cancelled, including the reservations. Distributors are requested for access right distribution when the payment is done. If the distribution fails, the back-office will manage the event with the concomitant after-sale actions. If the transaction succeeds, then the back-office delivers the documents and rights to travel to the end-user. There is a settlement mechanism to pay the respective actors according to their commercial agreements for their part of the transaction.
Main roles and actors	BoB: Ticket Vendor, Price & product service, Ticketing service, Customer OSDM 3.2: Fare Provider, Allocator, Customer, Passenger, Distributor TOMP-API: The TOMP-API can be applied on a many of the mentioned roles. It could be a FARE PRODUCT RETAILER, but also a TRANSPORT OPERATOR. It fully supports the two-phase-commit scenario. Transmodel: FARE PRODUCT RETAILER ROLE, FARE PRODUCT DISTRIBUTOR ROLE, FARE PRODUCT ATTRIBUTOR ROLE. TAP TSI: Carrier, Distributor, Attributor, Retailer, Customer, Passenger
Main objects	BoB: Product, Manifest, Ticket Bundle OSDM 3.3.0: Transmodel: TYPE OF REQUEST, TRAVEL PACKAGE, CUSTOMER ACCOUNT, TRAVEL PREFERENCES, FARE CONTRACT, CUSTOMER PURCHASE PACKAGE, CUSTOMER PURCHASE STATUS, OFFERED TRAVEL SPECIFICATION, TRAVEL DOCUMENT, FULFILMENT METHOD, LOCATABLE SPOT, SPOT RESERVATION SPOT RESERVATION, TRAVEL SPECIFICATION, TRAVEL PACKAGE DOCUMENT FULFILMENT REQUEST POLICY, MEDIUM ACCESS DEVICE, MEDIUM APPLICATION INSTANCE TOMP-API: Pending Package, Confirmed Package, Conditional Confirmed Package
Contributions	BoB: <i>This use-case is exactly the same as use-case 2, where is component or leg of the trip is requested in parallel from each public transport provider:</i> Product API: POST /product [retrieve products from each participant based on trip plan] Product API: POST /manifest [retrieve manifests for each leg] Ticket API: POST /ticket [request ticket based on manifest for each leg] Ticket API: GET /ticketbundle [retrieve ticket bundle for each leg] <i>The retailer can now combine each individual ticket bundle into one, delivering it to hands of the traveller.</i>

	OSDM 3.3.0: Transmodel: TRAVEL PACKAGE QUERY SPOT QUERY FULFILMENT QUERY TOMP-API 2.0: POST /processes/search-offers/execution POST /processes/purchase-offers/execution POST /processes/commit-purchase/execution POST /processes/rollback-purchase/execution ENTUR: TAP-TSI: ReservationRequest, CancellationRequest, PartialCancellationRequest FerryGateway: Use the returned information-set from the GetPriceResponse possibly also including its Token for reference. Amend the Passengers-list with needed details (Name, ID-card, etc.) and then (phase 1) post a ReservationRequest resulting in a ReservationResponse That assures that the purchase is possible, and the complete package is locked exclusively for a certain time span. (phase 2) post a ConfirmReservationRequest resulting in a ConfirmReservationResponse finalizing the purchase.
--	---

6.5.3.6 After-sale process from the retailer desk

This use case covers the processes followed when a TRANSPORT CUSTOMER comes to the retailer desk with a request for a change (i.e., permitted actions according to the after-sale conditions). In some cases, it can be the retailer that calls the TRANSPORT CUSTOMER for a proposal. This situation can be caused by the TRANSPORT CUSTOMER (i.e., by a personal event such as a change of plans) or by the transport operator (e.g., due to a service perturbation). The change can be proposed by the TRANSPORT CUSTOMER (within the limits of the after-sale parameters specified for their products) or by the operator (who may also include changes outside the after-sales service conditions).

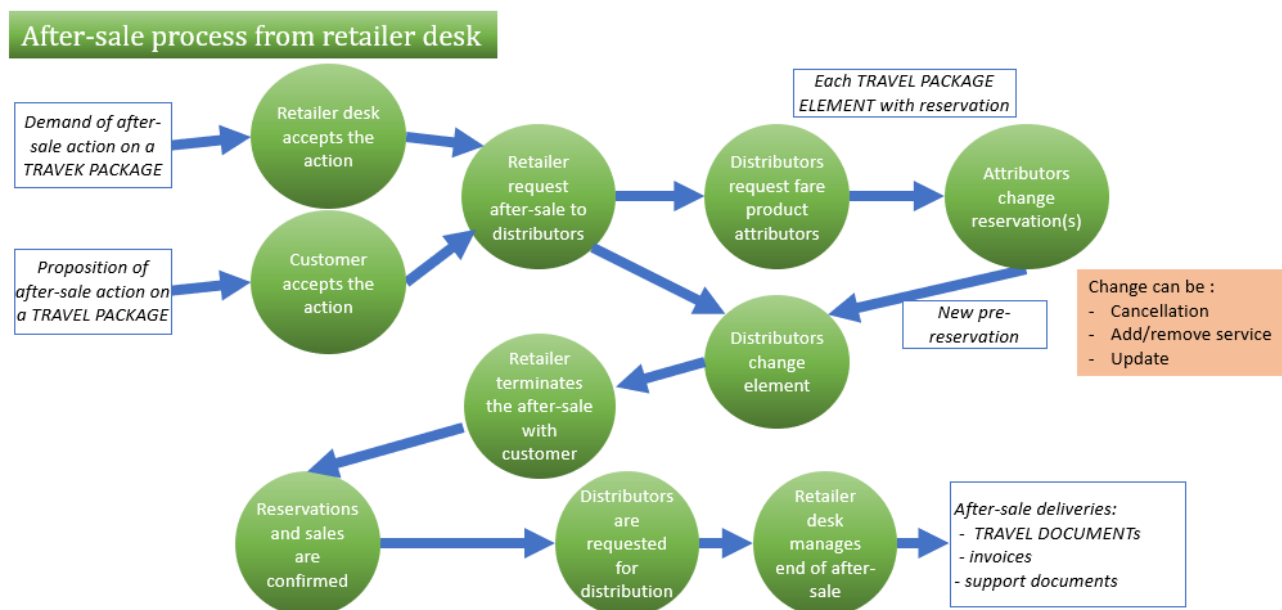


Figure 14 – Illustration of the use-case “After-sale process from retailer desk”

Table 18 – Details of the use-case “After-sale process from retailer desk”

Use Case 13 : After-sale process from retailer desk	
Summary	In the after-sale processes for a retailer, the after-sale actions on a TRAVEL PACKAGE with several elements can involve complex actions for distributors and attributors to change multiple reservations at the same time. The end of the process is managed like a purchase process, with an all-or-nothing commit of all the changes.
Description	It is assumed that the changes can be only to refund, exchange or provide compensation, as described for example, by the REFUNDING, EXCHANGING and REBATING usage parameters associated with the FARE PRODUCT - and also any TRAVEL REDRESSES associated with the TRAVEL GUARANTEES that have been triggered by a TRAVEL EVENT affecting the passenger. It is assumed that the actions are limited to those authorised for a given managed FARE PRODUCT for a retailer support desk. When a TRANSPORT CUSTOMER requests a change at a retailer desk, the retailer agent must accept the changes in line with the after-sale conditions on the TRAVEL PACKAGE, the evidence given by the TRANSPORT CUSTOMER, etc., and any relevant TRAVEL GUARANTEES and TRAVEL REDRESSES. When an operator proposes a change, a refund or a compensation, the TRANSPORT CUSTOMER must first accept it before the change is enacted. The proposal may be according to the after-sales conditions on the TRAVEL PACKAGE. Then the implementation of the after-sale change by the systems can start. The retailer contacts the distributors of impacted elements. The distributors, if they accept the after-sale action, request the change from the reservation attributors (if needed).

	With new pre-reservations, distributors change elements (i.e., start an internal sale transaction) and inform the retailer. The retailer desk ends the after-sale process with the TRANSPORT CUSTOMER, undertaking any necessary payment, update of customer account, etc. If the process is still ok, the retailer desk validates the new reservations and internal transactions and makes a request (if needed) for distribution. The retailer desk does the delivery of TRAVEL DOCUMENTs, invoices, etc., o the TRANSPORT CUSTOMER. Note that for period pass FARE PRODUCTs, such as season tickets, there may be specific conditions for partial refunds of the unconsumed access rights of the remaining period.
Main roles and actors	OSDM 3.2: Fare Provider, Allocator, Customer, Passenger, Distributor TOMP-API: Transport Operator Transmodel: FARE PRODUCT RETAILER ROLE, FARE PRODUCT DISTRIBUTOR ROLE, FARE PRODUCT ATTRIBUTOR ROLE. TAP TSI: Carrier, Attributor, Customer, Passenger, Distributor
Main objects	OSDM 3.3.0: Transmodel: TYPE OF REQUEST, TRAVEL PACKAGE, PASSENGER FARE OFFER, OFFERED TRAVEL SPECIFICATION, REQUESTED TRAVEL SPECIFICATION, FARE OFFER PRICE, CUSTOMER PURCHASE STATUS, TYPE OF RESALE, USAGE PARAMETERS (EXCHANGING, REFUNDING, RESELLING, REBATING, TRANSFERABILITY, SUSPENDING), TRAVEL GUARANTEE, TRAVEL REDRESS, CUSTOMER PURCHASE PACKAGE, TYPE OF TRAVEL DOCUMENT, TRANSPORT CUSTOMER, TRIP PATTERN, PASSENGER SPOT ALLOCATION, SERVICE JOURNEY, LOCABLE SPOT, SPOT AVAILABILITY STATUS TOMP-API: Package, Leg, Product, Payment, Refund, Deposit, Reimburse, Cancel package, Cancel leg
Contributions	<i>BoB: With BoB, the customer always has to turn to the retailer through which the ticket was bought to receive after-sale support. Hence this use-case is identical to use-case 5.</i> OSDM 3.3.0: Transmodel: TRAVEL PACKAGE ALTERATION / CANCELLATION / REFUND QUERY TRAVEL PACKAGE SPOT CANCELLATION QUERY TOMP-API 2.0: GET /collections/payments/items POST /processes/cancel-package/execution POST /collections/redress-options/items POST /processes/claim-refund-option/execution POST /processes/confirm-refund-option/execution

	POST /processes/assign-ancillary/execution POST /processes/assign-asset/execution ENTUR: TAP-TSI: FerryGateway: Use the BookingReference supplied in the ConfirmReservationResponse and post a RecallBookingRequest (BookingReference) resulting in a RecallBookingResponse providing the expanded information-set. This information-set is then used as a basis for the ticketing provider to produce the travel documents.
--	--

6.5.3.7 After-sale process from CUSTOMER ACCOUNT

This use case covers the processes involved between actors when a TRANSPORT CUSTOMER performs an after-sale action on his CUSTOMER ACCOUNT with a mobile phone.

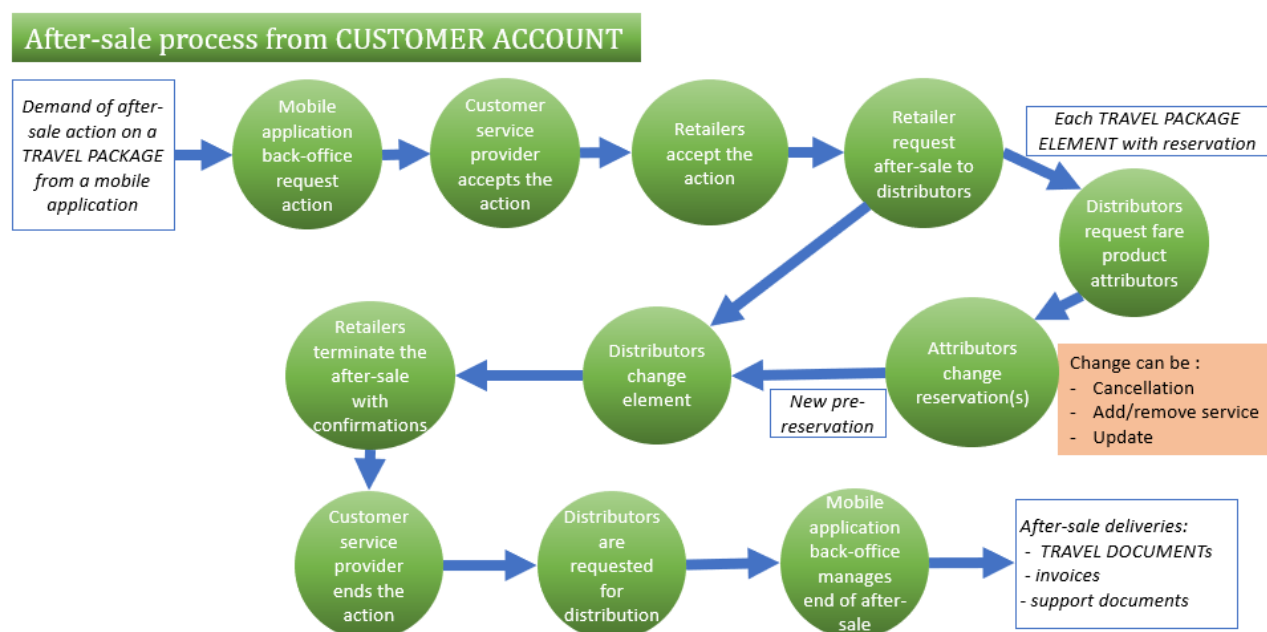


Figure 15 – Illustration of the use-case “After-sale process from customer account”

Table 19 – Details of the use-case “After-sale process from customer account”

Use Case 14 : After-sale process from CUSTOMER ACCOUNT	
Summary	In the after-sale process on a CUSTOMER ACCOUNT, the after-sale actions on a TRAVEL PACKAGE with several elements can involve complex actions for distributors and attributors on reservations. The end of the process is managed like a purchase process made as an all-or-nothing transaction.

Description	It is assumed that the TRANSPORT CUSTOMER has access to his account with a mobile or web device. It is assumed that the changes can only be a full, partial or free refund, to revoke a TRAVEL DOCUMENT or to contract or modify a reservation (or a service in the reservation). It is assumed that the actions are limited to authorised operations on TRAVEL PACKAGE and managed by the customer service account by mobile phone or web device. When a TRANSPORT CUSTOMER requests a change on his CUSTOMER ACCOUNT, the customer service provider must pre-accept the change following the after-sale conditions, the evidence given by the TRANSPORT CUSTOMER, etc . The customer service account requests the impacted retailers to make the action. If they all accept, the after-sale process on their systems can start. The retailers contact the distributors of impacted elements. The distributors, if they accept the after-sale action, request the change from the reservation attributors (if needed). With new pre-reservations, distributors change elements (start an internal sale transaction) and inform the retailers. The retailers end the after-sale process with the TRANSPORT CUSTOMER (payment, update of customer account, etc.) and if the process is still ok, the retailers validate reservations and internal transactions. The customer service provider manages the end of the process and request (if needed) for distribution. The mobile application back-office does the delivery of documents to the customer.
Main roles and actors	OSDM 3.2: Fare Provider, Allocator, Customer, Passenger, Distributor TOMP-API: Transport Operator. The general idea is that the MaaS provider (reseller) acts on behalf of the traveller/customer. These end users NEVER act directly with the Transport Operator. From this perspective, this use case is not different than the previous ones. Transmodel: FARE PRODUCT RETAILER ROLE, FARE PRODUCT DISTRIBUTOR ROLE, FARE PRODUCT ATTRIBUTOR ROLE, CUSTOMER SERVICE PROVIDER ROLE.
Main objects	BoB: <i>Not applicable, see use-case 7.</i> OSDM 3.3.0: Transmodel: TYPE OF REQUEST, TRAVEL PACKAGE, PASSENGER FARE OFFER, TRIP PATTERN, OFFERED TRAVEL SPECIFICATION, REQUESTED TRAVEL SPECIFICATION, FARE OFFER PRICE, TYPE OF RESALE, USAGE PARAMETERS (EXCHANGING, REFUNDING, RESELLING, REBATING, TRANSFERABILITY, SUSPENDING), TRAVEL GUARANTEE, TRAVEL REDRESS, CUSTOMER PURCHASE PACKAGE, TYPE OF TRAVEL DOCUMENT, TRANSPORT CUSTOMER, CUSTOMER PURCHASE STATUS, PASSENGER SPOT ALLOCATION, SERVICE JOURNEY, LOCABLE SPOT, SPOT AVAILABILITY STATUS, TRAVEL

	DOCUMENT, MEDIUM APPLICATION INSTANCE, SECURITY LISTING, SECURITY LIST TOMP-API: Customer, Traveller, Transport Operator
Contributions	OSDM 3.3.0: Transmodel: TRAVEL PACKAGE ALTERATION / CANCELLATION / REFUND QUERY TRAVEL PACKAGE SPOT CANCELLATION QUERY TRAVEL PACKAGE REVOKE FULFILMENT QUERY TOMP-API 2.0: N/A TOMP-API 2.0: GET /collections/payments/items POST /processes/cancel-package/execution POST /collections/redress-options/items POST /processes/claim-refund-option/execution POST /processes/confirm-refund-option/execution POST /processes/assign-ancillary/execution POST /processes/assign-asset/execution ENTUR: TAP-TSI:

7 Existing Transmodel Elements relevant for a multimodal travel purchase API specification

7.1 Introduction

This section outlines the existing Transmodel elements relevant for trip planning and fares. This includes both existing data model elements and existing API message pairs to perform service functions such as trip planning and fare finding that are covered in the existing Transmodel Passenger Information Service.

A following section [8] identifies additional concepts that could be added to Transmodel to represent a full multimodal travel purchase API with purchase and reservation capabilities. This also involves both extensions to the data model and some new API message pairs to perform the additional functions of the multimodal travel purchase API.

Thus, the representation of a Multimodal travel purchase/reservation API in Transmodel has two very distinct aspects:

The conceptual models of the data elements. That is, locations, trips, fare offers, seat availability, passengers, reservations, etc., that represent the data elements which a purchase/reservation API needs to operate upon in order to implement the desired trip planning, booking and after sales use cases. This will include both reference data such as products and prices with which the system is prepopulated, and then transaction data such as individual trip plans, fare offers and reservations that are generated by the users' interaction with the system.

The conceptual model of a multimodal travel purchase/reservation API as a set of request / response messages that exchange the data required to find a trip and make a booking (i.e., purchase rights to travel with or without a reservation). This will follow a common abstract architecture for messages, with specific functional message pairs for each step in the process. For example, finding locations, finding trips between those locations, finding fare offers for those trips, making a purchase with or without reservation a specific trip and fare offer, fulfilling a ticket for a purchased trip, modifying it, making a complaint about it, etc.

The API messages exchange data elements defined in the underlying conceptual model, along with parameters to specify or qualify the results; the messages can thus be regarded as views over the underlying conceptual model.

The API messages may be optimised for efficient and scalable exchange in real-time; e.g., by omitting certain elements and flattening or joining structures to simplify the processing of client systems. Certain data elements such as conditions may be rendered as simple textual information rather than as structured elements.

A concrete API that implements the conceptual API may reorder or rename elements but should nonetheless have data elements corresponding to all the necessary elements in each message.

The existing Transmodel has conceptual models for describing the public transport network and services, and for describing fares and fare prices, including complex conditions of use. It can also describe specific trip plans for a customer, specific fare offers of products for a trip, and the details of the customer's fare purchases if they choose a particular offer. It lacks certain elements to represent the purchase and reservation, in particular details of the purchase and any reservation as held by the booking system and also lacks certain constraints on what changes may be made to a booking, as well as limits as to what actions may be made by a specific distribution channel (such as by an app as opposed to a support desk). Transmodel also does not specify rules for combining products by aggregators and distributors.

NOTE: The existing Transmodel fare model covers a very wide range of FARE PRODUCTS, including complex zonal, stage and multimodal fares, season passes, and capped fares (i.e., usage discounts within a given period); not just the simple single point-to-point trip fares traditionally considered by "purchase and reservation APIs".

The existing Transmodel also has a conceptual model describing API request/delivery messages for trip planning and for finding the available fares, but not for purchase and reservation of individual trips, nor for after sales support and compensation

All Transmodel data elements are modularised into separate functional models; some new models are added for the new functions.

Note that the following section does not in any way provide an extensive description for the existing Transmodel elements, or for the proposed extension, but is merely intended to give an overview.

7.2 Using Transmodel to model these use cases

The following table describes the Transmodel coverage for each use-case and the extension required to cover the gaps.

Table 20 – Use Cases vs coverage by Transmodel v6.2

Use Case §6.4	summary	(Sub)functions	Conceptual model	API requests
Use Case 1: Choice of a solution by FARE PRODUCT catalogue consultation 6.5.2.1	Using the FARE PRODUCT catalogue, the TRANSPORT CUSTOMER chooses one or more catalogue items and enters additional information to allow the determining of the best offer (FARE PRODUCT, SALES OFFER PACKAGE, the related FARE PRICES and TRAVEL GUARANTEES).	<i>Filter/order the product set according to different parameters:</i> <i>customer profile,</i> <i>travel specification,</i> <i>price,</i> <i>guarantee,</i> <i>O/D zone, etc)</i>	Covered CUSTOMER ACCOUNT, TRAVEL PREFERENCES, CUSTOMER PROFILE, TRAVEL SPECIFICATION SALES OFFER PACKAGE, FARE PRODUCT, FARE PRODUCT PRICE, VALIDITY PARAMETER, USAGE PARAMETER, TEMPORAL VALIDITY CONDITION, SERVICE JOURNEY, TYPE OF SERVICE, TYPE OF PRODUCT CATEGORY, FACILITY SET, LINE TARIFF ZONE), TYPE OF TRAVEL DOCUMENT, FULFILMENT	To be updated according to model extension Stop Fare Query, Fare Product Query

			METHOD, DISTRIBUTION CHANNEL, DISTRIBUTION ASSIGNMENT+ Extensions (TRAVEL GUARANTEE)	
(Prerequisite Use case) Trip Planning	Find possible trips to travel between two points on a specified date,	<i>Specify customer origin, destination, preferences and preferred optimisation.</i> <i>Present results prioritised according to request criteria.</i>	Covered TRAVEL PREFERENCES, CUSTOMER PROFILE, TRAVEL SPECIFICATION TRIP, TRIP LEG,	Covered (used in OJP) Trip Query
Use Case 2: Information on and choice of a fare solution for a specific trip 6.5.2.2	After having received trip proposals from the trip planner, the TRANSPORT CUSTOMER chooses one of them and requests information on the corresponding TRAVEL PACKAGES (FARE PRODUCTS, SALES OFFER PACKAGES, etc., and the related PRICES and TRAVEL GUARANTEES).	<i>Select travel options for trip.</i> <i>Revise/Extend PASSENGER FARE OFFERS making up a TRAVEL PACKAGE in certain ways, including:</i> <i>Change travel options (e.g., mode, class, earlier/later schedule).</i> <i>Propose available add-on services (meals, luggage transport, etc.).</i> <i>Propose possible types of travel documents and fulfilment methods for a chosen solution.</i>	Covered CUSTOMER ACCOUNT, OFFERED TRAVEL SPECIFICATION, TRIP, TRIP LEG, PASSENGER FARE OFFER, SALES OFFER PACKAGE, FARE PRODUCT, FARE PRICE, SUPPLEMENT PRODUCT, SALE DISCOUNT RIGHT, FARE STRUCTURE ELEMENT, USAGE PARAMETER (USER PROFILE, TYPE OF CONCESSION,	To be updated according to model extension Single Trip Fare Query, Repeated Trip Fare Query, Mobility Service Query + Extensions

		<i>Display pre-requisite and available discount rights and usage points to be earned.</i> <i>Display after-sales conditions and guarantees.</i> <i>Propose trip repetitions on specific day types for a specific period.</i>	ENTITLEMENT, LUGGAGE ALLOWANCE, etc.) Scoping Parameters (TARIFF ZONE, TOPOGRAPHIC AREA, etc.), SERVICE JOURNEY, TYPE OF TRAVEL DOCUMENT, FULFILMENT METHOD, DISTRIBUTION CHANNEL, DISTRIBUTION ASSIGNMENT, + Extensions (TRAVEL GUARANTEE, TRAVEL PACKAGE) OFFER RULE	Travel Package CRUD: • (Travel Package Creation Query), • (Travel Package Alteration Query), • (Travel Package Cancellation Query)
--	--	---	--	---

Use Case 3: Getting availability with or without reservation			Covered	Existing
6.5.2.3	When choosing a TRAVEL PACKAGE for a specific trip, with or without making a reservation, the TRANSPORT CUSTOMER wants to be informed about the availabilities of each solution and their respective options before making a purchase. May or may not place a reservation, temporary or confirmed. May be done in advance to preplan (e.g., to check availability of product or seat) or also at time of travel to check current availability of assets.	<i>Check & display spaces left (seats, spots) displaying the deck plan & all guarantees that can be used and selected.</i> <i>Check& display available assets (e.g., scooters, shared bikes, parking places, luggage spots).</i> <i>Check& display available ancillary products and features.</i> <i>Check & display likely / actual availability of quota-based products.</i> <i>Inform on services that are not available.</i> <i>Inform on the allowed modifications / forbidden modifications (i.e., that require cancellation / restart).</i> <i>Inform on the distribution channels of travel documents.</i> <i>Inform on travel document types.</i>	PASSENGER FARE OFFER, FARE PRODUCT, SALES OFFER PACKAGE, FARE PRICE, DECK PLAN, PASSENGER SPOT, SCHEMATIC MAP, LOCATABLE SPOT, SPOT ALLOCATION, SPOT AVAILABILITY STATUS, ENTITLEMENT, SUPPLEMENT PRODUCT, SALES OFFER PACKAGE, PARKING, PARKING BAY, VEHICLE, PARKING BAY STATUS, QUALITY STRUCTURE FACTOR, AVAILABILITY CONDITION, TYPE OF TRAVEL DOCUMENT, FULFILLMENT METHOD, DISTRIBUTION ASSIGNMENT, TRIP SPOT AVAILABILITY ENTRY,	Fare Product Query, Mobility Service Query, To be developed (Travel Package Spot reservation: • (Travel Package Spot Availability Query), • (Travel Package Spot Reservation Query), • (Travel Package Spot Cancellation Query) (Travel Package Availability Query)

			CUSTOMER RESERVATION ENTRY CUSTOMER PURCHASE PACKAGE, TRAVEL DOCUMENT, MEDIUM ACCESS DEVICE, SMARTCAR, EMV CARD + Extensions Availability of ancillaries (other than spot reservation) (TRAVEL GUARANTEEs, TRAVEL PACKAGE) (FARE OFFER RULE: Forbidden modifications Allowed Modifications)	
Use Case 4: Purchase process 6.5.2.4	When a TRANSPORT CUSTOMER has selected one or more a PASSENGER FARE OFFERs and any required choices for a TRAVEL PACKAGE, he proceeds to the purchase.	<i>Create/register a travel basket.</i> <i>Hold/lock the resources (to finalise the purchase) – optional.</i> <i>Lock (?)</i>	Covered CUSTOMER PURCHASE PACKAGE, TRAVEL DOCUMENT, FULFILMENT METHOD,	To be developed (Travel Package Purchase Query with time limited hold, then confirmation or rollback)

		<i>Get final agreement from the customer (on requested options and guarantees).</i> <i>Fix reservations (for all trip legs) and provide the result to the customer.</i> <i>Rollback the purchase in case of unexpected issue.</i>	SALES TRANSACTION, FULFILMENT ENTRY, + Extensions (TRAVEL GUARANTEES, TRAVEL PACKAGE), Add model state to support 2-phase commit.	(NB Payment step will use standard commercial online methods) (Travel Package Fulfillment Query)
--	--	--	--	--

Use Case 5: Change before the travel	A TRANSPORT CUSTOMER can request an after-sale action at any moment of the travel, but the focus here is on the process before the travel.	Request for Redress & Reimbursement (full, partial, free), Compensation, Exchange of products. Request to change access rights and travel documents Request to change on reservation, and on ancillary products.	Covered	To be developed
UC5.1 After-sale request by the traveller UC5.2 Change under the responsibility of any providers involved in the trip for a through ticket (not the traveller) → re-accommodation 6.5.2.5			TRANSPORT CUSTOMER, FARE CONTRACT, SALES TRANSACTION, TRAVEL SPECIFICATION, CUSTOMER PURCHASE PACKAGE, TRAVEL DOCUMENT, CUSTOMER RESERVATION, SPOT ALLOCATION, LOCATABLE SPOT, REFUNDING, EXCHANGING, REPLACING, REBATING, SUSPENDING, TRAVEL DOCUMENT, FULFILMENT ENTRY,	(Travel Package Exchange Query), (Travel Package Refund Query), (Travel Package Refund Offer Query), (Travel Package Spot Reservation Query),

			+ Extensions (TRAVEL GUARANTEE, REDRESS, TRAVEL PACKAGE), TRAVEL PACKAGE REFUND ENTRY, CUSTOMER REDRESS ENTRY	(Travel Package Reservation Cancellation Query), (Travel Package Revoke Fulfilment Query), (Travel Package Fulfilment Query)
Use Case 6: Change during the travel UC 6.1 : change by the traveller UC6.2 : Change under the responsibility of any providers involved in the trip (not the traveller)	When a traveller starts his travel, he possesses TRAVEL DOCUMENTs which provide evidence of his purchased TRAVEL PACKAGES. Disruptions (e.g., accidents) and trip modifications made by the TRANSPORT CUSTOMER or OPERATOR may modify the trip	<i>Request by the user and/or proposed by the system for permitted adjustments/alterations to their TRAVEL PACKAGE as a result of service disruptions.</i> <i>Display information on disruptions possibly affecting TRAVEL PACKAGE elements.</i> <i>Display available Easements.</i>	Covered TRANSPORT CUSTOMER, FARE CONTRACT, VALIDITY PARAMETER, TRAVEL DOCUMENT, CUSTOMER PURCHASE PACKAGE, TRAVEL SPECIFICATION, SITUATION, EASEMENT,	To be developed • (Travel Package Alteration Query), • (Travel Package Cancellation Query), • (Travel Package Refund Offer Query)

6.5.2.6	impacting the SALES OFFER PACKAGES, reservations, final prices, guarantees and TRAVEL DOCUMENTS.	<i>Offer any alterations proposed by operator.</i> <i>Fulfil any modifications to travel Documents.</i> <i>Record failures of guaranteed connections and other Guarantees.</i>	FARE PRODUCT EXCHANGE ENTRY, TRAVEL COMPENSATION ENTRY, + Extensions (TRAVEL GUARANTEE, REDRESS, TRAVEL PACKAGE)	
Use Case 7: Customer account consultation 6.5.2.7	A TRANSPORT CUSTOMER can consult his CUSTOMER ACCOUNT and perform some actions on the FARE PRODUCTS –and SALES OFFER PACKAGES he has purchased.	<i>Creation and management of accounts and account details (contracts, media, payment methods).</i> <i>Search and display Customer 's personal Transaction history, e.g., travel consumption, previous interactions such as purchases, reservations, after sales actions, (payments), acquired travel documents.</i> <i>Display Information on the organisation in charge of the account management and account transactions.</i> <i>NOTE: Purchases, TRAVEL DOCUMENTs and other data</i>	Covered CUSTOMER ACCOUNT, CUSTOMER PAYMENT MEANS, FARE CONTRACT CUSTOMER ACCOUNT REGISTRATION ENTRY, CUSTOMER ACCOUNT DEREGISTRATION ENTRY, CUSTOMER MODIFY PROFILE ENTRY CUSTOMER MEDIA REGISTRATION ENTRY, CUSTOMER PAYMENT MEANS ENTRY FARE CONTRACT ENTRY, SALES TRANSACTION, etc. ROLE	To be developed (CRUD for CUSTOMER ACCOUNT creation & modification), <i>NOTE: This may use a standard Account Management API.</i> (CUSTOMER ACCOUNT History Query), Information about the responsible parties for transactions on the account:

		<i>associated with the account are created by transactions against the account.</i>	+ Extensions (TRAVEL PACKAGE)	(FARE CONTRACT Entry Responsible Organisation Query)
--	--	---	---	--

Use Case 8 : Retailer parameterisation	Periodically, a retailer must receive the up-to-date list of FARE PRODUCTS that can be offered for sale to TRANSPORT CUSTOMERS.	<i>Exchange information on all existing FARE PRODUCTS and SALES OFFER PACKAGES from a particular retailer.</i>	Covered FARE PRODUCT, SALES OFFER PACKAGE, USAGE PARAMETER, FARE PRICE, etc., VERSION FRAME, VALIDITY PERIOD, VERSION, OPERATOR, DISTRIBUTION CHANNEL, + Extensions (TRAVEL GUARANTEE, OFFER RULE – for combining and or modifying offers).	To be updated according to model extension + To be developed (Fare Product Query), (Offer Rule Query),
Use Case 9 : Collecting FARE PRODUCTS information	When a retailer wants to present the catalogue of the possible SALES PACKAGES required by the end-user for a specific trip, all the actors are required to provide their information.	<i>Static or Dynamic aggregation of data from different suppliers into a coherent product set to suit a specific trip.</i>	Covered SALES OFFER PACKAGE, FARE PRODUCT, FARE PRICE, etc., + Extensions (TRAVEL GUARANTEE, Tagging for marketing).	To be updated according to model extension Fare Product Query + To be developed (Travel Package Creation Query),

Use Case 10: Detailed information for a reliable offer	A request for the availability of an offer composed by many SALES OFFER PACKAGES addressed by a trip planner or by a traveller.	<i>Confirmation of availability of SALES OFFER PACKAGES for a specific trip and travel choices as selected by a TRAVEL PACKAGE.</i>	Covered SALES OFFER PACKAGE + Extensions (e.g., TRAVEL PACKAGE, SALES OFFER PACKAGE AVAILABILITY, TRAVEL GUARANTEES)	To be developed (Travel Package Availability Query)
Use Case 11 : Calculate the price of a reliable offer	When there is a demand for the real price of a complex TRAVEL PACKAGE, each element of the offer must be priced. The final answer must be also managed.	<i>Confirmation of pricing of existing FARE PRODUCTS and SALES OFFER PACKAGES for a specific trip and travel choices as selected by a TRAVEL PACKAGE.</i>	Covered Dynamic FARE PRICE + Extensions (e.g. TRAVEL PACKAGE)	To be developed (Travel Package Price Query)
Use Case 12 : Distributed purchase process	When there is a demand for the overall sale of a complex offer, each element of the TRAVEL PACKAGE must be “sold internally” with each actor. Each single reservation and sale must be synchronized: The final	<i>Confirmation of a given transaction (i.e., two-phase commit with rollback if partial failure).</i>	Covered + Extensions	To be developed

	sale of all elements must be managed to be all-or-nothing.		(e.g., TRAVEL PACKAGE, confirmation state)	(Travel Package Confirmation Query)
Use Case 13 : After-sale process from retailer desk	In the after-sales process for a retailer, the after-sales actions on a TRAVEL PACKAGE with several elements can necessitate complex interactions by distributors and attributors on purchases and reservations. The completion of the process is managed like a purchase as an all-or-nothing process.	<i>Two-phase commit with rollback if partial failure</i>	Covered + Extensions (e.g. TRAVEL PACKAGE)	To be developed
Use Case 14 : After-sale process from CUSTOMER ACCOUNT	In the after-sales process from a CUSTOMER ACCOUNT, the after-sales actions on a TRAVEL PACKAGE with several elements can involve complex interactions by distributors and attributors on purchases and reservations. The end of the process is managed like a purchase process as an all-or-nothing process.	<i>Refusal / failure of a complex after-sales offer would be shown as reversals of previous unconfirmed offers so that history is apparent to user.</i>	Covered CUSTOMER ACCOUNT, FARE CONTRACT ENTRY, + Extensions (e.g. TRAVEL PACKAGE)	To be developed (Travel Package Confirmation Query)

7.3 Overview of existing Transmodel Model for trips and fares

7.3.1 The Network

The available transport network for which fares can be purchased is described in several different Transmodel sub models.

Access points to the network are described as SCHEDULED STOP POINTs; designated points at which a passenger may board or disembark from a public transport.

Paths through the network can be described as a SERVICE PATTERN of SERVICE LINKs between SCHEDULED STOP POINTs.

NOTE: Transmodel distinguishes between the stops as they appear as logical points in the timetable (i.e., SCHEDULED STOP POINTs), and the physical stop or STOP PLACE representing a transport interchange which may in fact be made up of multiple components such as platforms (QUAY)s, boarding points on a platform (BOARDING POSITION), as well as other areas not directly adjacent to the passenger vehicle such as ticket halls, corridors, etc.. Timetabled SCHEDULED STOP POINTs can be related to STOP PLACES and QUAYs by a PASSENGER STOP ASSIGNMENT. This separation of concern allows the flexible combination of different data sets for timetable and interchanges and allows detailed physical models of stations to be provided. It is typically the SCHEDULED STOP POINT and its specialisation (FARE POINT) that is primarily of relevance for fares and fare purchase.

NOTE 2: Just because a path between two SCHEDULED STOP POINTs exists, it doesn't mean that a given fare allows it to be traversed; a separate layer of DISTANCE MATRIX ELEMENTs can specify the permitted origin / destination stop pairs and their notional distances (for distance based fares). Specific fare determined routing constraints between the origin and destination of a trip can be specified using SERIES CONSTRAINTs; see below.

Other concepts are also relevant for describing the network. LINEs can be used to label a collection of related routes, each route representing a different branch. TARIFF ZONEs and FARE SECTIONs can be used for calculating prices. MODEs describe the method of transport, bus, rail, ferry, etc.

Network elements have spatial coordinates and can also be associated with TOPOGRAPHICAL PLACEs to indicate the towns, cities, villages, etc., for which they provide access.

7.3.1.1 Overview of Stops

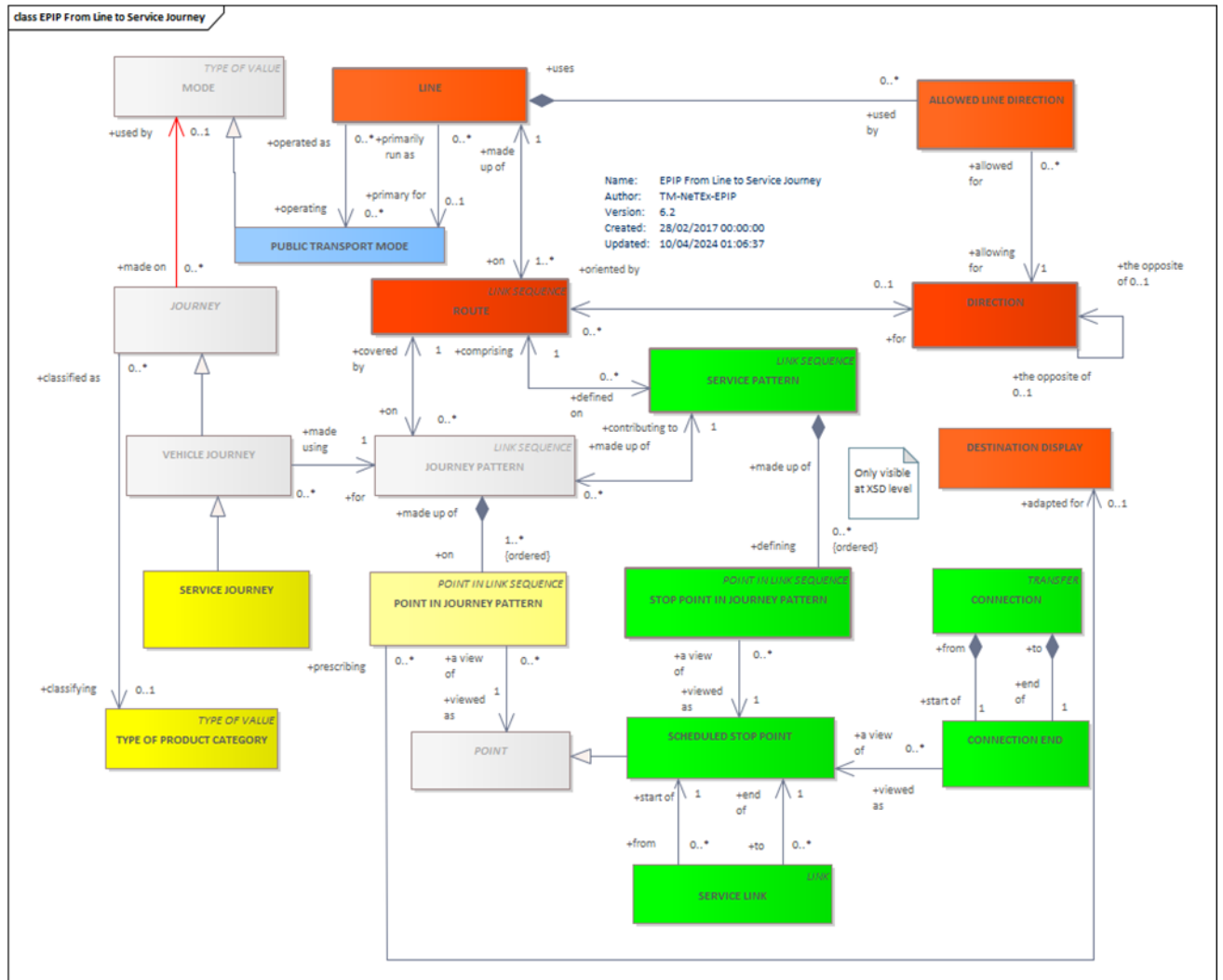


Figure 16 – UML diagram providing an overview of stops

7.3.1.2 Overview of Stop places

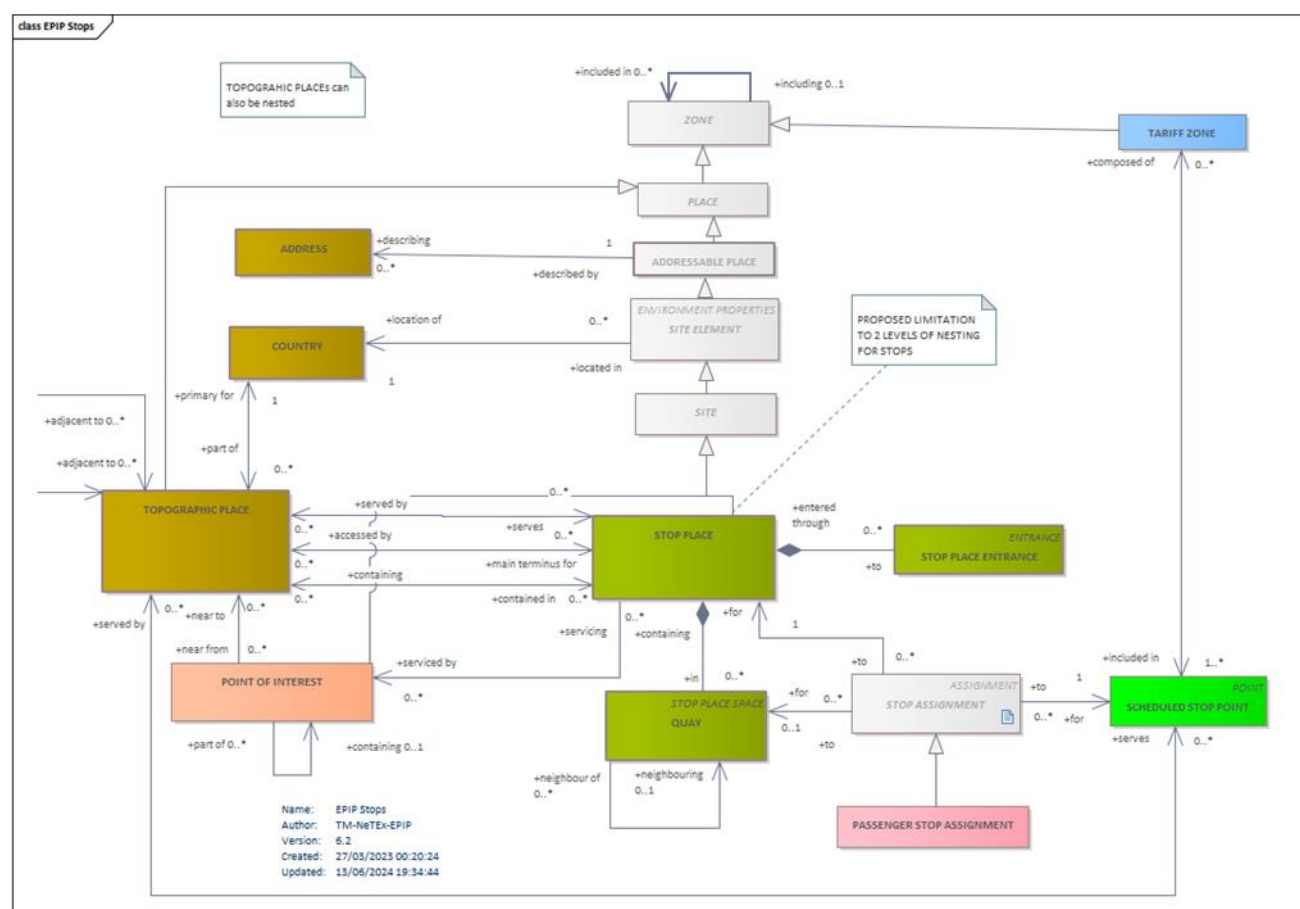


Figure 17 – UML diagram providing an overview of stop places

7.3.2 Journeys by Public Transport Vehicles

Available public transport services over the network are described as SERVICE JOURNEYS, each with a specified departure time on a specified DAY TYPE (e.g., weekday, Sunday, Public Holiday, etc). A Service CALENDAR relates DAY TYPEs to actual dates.

SERVICE JOURNEYS follow a SERVICE PATTERN from an origin to a destination stop; Each stop in the journey is represented by a POINT IN JOURNEY PATTERN and may have an associated PASSING TIME for the journey as well as other information such as the STOP ASSIGNMENT, DESTINATION DISPLAY, SERVICE INTERCHANGES, etc.. Note that some APIs use a simplified view that combines POINT IN JOURNEY PATTERN, PASSING TIME, etc., into a single CALL element.

SERVICE JOURNEYS are made by a VEHICLE or a TRAIN, which may have a DECK PLAN associated with it giving the seating layout, etc.

Note: A SERVICE JOURNEY effectively species a generic journey that may be repeated many times – e.g., Train 702 between Paris and Berlin runs every Monday at 12:03. For each date in the calendar that the SERVICE JOURNEY, runs a NORMAL DATED JOURNEY will be created, allowing unique identifiers and other details to be assigned to the individual journey on a specific date for both operational and reservation purposes. Typically, the same fares will apply to all the NORMAL DATED JOURNEYS of a SERVICE JOURNEY within a current validity period.

7.3.2.1 Overview of Service Journeys

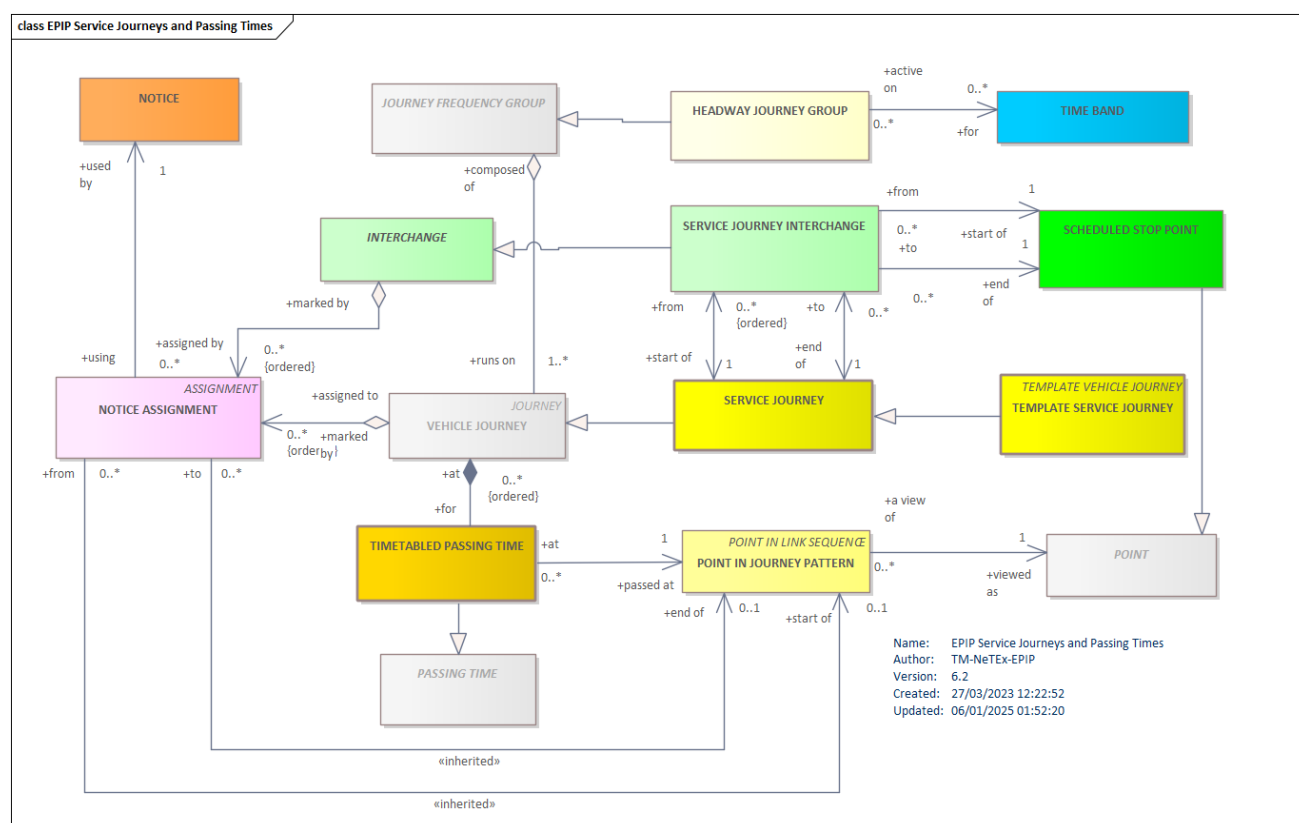


Figure 18 – UML diagram providing an overview of service journey

7.3.3 Fares for Public Transport

In order to be able to describe many different types of fares, and support the many different ways fares may be marketed to users, Transmodel breaks its representation of fares down into a number of different aspects; the fare structure, the fare price, the fare product, the sales offer used to package one or more products for purchase, the ticket or travel document that evidence the purchase, etc.

7.3.3.1 Fare structures and access rights

To relate the fare to the underlying network and consumption of services (including non-travel services such as parking, meals, etc), that a given fare offers, Transmodel uses the concept of 'access rights'. Access rights are specified in terms both of (i) references to network and service elements (such as LINES, TARIFF ZONEs specific SERVICE JOURNEYS, etc) as described above, on which travel can be made, and (ii) with further additional parameters describing temporal and other usage limitations, such as whether interchanges can be made, whether the journey can be broken and when the user can travel using a particular FARE PRODUCT. Each FARE STRUCTURE ELEMENT specifies one or several access rights that may be offered. A TARIFF, groups multiple FARE STRUCTURE ELEMENTs together that may be used variously to define the rights for a FARE PRODUCT or set of products. When a passenger travels, they 'consume' one or more access rights, recorded by the validation and marking of their TRAVEL DOCUMENTs.

A FARE STRUCTURE ELEMENT may specify different types of access rights and conditions.

Point-to-point fares are described by reference to DESTINATION MATRIX ELEMENTs. Each DESTINATION MATRIX ELEMENT specifies an origin and destination pair of SCHEDULED STOP POINTs

and may have price parameters and routing constraints (SERIES CONSTRAINTs) for travel between the two points associated with it.

GENERIC PARAMETER ASSIGNMENTs allow a wide variety of other parameters to be specified for a FARE STRUCTURE ELEMENT. For example:

Zonal fares are described by reference to one or more TARIFF ZONEs for which the FARE is valid.

Different temporal conditions specify the possible durations of a ticket once activated (as TIME INTERVALs for a number of hours, days weeks, etc.). A USAGE VALIDITY PERIOD specifies the period within which a ticket must be used once purchased. Thus, for example, a DAY PASS with a TIME INTERVAL of 24 hours (once activated), might be usable on any one chosen day within a USAGE VALIDITY PERIOD of one year. A FARE DEMAND FACTOR specifies the day types and time bands within which a ticket can be used (peak, off-peak, etc), including the possibility of different start times at different individual stops.

Condition of use and conditions of sale are specified with a wide variety of USAGE PARAMETERs. These may control travel use (e.g., RETURN TRIP, INTERCHANGING and STEP LIMIT to specify the whether the journey can be interrupted, the number of changes allowed, etc.), and the type of user eligible to buy it (e.g., USER PROFILE specifies restrictions on eligibility to a particular user type such as children students, seniors; GROUP TICKET specifies group travel conditions (including the number and types of user that may make up a group); COMMERCIAL PROFILE specifies corporate discount schemes, etc.). Permitted baggage is described by a LUGGAGE ALLOWANCE. Other USAGE PARAMETERs specify commercial conditions (e.g., a PURCHASE WINDOW, indicates when the fare can be bought relative to travel, and whether the fare can be refunded or exchanged (REFUNDING, EXCHANGING), or in the cases of a season ticket, suspended for a while (SUSPENDING). TRANSFERABILITY indicates if the purchase may be used by someone other than the nominated user.

GENERIC PARAMETER ASSIGNMENTs can be combined logically to create compound conditions – for example, to define peak and off-peak travel differently for different directions of travel.

There are many further complexities that may be involved, such as the requirement to consume certain rights in a particular order (FARE STRUCTURE ELEMENT IN SEQUENCE), and that can be represented precisely in Transmodel

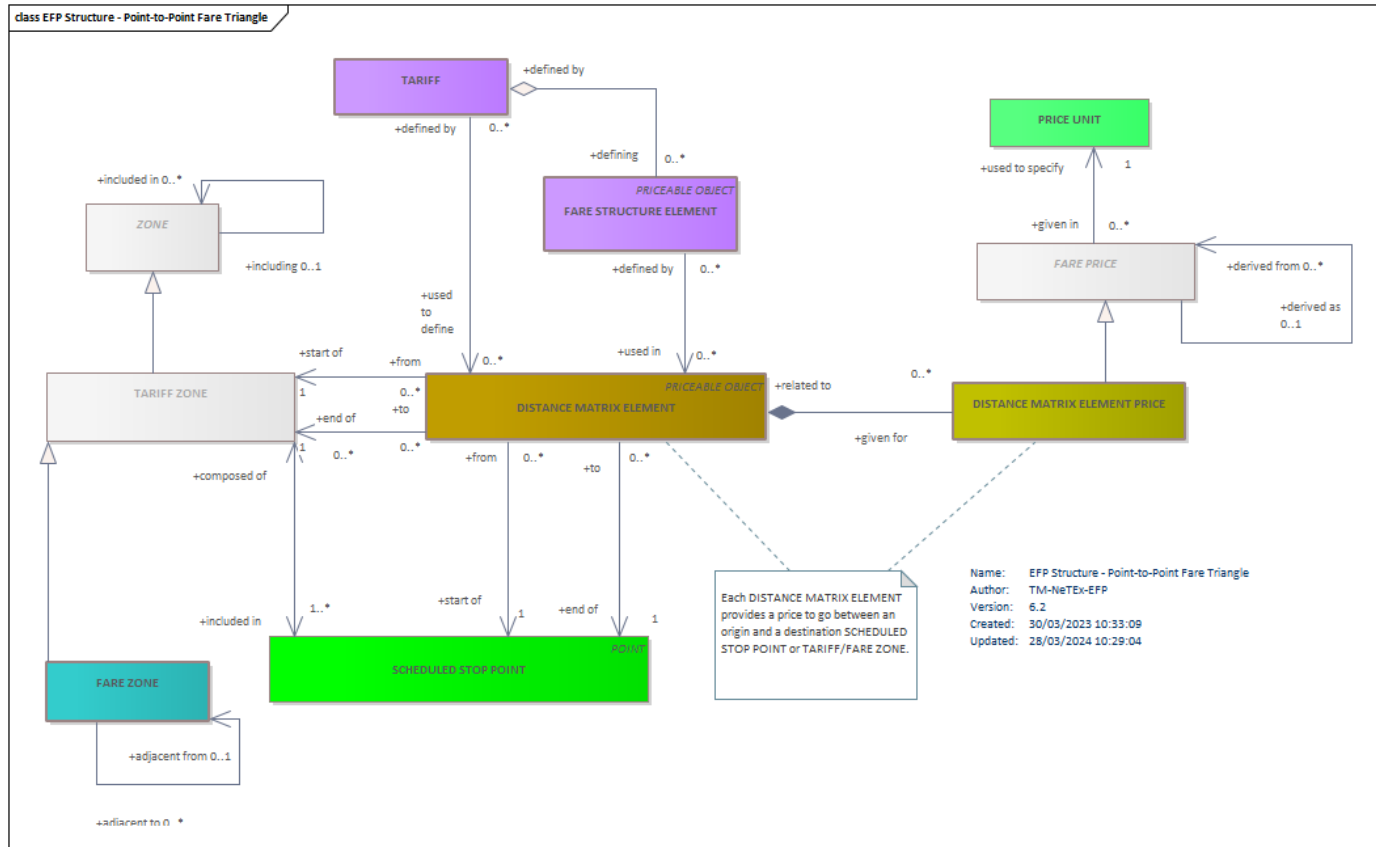


Figure 19 – UML diagram of Point-to-Point fare triangle

A similar structure can be used for zone-to-zone fares

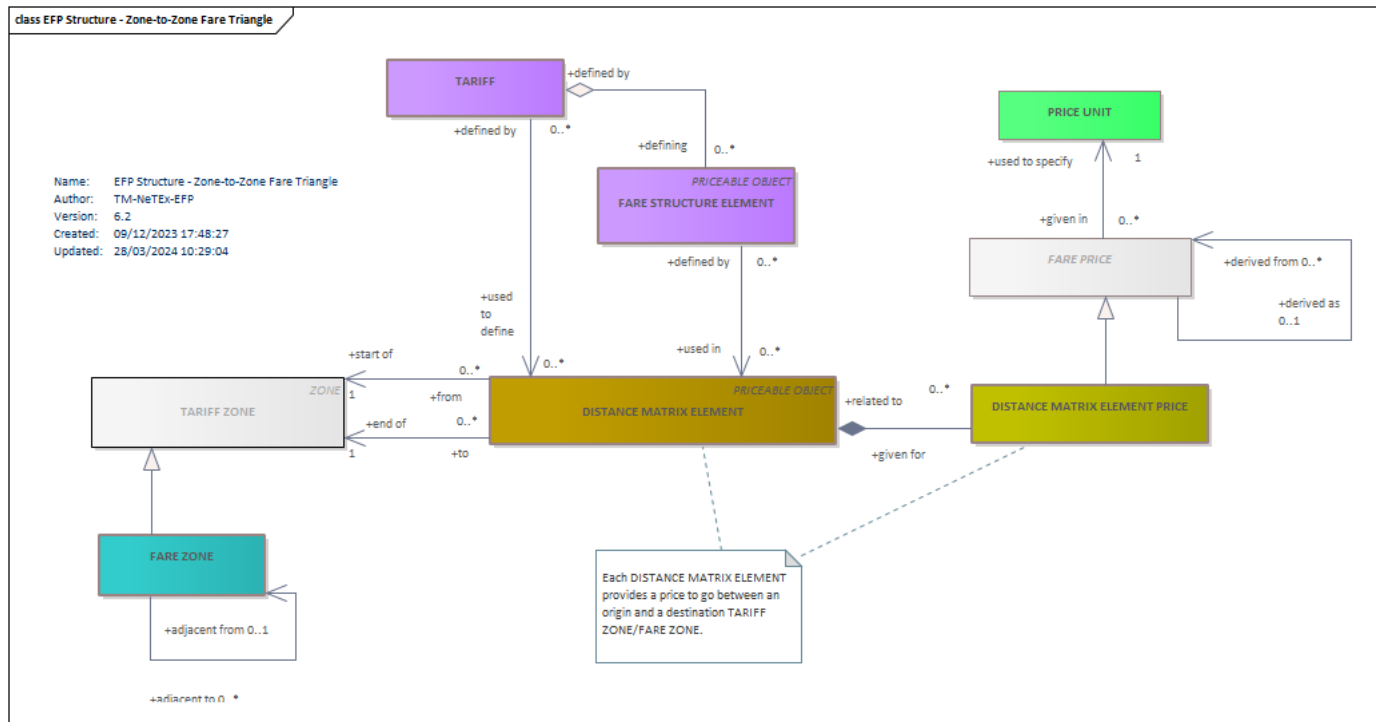


Figure 20 – UML diagram of Zone-to-Zone fare triangle

If the fare grants access rights to travel for a given duration, this is represented with a TIME INTERVAL

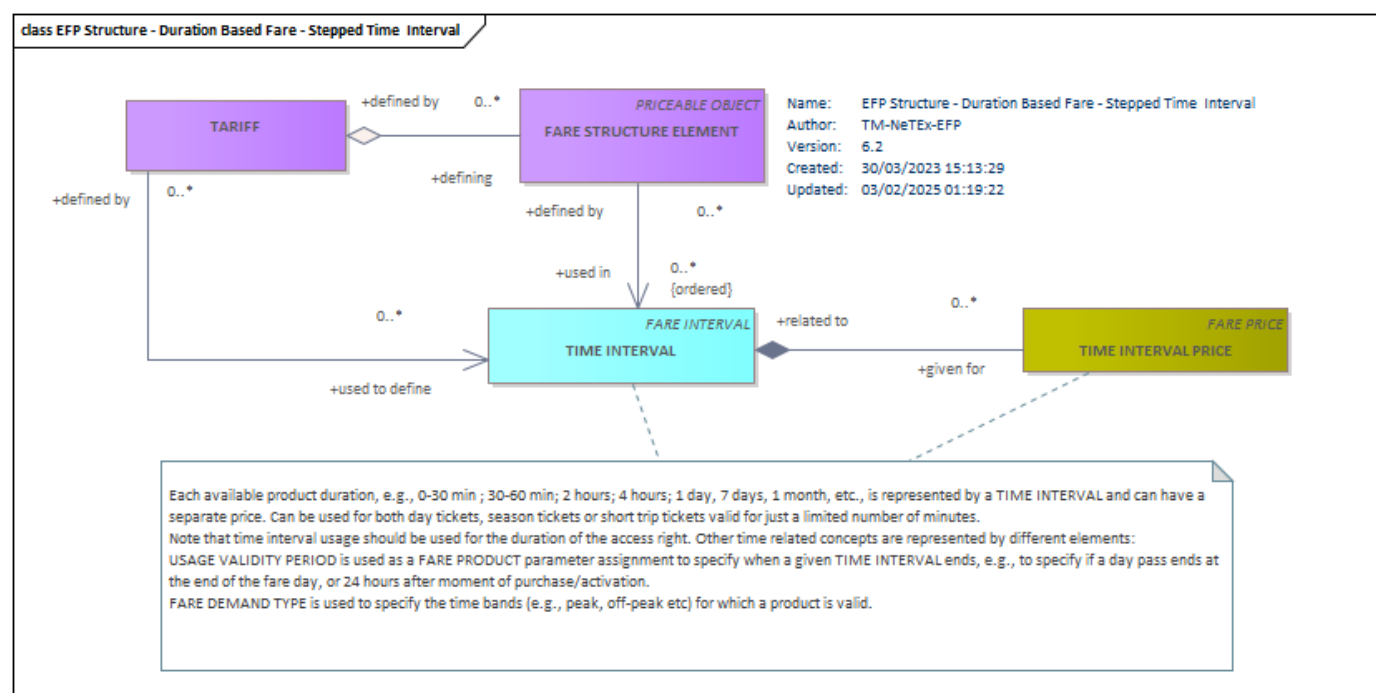


Figure 21 – UML diagram of Duration based fare

7.3.3.2 Fare Products

To represent the sets of access rights that a use can buy as a single purchase, Transmodel uses FARE PRODUCTS. Each FARE PRODUCT assembles a set of access rights and conditions on using those access rights as VALIDABLE ELEMENTs which reference the FARE STRUCTURE ELEMENTs. There are a number of different types of FARE PRODUCTS, two common types are:

- PREASSIGNED FARE PRODUCTS provide access to travel on the network. For example, a single or return trip, or a season pass.
- SUPPLEMENT PRODUCTS define ancillary products such as extra luggage and reservations that are sold alongside a PREASSIGNED FARE PRODUCT.

In addition, Transmodel defines separate FARE PRODUCTS to describe other types of rights that are often sold or bundled together; a SALES DISCOUNT RIGHT is a FARE PRODUCT that gives the right to buy other products at a discount (as say for a travel card); a USAGE DISCOUNT RIGHT defines a rebate or 'frequent flier miles' for "travel distance" used; a CAPPED DISCOUNT RIGHT allows a fare cap to be set for multiple journeys by the same user such that consumption within a given period is capped to the price of a period pass or other amount.

The FARE PRODUCT will usually have generic options for its parameters, that is, it includes all the possible choices allowed for a specific sales proposition. For example, a single trip fare product for a network with many stops may have multiple different DISTANCE MATRIX elements associated with it to describe each of the possible origins and destinations, each with a different price, and only one of which will be selected for a given offer to a passenger making a specific trip; Certain parameters may be fixed which have no choice, for example, a product might be limited to a child USER PROFILE, or time of travel may be limited to an off-peak time window by means of a FARE DEMAND TYPE.

A FARE PRODUCT may specify constraints on the purchase, use and after sales of a product, for example, conditions of travel use (STEP LIMIT, INTERCHANGING, MINIMUM STAY, ROUND TRIP, USAGE VALIDITY

PERIOD, etc.), eligibility of different user types to buy the product (USER PROFILE, GROUP TICKET), when and how it may be purchased (PURCHASE WINDOW, SUBSCRIBING, etc), and after sales (EXCHANGING, REFUNDING), etc.

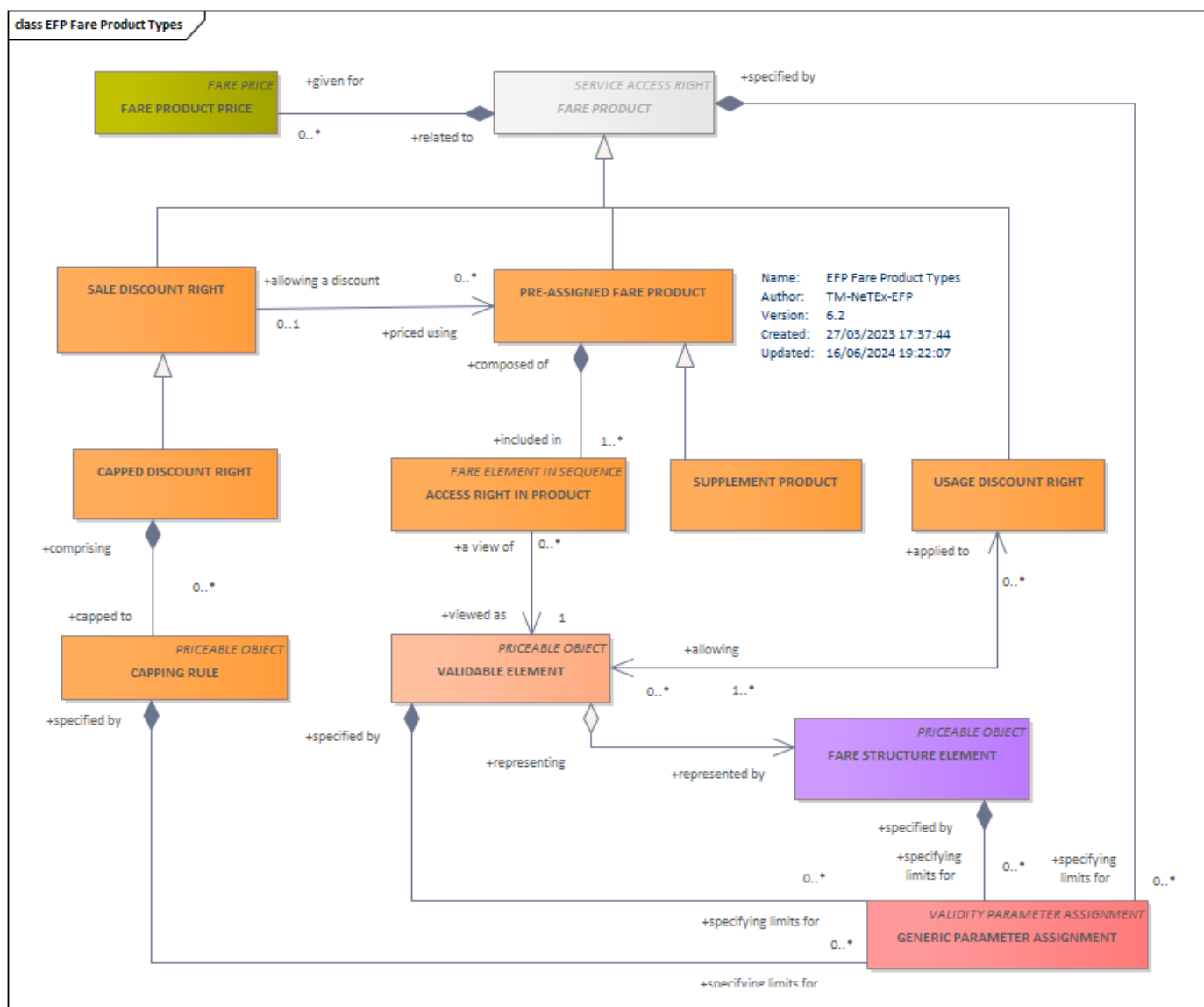


Figure 22 – UML diagram of Fare product types

This flexible approach reflects the very varied ways operators construct and market products in the real world. Thus, for example, an operator might choose to create separate first class and second-class single trip products, while another might have just one single trip fare product with first class and second-class as options.

7.3.3.3 Fare Product Parameters

The access rights and conditions of use of a FARE PRODUCT can be made conditional on a wide range of parameters using the Transmodel ACCESS RIGHT PARAMETER element (some examples are shown below). Some of these scoping parameters such as which LINE, OPERATOR, CLASS OF SERVICE, etc., may be used are based on the network. Time related constraints (TEMPORAL VALIDITY PARAMETERS) such as a DAY TYPE (e.g., Monday, Sunday, Public Holiday etc) or a TIME BAND (e.g., 9:00 tp 5:00) can also be specified.

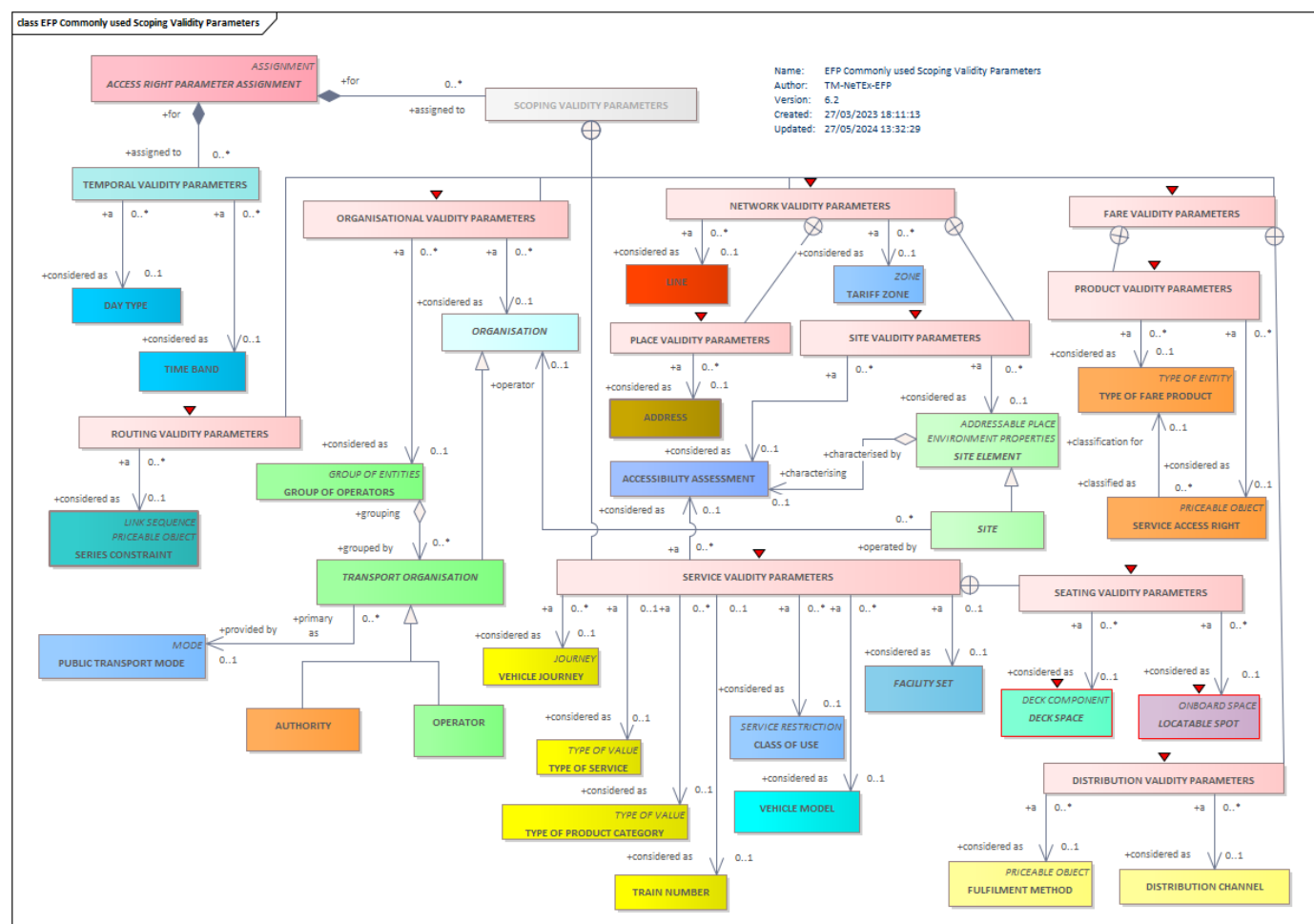


Figure 23 – UML diagram of Commodity and scoping validity parameters

Fare USAGE PARAMETERS specify fare related constraints on a product, such as, who can use it (eligibility: USER PROFILE, COMMERCIAL PROFILE, RESIDENTIAL QUALIFICATION, etc); how often it can be used, how it can be USED (FREQUENCY OF USE, INTERCHANGING, etc.); when it must be purchased (PURCHASE WINDOW, etc.).

Some of the most common usage parameters are shown in the following diagram. Additional parameters may be added to meet further business needs.

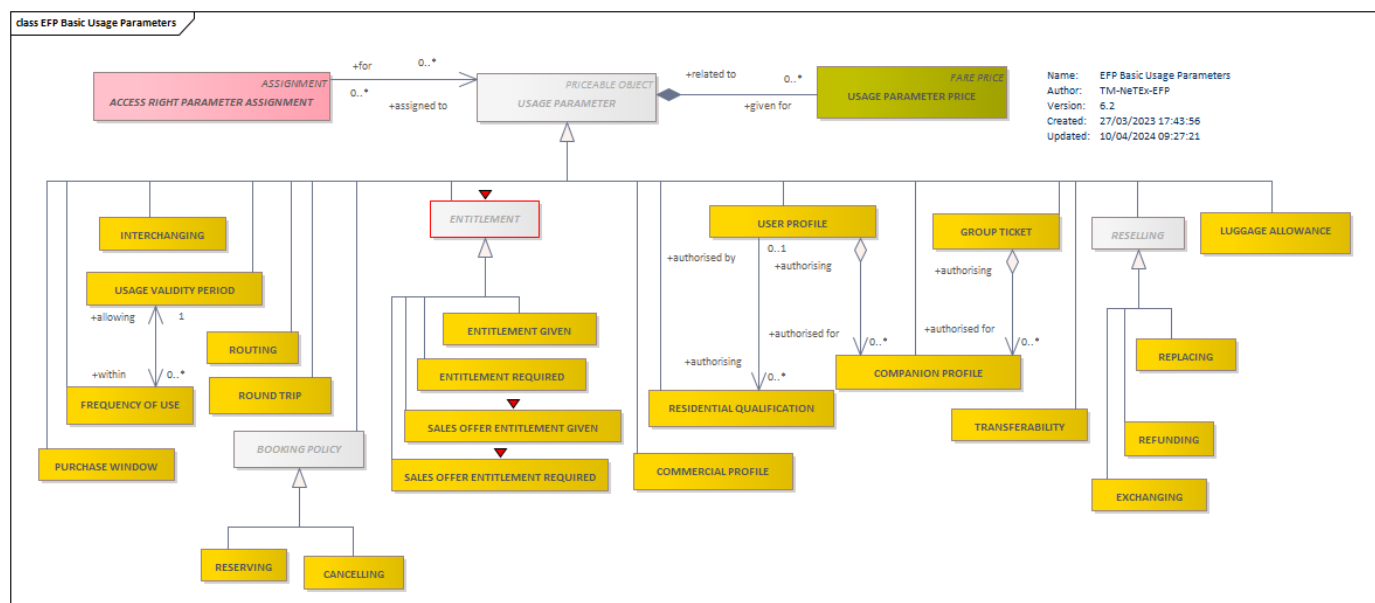


Figure 24 – UML diagram of Basic usage parameters

Dependencies between FARE PRODUCTS and SALES PACKAGES can be described using ENTITLEMENTS, for example, if a passenger must have a travel card in order to purchase a particular product. Or if purchase of a particular product gives him the right to purchase another.

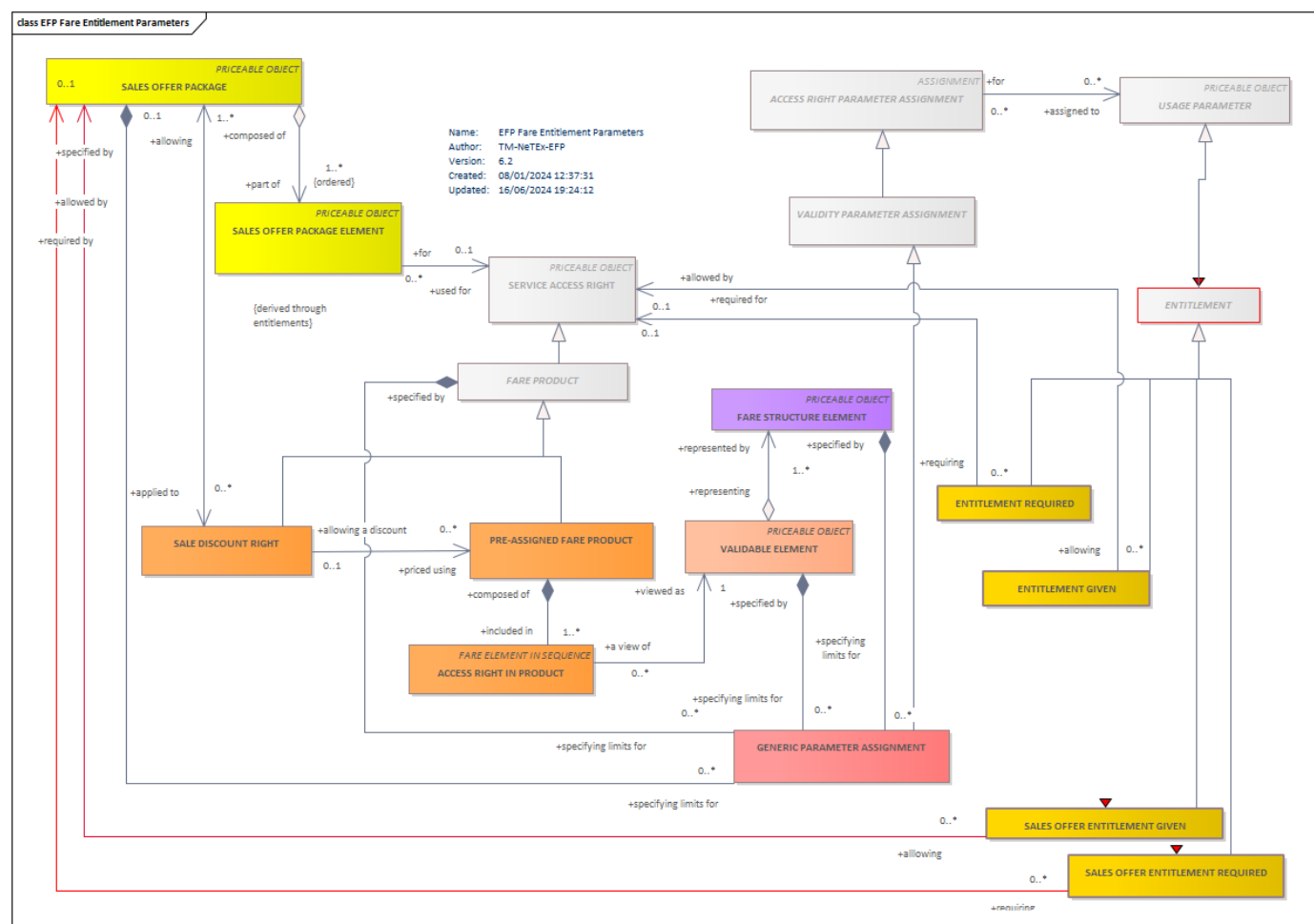


Figure 25 – UML diagram of Fare entitlement parameters

7.3.3.4 Sales Packages

A Transmodel FARE PRODUCT is independent of how it is materialised as a ticket or other proof or purchase and of how it is distributed.

To be purchasable by a user, a FARE PRODUCT must be further associated with a media (TYPE OF TRAVEL DOCUMENT) such as paper, self-print or a mobile app, and by packaging it in a SALES OFFER PACKAGE.

The SALES OFFER PACKAGE can also specify how the product may be distributed, using FULFILMENT METHODS (collect, preprint, online etc) and permitted DISTRIBUTION CHANNELS (online, over counter, travel agent etc), and DISTRIBUTION ASSIGNMENT elements.

A SALES OFFER PACKAGE may bundle together multiple different FARE PRODUCTS as a single purchase. For example, the rights to travel between two points on the network (a PREASSIGNED FARE PRODUCT) might be bundled with other ancillary products such as for a paid reservation, a meal, parking at the departure station, etc., (as, say, SUPPLEMENT PRODUCTS), or a metro day pass for the destination city other (another PREASSIGNED FARE PRODUCT), or to include discounts, (SALES DISCOUNT RIGHT, USAGE DISCOUNT RIGHT) on further purchases.

A SALES OFFER PACKAGE may additionally specify further constraints that are specific to the package rather than the FARE PRODUCT on which it is based to constrain purchase, use and after sales of a product, for example, conditions of use (STEP LIMIT, INTERCHANGING, MINIMUM STAY, ROUND TRIP,

USAGE VALIDITY PERIOD, etc.), eligibility of different user types (USER PROFILE, GROUP TICKET), purchase (PURCHASE WINDOW, SUBSCRIBING, etc.), and aftersales (EXCHANGING, REFUNDING), etc.

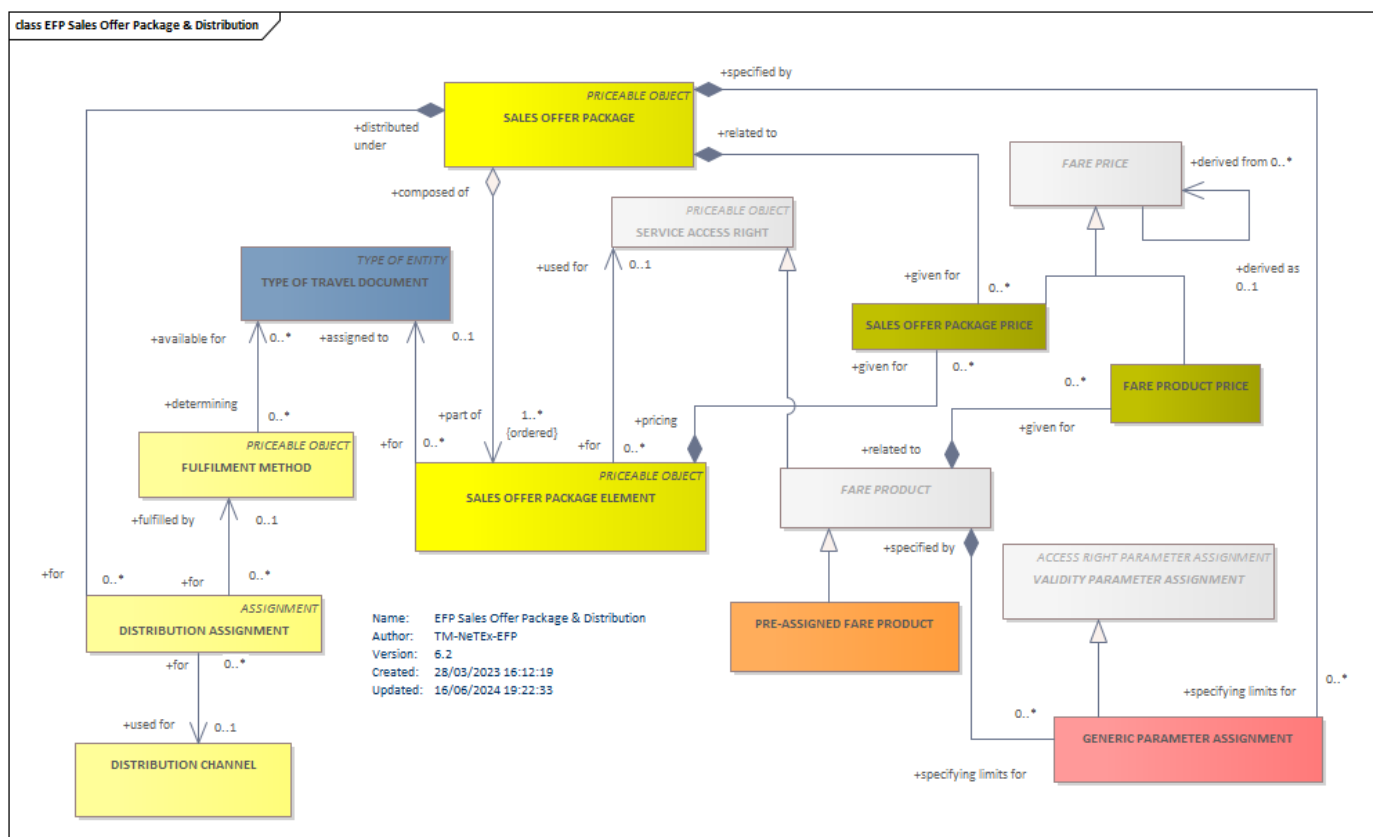


Figure 26 – UML diagram of Sales offer package & distribution

Thus, one can summarise the Transmodel representation of a fare as a set of low-level rules and constraints (specified by FARE STRUCTURE ELEMENTs, and by VALIDITY PARAMETERs that make ACCESS RIGHT PARAMETER ASSIGNMENTs to various elements as parameters) that may be combined very flexibly to make any type of FARE PRODUCT. The FARE PRODUCTs may then be packaged into a SALES OFFER PACKAGE, with further conditions (specified by the same mechanisms) if needed.

Overall, Transmodel separates the many different aspects of fare definitions into separate concerns by means of separate components for each functional consideration. These are then combined systematically to create purchasable offerings. This means existing elements to define the network and trips and conditions can be reused and the model can be extended to add future requirements without inadvertent side-effects.

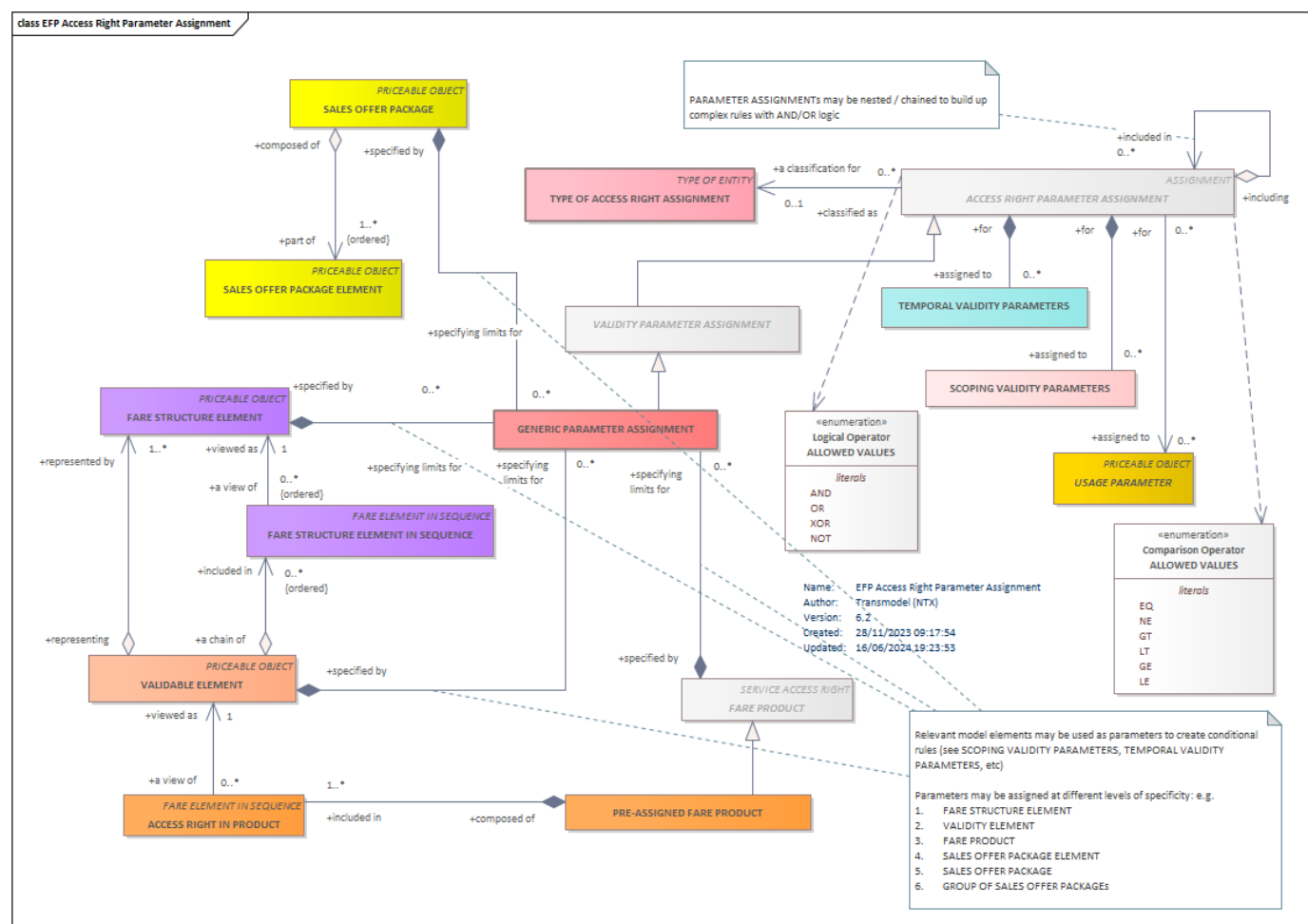


Figure 27 – UML diagram of Access rights parameter assignment

7.3.3.5 Fare Prices

In the Transmodel representation, prices are not an attribute of products, but separate elements in their own right that may be associated with different priceable objects at different levels. Thus, for example, for a FARE PRODUCT describing a single fare on the network with many stops and progressive pricing according to distance, each possible Origin / Destination pair is described by a DESTINATION MATRIX ELEMENT representing the access rights to travel between the two stops. Each such DESTINATION MATRIX ELEMENT could have its own FARE PRICE, either arbitrary or according to a distance-based rule. A reservation could be represented by SUPPLEMENT PRODUCT with a single FARE PRICE representing the flat reservation fee. Both products can be bundled into a SINGLE SALES PACKAGE that can be purchased as one. To price a trip, a fare engine would simply add together the FARE PRICE of the selected DESTINATION MATRIX ELEMENT and the FARE PRICE of the reservation to arrive at a single price. Alternatively, the same two products could be packaged individually as two separate SALES OFFER PACKAGES, so that the travel or just a reservation can be purchased separately.

Classically, a number of different pricing algorithms are used to reduce the amount of price data that needs to be exchanged. Common examples are stage fares, where the price is typically based on the number of unit stages consumed, i.e., passed; or time-based fares where the access is based on duration steps, rather than distance.

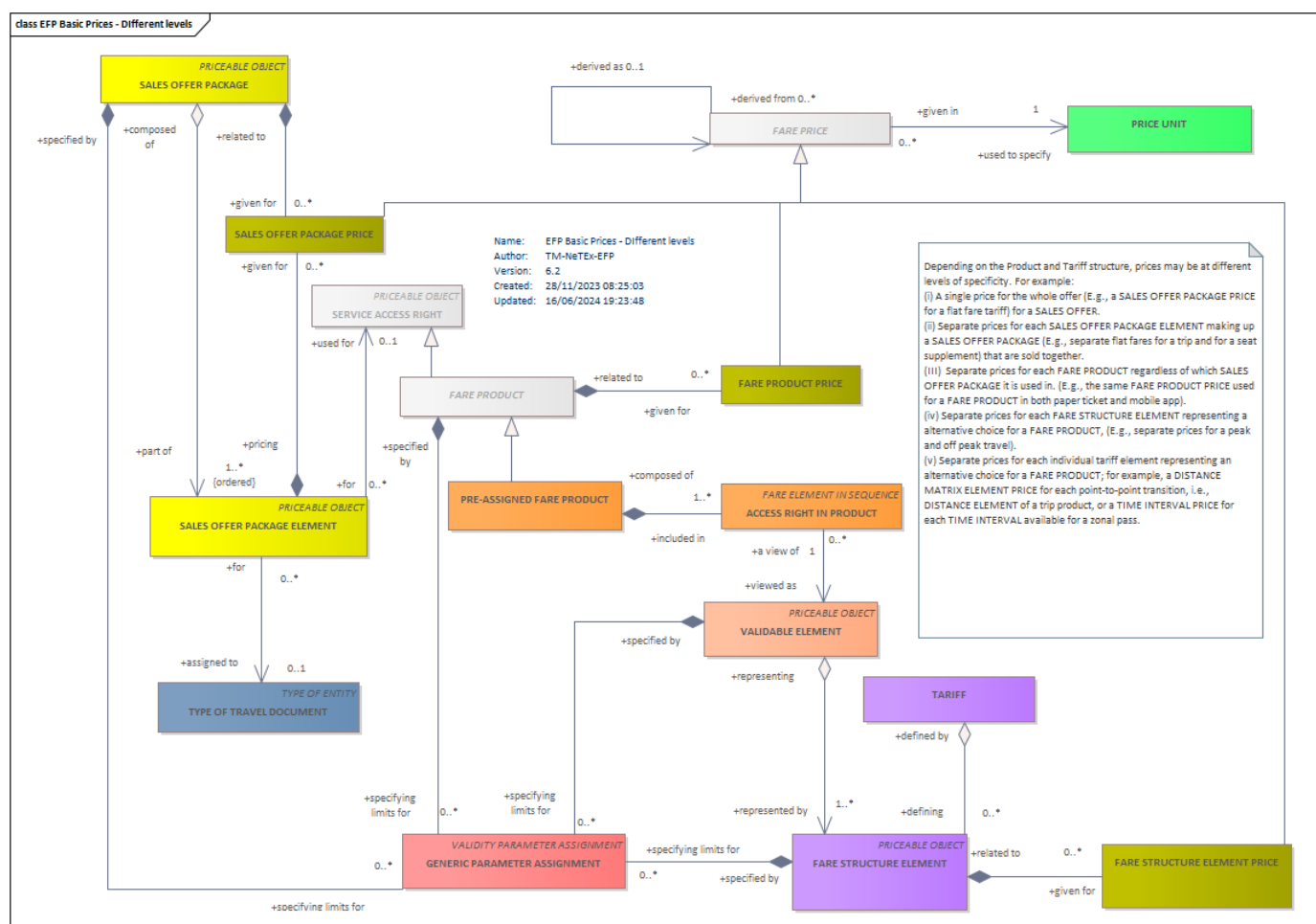


Figure 28 – UML diagram of Basic prices – different levels

Prices may be derived from other prices using a PRICING RULE, along with parameters governing how the result should be rounded. For example, child fares prices could be defined as half the adult fare price. A system need then only exchange the base adult prices as absolute amounts and the parameters and PRICING RULEs needed to calculate the derived prices from the base prices.

For a Multimodal Travel API, the pricing rules and fare calculation will typically be opaque to the user, the fare engine having calculated an absolute price for each trip offer before presenting it to the user. However, it may be necessary to give a justification to the user, e.g., Senior price is 75% of adult price plus 12% VAT so the parameters may be relevant for information purposes.

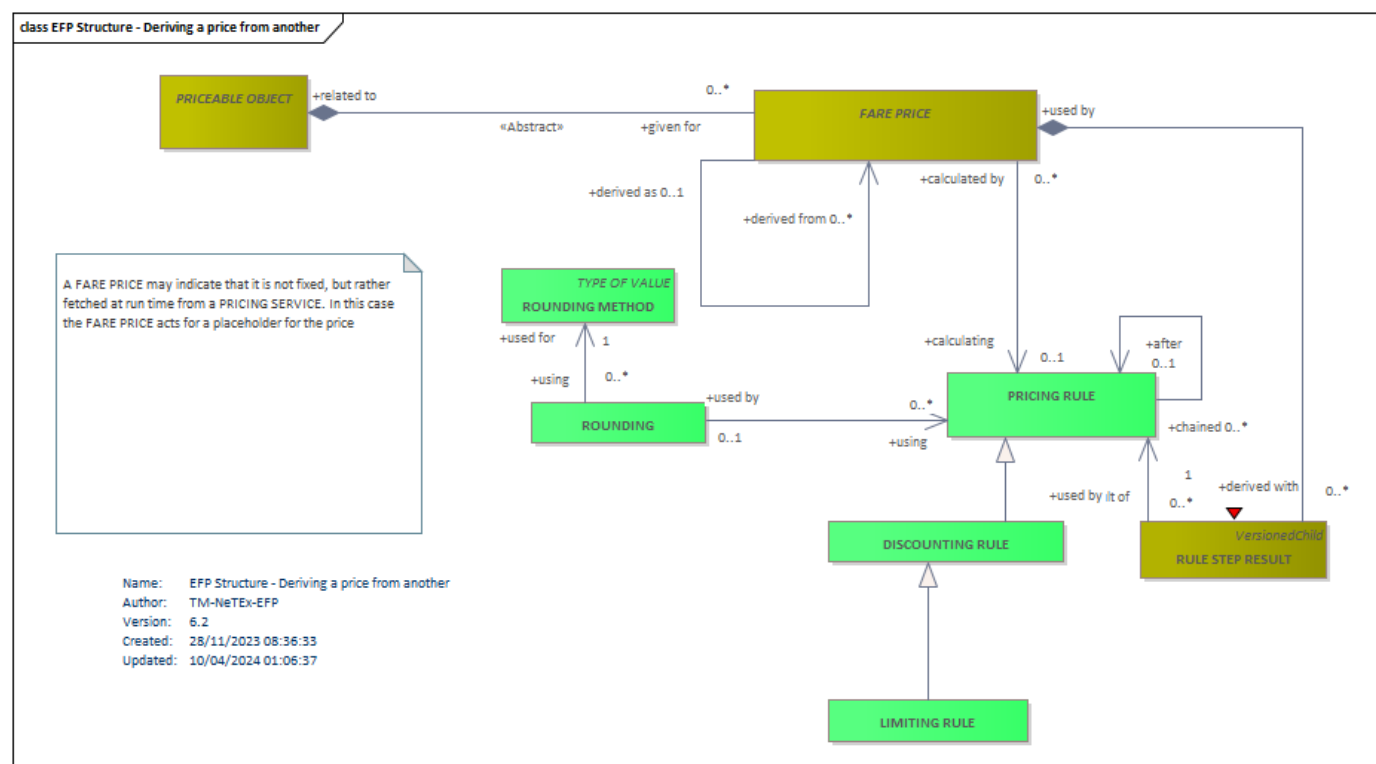


Figure 29 – UML diagram of deriving a price from another

Having a FARE PRICE as a separate element makes it straightforward to change prices without otherwise altering the fare tariff and product structures.

It also makes it easy to accommodate dynamic pricing models, where the price is computed at run time according to yield management or other criteria. In this case, the FARE PRICE in the static model does not specify an actual amount, but is a placeholder indicating the dynamic PRICING SERVICE to be used to fetch the price at run time.

Again, for a Multimodal Travel API, the resolving of dynamic prices will typically be opaque to the user, however the information about the pricing service may be relevant for multi-tier distribution where the prices may be needed to be recalculated at run-time. It may also be a requirement to show a justification of the price (for example in terms of number of fare stages needed or the current prices of specific legs).

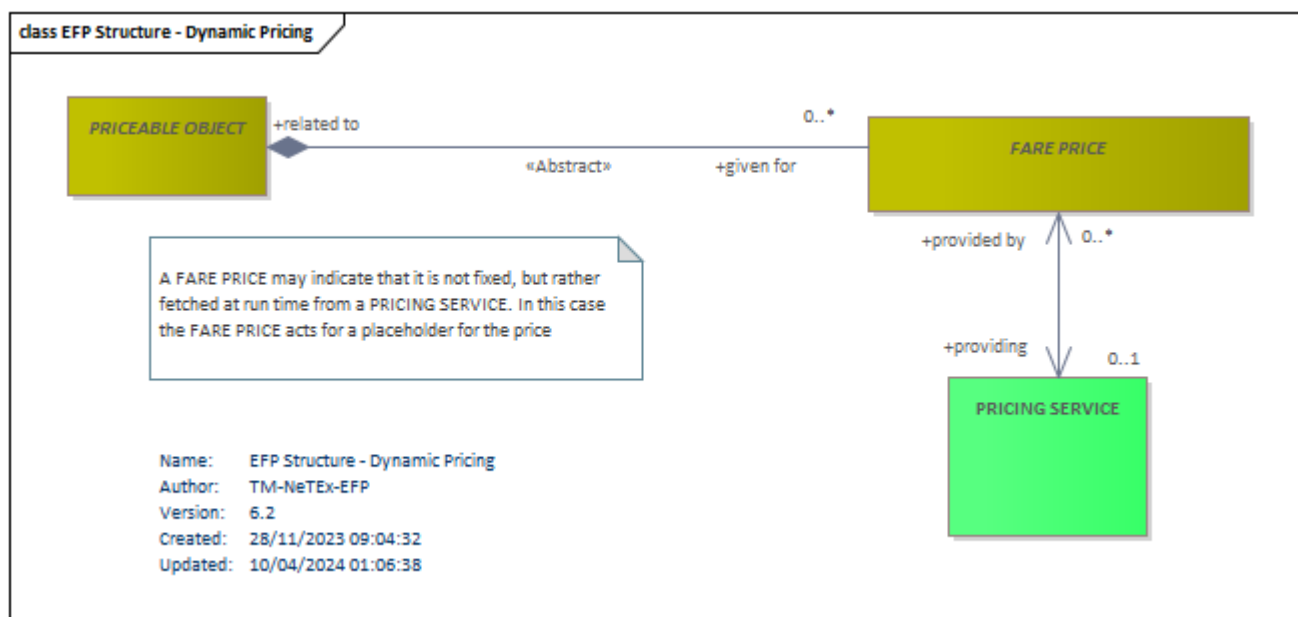


Figure 30 – UML diagram of dynamic pricing

7.3.4 Trips & Journey planning

Given an origin and destination, a journey planning engine will compute and propose a Passenger TRIP made up of one or more TRIP LEGs;

Different types of TRIP LEG – PT RIDE, CONNECTION LEG, ACCESS LEG are possible. Each PT RIDE trip leg describes the actual travel and identifies a boarding and alighting SCHEDULED STOP POINT for a SERVICE JOURNEY on public transport by the passenger to make the desired trip (i.e. a consumption of purchasable access rights). CONNECTION LEGs describe connections at an interchange; a planned interchange between two SERVICE JOURNEYS is represented by a SERVICE JOURNEY INTERCHANGE. ACCESS LEGs describe how to reach the start and destination SCHEDULED STOP POINTs if they are different from the SCHEDULED STOP POINTs.

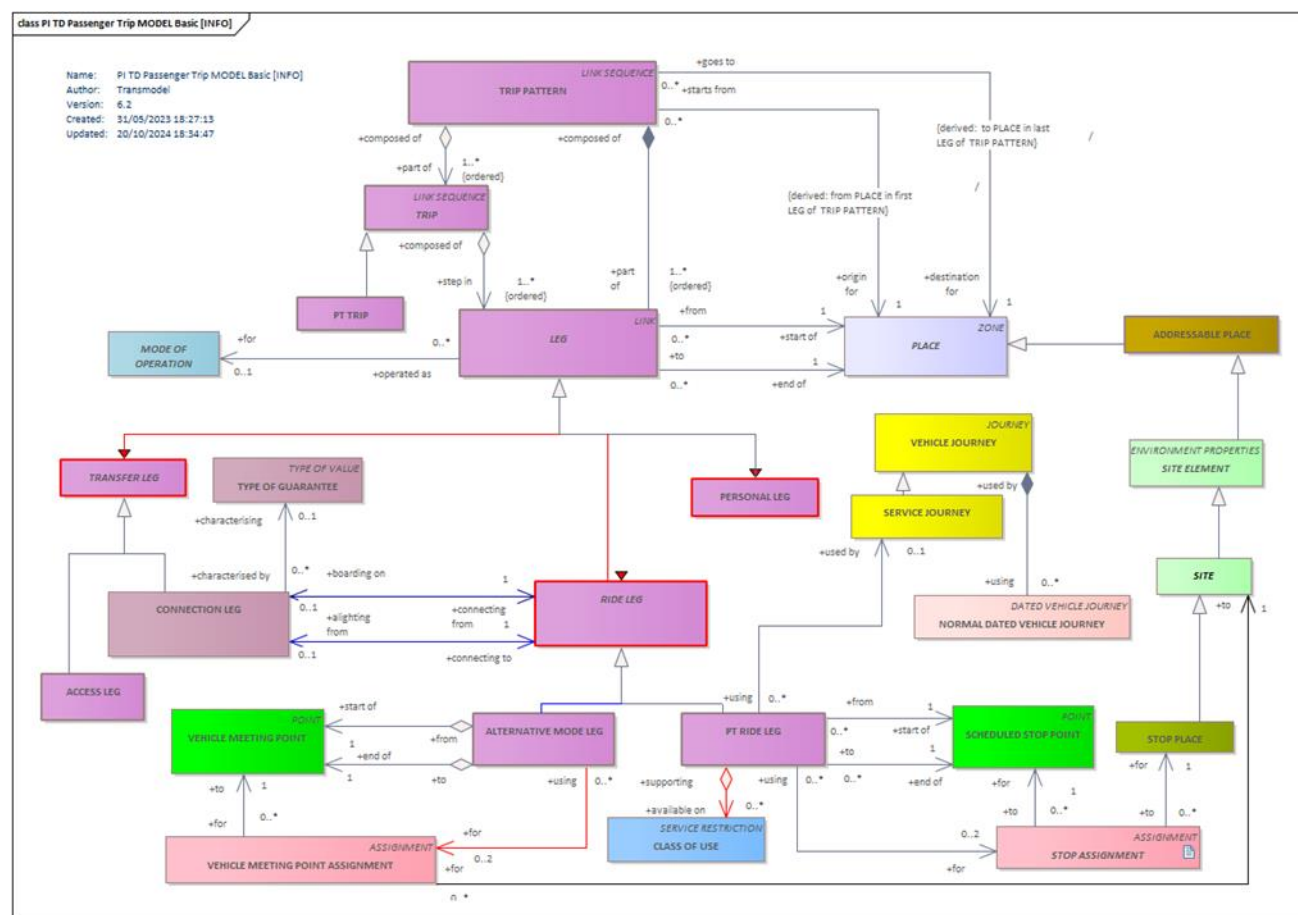


Figure 31 – UML diagram of passenger trip

7.3.5 Fare Finding

Once a possible trip has been identified, a trip planner can search for the fares available for the TRIP LEGs (and for the overall journey if they are different). These may be filtered by further criteria such as the user type, the passenger's entitlements, etc. The available fares are presented as PASSENGER FARE OFFERS, that is trip-specific offers for a particular passenger with an overall FARE PRICE for the offer.

7.3.6 Sales

7.3.6.1 Customers and Accounts

The purchase of a ticket in effect entails a contract between the operator and the customer to provide/obtain services. In practice, the customer may be anonymous (i.e., the operator doesn't know their identity) but nonetheless is subject to passenger rights and regulations.

Transmodel therefore represents all product purchases as SALES TRANSACTIONS on a FARE CONTRACT for a TRANSPORT CUSTOMER (who may be anonymous). The SALES TRANSACTION records details of the sale.

The TRANSPORT CUSTOMERs may have a CUSTOMER ACCOUNT, corresponding to their persistent identity for online transactions and against which transactions can be logged, including not just the purchase, but fulfilment (FULFILMENT ENTRY) of any necessary TRAVEL DOCUMENTs, validation and control (VALIDATION ENTRY, REVENUE PROTECTION ENTRY) and account management (CUSTOMER ACCOUNT ENTRY), and also complaints and travel redress events.

The CUSTOMER PURCHASE PACKAGE specifies which specific parameters have been selected from those available in the SALES OFFER PACKAGE and PASSENGER FARE OFFER that have been purchased. A CUSTOMER PURCHASE PACKAGE is made up of one or more CUSTOMER PURCHASE PACKAGE ELEMENTS, each referencing a SALES OFFER PACKAGE ELEMENT and thus a FARE PRODUCT. The package has CUSTOMER PURCHASE PACKAGE ASSIGNMENTS indicating which options from the package have been selected, as well as the invariant conditions that apply.

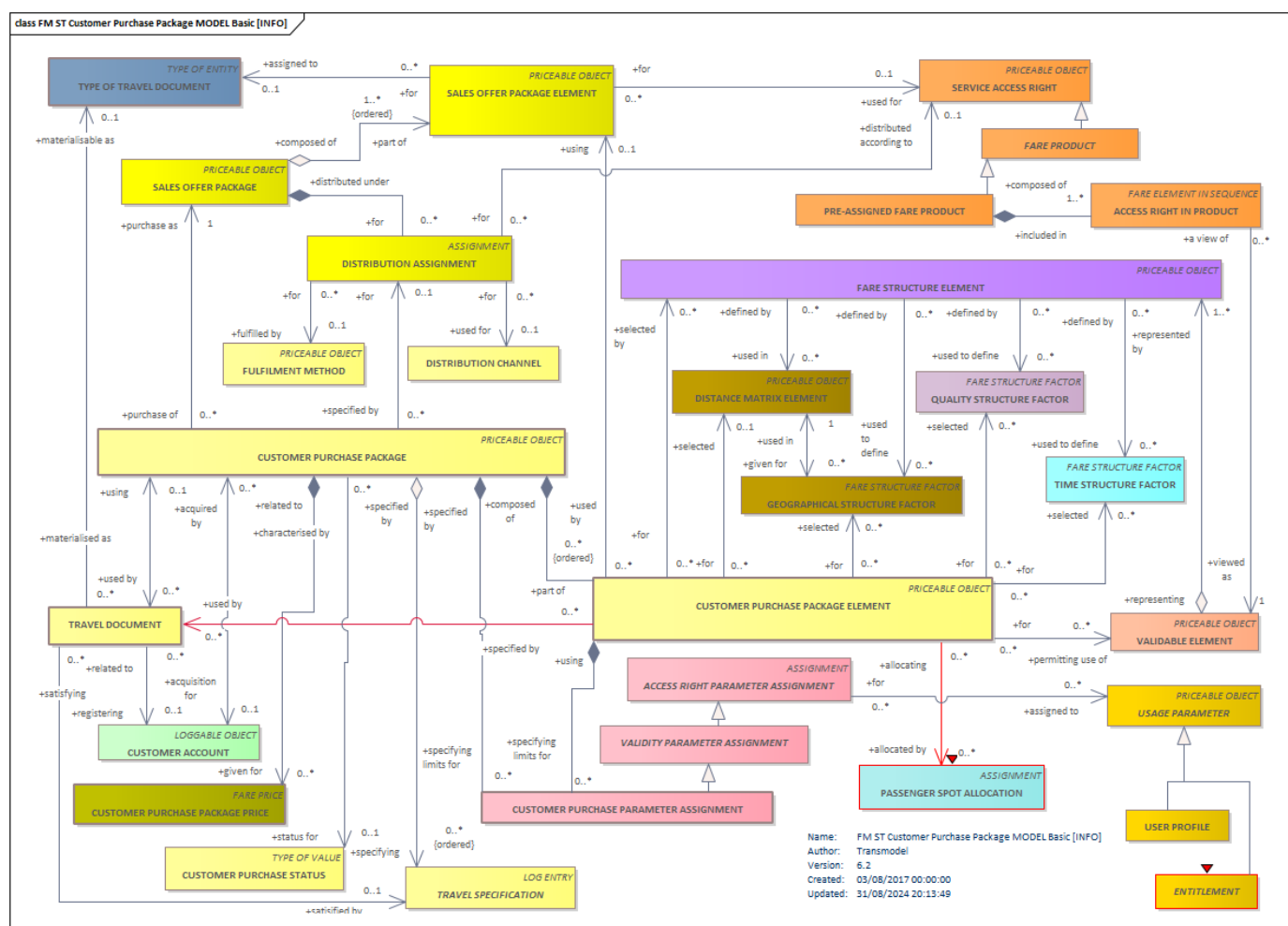


Figure 34 – UML diagram of customer purchase package

7.3.7 Common elements

Transmodel also has a library of low-level components used in many different places. These include:

- NOTICE elements, that allow text to be associated with any element using NOTICE ASSIGNMENTS,
- INFOLINK elements, that allow URLs to be associated with elements,
- SCHEMATIC MAP elements, which can be used to provide an interactive visualisation of elements, for example, for a station map, line map, seating plan, etc,
- FACILITY definitions, which name the amenities available onboard and/or at interchanges,
- ORGANISATION elements, representing companies and authorities providing transport services. Carriers are identified as OPERATORS, a specific type of TRANSPORT ORGANISATION.
- MODEs identify the means of transport (bus, train, car bicycle, etc.), being run according to a MODE OF OPERATION (scheduled, flexible, personal etc).

7.4 Overview of existing Transmodel queries for Trip Planning and Fare Finding

This section summarises the existing Transmodel queries for Trip Planning and fare finding that make use of the conceptual model elements described above to exchange data to carry out the various steps of trip planning, and purchase, etc.

The PI Query Model indicates how the Transmodel data elements relate to APIs and web services that deliver transport data to the end user, for example, journey planners, fare engines, etc. The model does not seek to define a definitive set of services, nor to set out an actual exchange format and syntax, but to give guidance as to which Transmodel elements are relevant for typical passenger information queries, and to identify useful criteria for elements to be the parameters for a query

The PI Query model uses a common framework comprising an abstract query made up of a request/response pair. The framework is the same as that used in SIRI functional messages.

Specific sub models describe each type of functional query as a specialisation of the common model (e.g., trip planning, stop finding, stop departures, etc.). It is assumed that an additional set of messages can be defined for a multimodal travel purchase/reservation API based on a common design pattern such as the PI Query model.

7.4.1 Common Passenger Information Query MODEL

The Common Transmodel PI Query model assumes a conventional request/response architecture. A PI REQUEST is made with specified search criteria to a PARTICIPANT SYSTEM able to support the request type, which returns a PI DELIVERY with results or error codes. The PI REQUEST may be accompanied by a PI REQUEST FILTER which specifies parameters constraining the query and the contents of the results and a PI REQUEST POLICY, which specifies parameters affecting the computation of the results.

The PI DELIVERY provides a container for the elements returned by the query, along with the original query (or queries) itself, and logs attributes useful for audit such as the start time and end time.

Each PT DELIVERY (corresponding to the reply from an http request or remote procedure call) may contain the responses to one or more PI REQUESTs. There are different types of concrete request (e.g., LOCATION REQUEST, TRIP REQUEST, etc.) each with its own corresponding delivery type for the results (LOCATION DELIVERY, TRIP DELIVERY, etc.).

7.4.2 Different types of Trip Planning Query

Specific functional sub models describe commonly found queries for trip planning:

- **The Location Query Model** indicates the elements relevant for a query to find public transport stops.
- **The Trip Query Model** indicates the elements relevant for a query to find public transport journeys for a passenger between given locations.
- **The Single Journey Query Model** indicates the elements relevant for queries to find the details about a planned movement of any type of vehicle from the start point to the end point of a JOURNEY on a specified route.

7.4.3 Different types of Fare Planning Query

Several different types of fare query are defined, also as specialisations of the common model. The queries are concerned with finding the fare products and prices that are available for a specified trip.

- **The Stop Fare Query Model** indicates the elements relevant for queries to find the tariff zones for a given stop.
- **The Fare Product Query Model** indicates the elements relevant for queries to find the details about a specific fare product regardless of any specific trip.
- **The Single Trip Fare Query Model** indicates the elements relevant for queries to find the fares for public transport journeys between given locations.

- The **Repeated Trip Fare Query Model** indicates the elements relevant for queries to find the fares for public transport season passes involving repeated journeys at a given frequency between given locations or within specified zones.

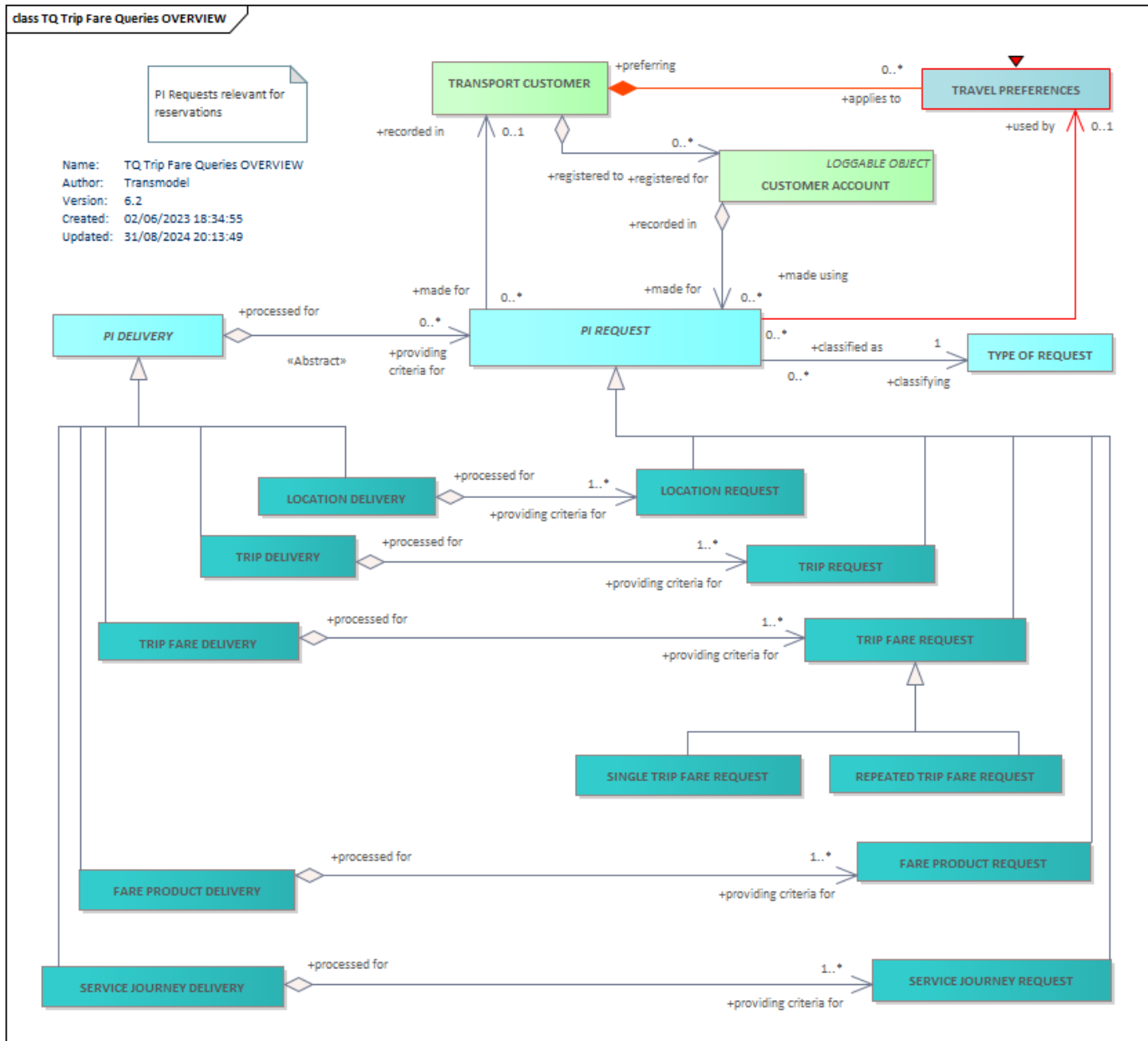


Figure 35 – Passenger Information Common Query Model

7.4.4 Different types of Ancillary Query

- The **Organisation Query Model** indicates the elements relevant for a query to find organisations associated with any aspect of public transportation.
- The **Schedule Query Model** indicates the elements relevant for a query to find public transport timetables.

- The **Service Journey Query Model** indicates the elements relevant for queries to find the details about a specific service journey.
- The **Situation Query Model** indicates the elements relevant for a query to find situation data about disruptions affecting public transport.
- The **Vehicle Type Query Model** indicates the elements relevant for a query to find a classification of public transport vehicles according to the vehicle scheduling requirements in mode and capacity (e.g., standard bus, double-decker, etc.).

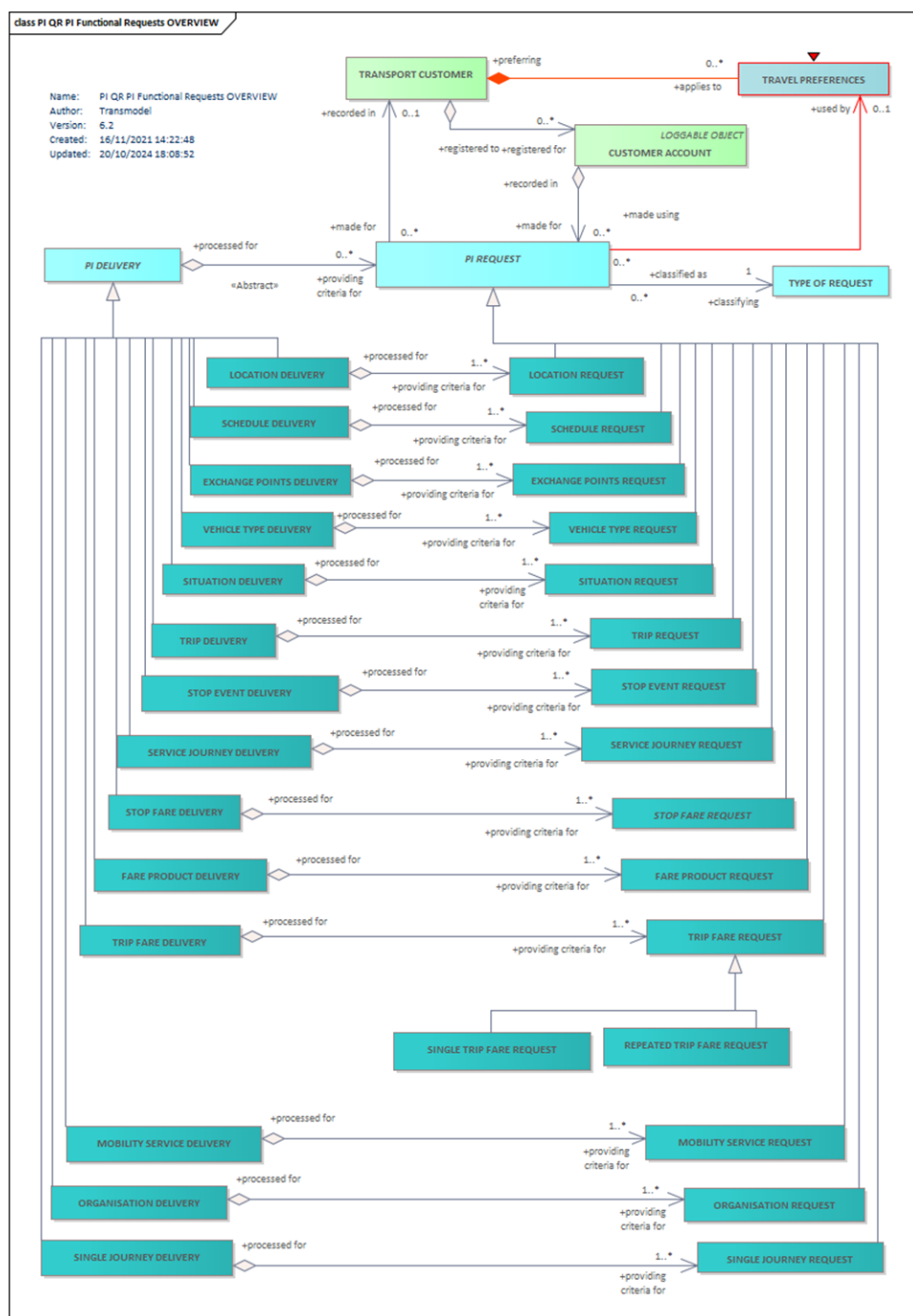


Figure 36 – Functional requests overview

8 New Transmodel concepts to support Multimodal Travel Purchase/Reservation APIs

Support for Multimodal travel purchase APIs requires extending Transmodel to add some new functional sub models. This section outlines some of the additional elements relevant for supporting a purchase/reservation API. Discussion is broken down under two headings:

- d) Additions to the Transmodel conceptual model to support Multimodal Travel Purchase/Reservation concepts.
- e) New request/response message pairs needed to support a Multimodal Travel Purchase/Reservation API with which to make purchases of travel products and reservations.

Please note that these additions do not constitute a final or definitive proposal for extending Transmodel, but rather an incomplete discussion “strawman” to show how such additional features might be added.

8.1 Extensions to the Transmodel Trip Model

The current Transmodel Trip model does not include a detailed stop itinerary, i.e., details on the boarding, intermediate and alighting stops and times for an individual passenger trip, as it assumes these can be derived from the SERVICE JOURNEY using the TRIP LEGs. It will be desirable to include the full itinerary (equivalent to OSDM Leg Stop elements, i.e., **Board, Alight. TripVia, NotVia** elements) in order to make the trip definition of a CUSTOMER PURCHASE PACKAGE capable of being used as a stand-alone artefact without recourse to the trip planner.

This might be done by adding a general STOP POINT IN RIDE LEG for each stop in the trip leg, specialised as BOARDING STOP, INTERMEDIATE STOP and ALIGHTING STOP elements.

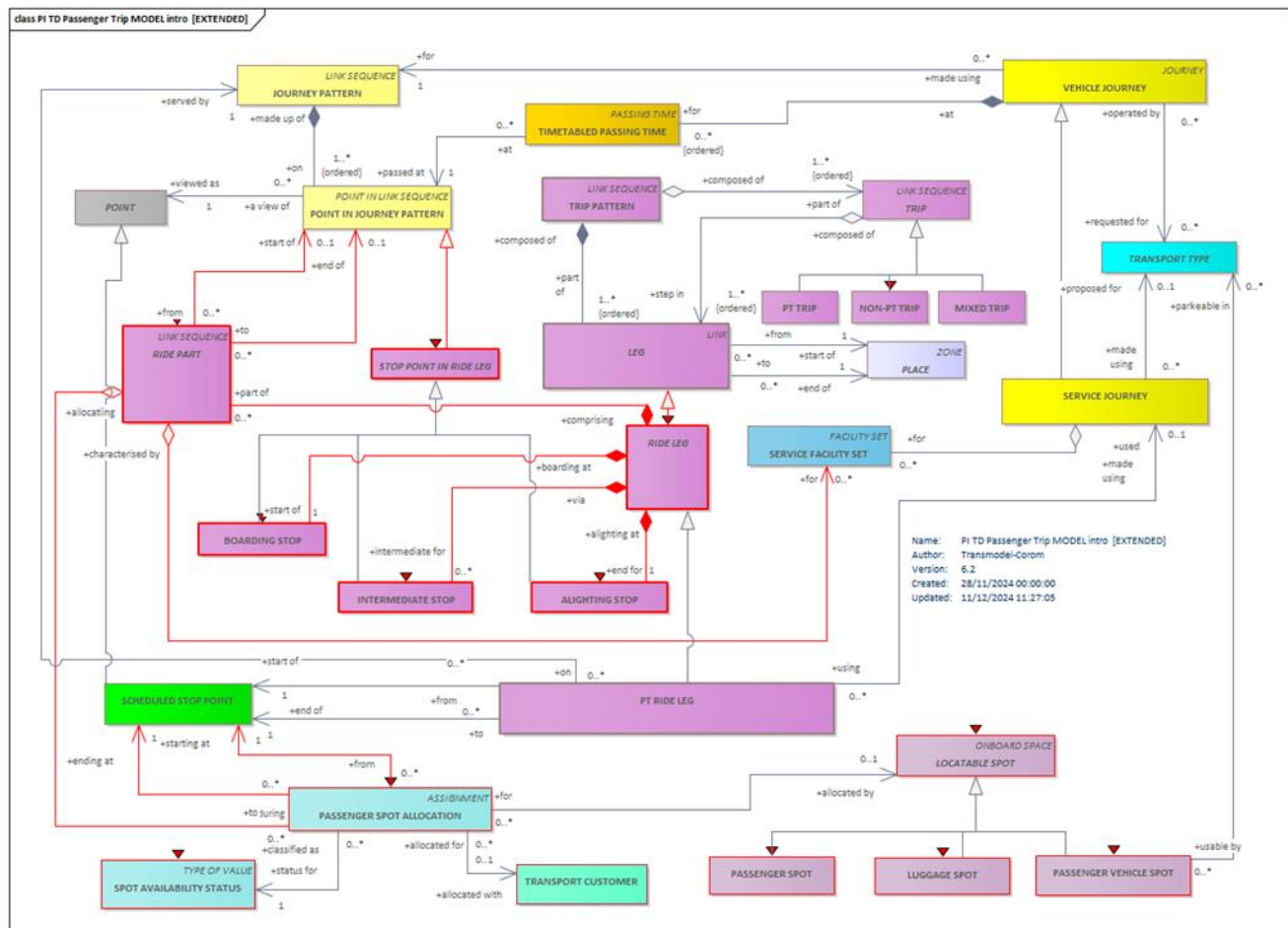


Figure 37 – UML diagram of passenger trip

8.1.1 Travel Package

A new TRAVEL PACKAGE element is proposed to help simplify the API:

The number of separate parameters that need to be passed on a message is reduced – instead once a TRAVEL PACKAGE is created, it is passed around as the primary parameter.

The API can be considered to be a set of REST Like operations on the TRAVEL PACKAGE – CREATE, CANCEL, MODIFY, PAY, etc

It is possible to perform certain operations in an alternate order – e.g., to reserve seat before or after purchase, thus supporting alternate use cases.

The atomic (i.e., all or nothing) purchase of a set of multiple products/offers is made possible.

It is easy to specify TRAVEL GUARANTEES and TRAVEL REDRESSES that apply to all or part of the purchase of access rights.

It allows products from different SALES OFFER PACKAGES to be combined as one.

A TRAVEL PACKAGE is created by a purchase request and passed through other API requests for reservation, modification, payment, fulfilment, etc.– it can be conceptualised to the customer as a “shopping basket” which can be assembled and then paid for. Timeout validities can be set on the contents to hold reservations for a limited period before confirmation.

8.1.1.1 Travel Package Overview

A TRAVEL PACKAGE describes an offer for a specific trip or trips that are being purchased as a whole; it is made up of TRAVEL PACKAGE ELEMENTs, each specifying a component of the offer.

8.1.1.2 Travel Package Definitions

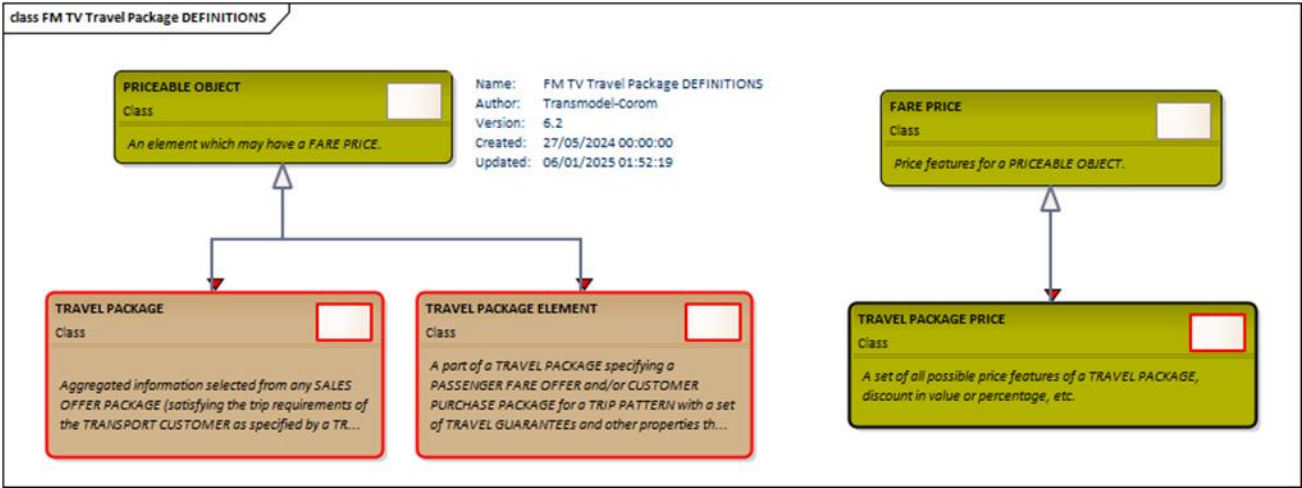


Figure 38 – Definitions of Travel Package

TRAVEL PACKAGE: Aggregated information selected from any SALES OFFER PACKAGE (satisfying the trip requirements of the TRANSPORT CUSTOMER as specified by a TRIP SPECIFICATION) presented to the TRANSPORT CUSTOMER and enriched by specific information; (i) on the TRAVEL GUARANTEE; (ii) on the organisation providing the guarantee.

NOTE: TRAVEL PACKAGE represents a contract between the TRANSPORT CUSTOMER and the organisation providing the TRAVEL GUARANTEE. TRAVEL PACKAGE ELEMENT: A part of a TRAVEL PACKAGE specifying a PASSENGER FARE OFFER and/or CUSTOMER PURCHASE PACKAGE for a TRIP PATTERN with a set of TRAVEL GUARANTEES and other properties that apply specifically to the part.

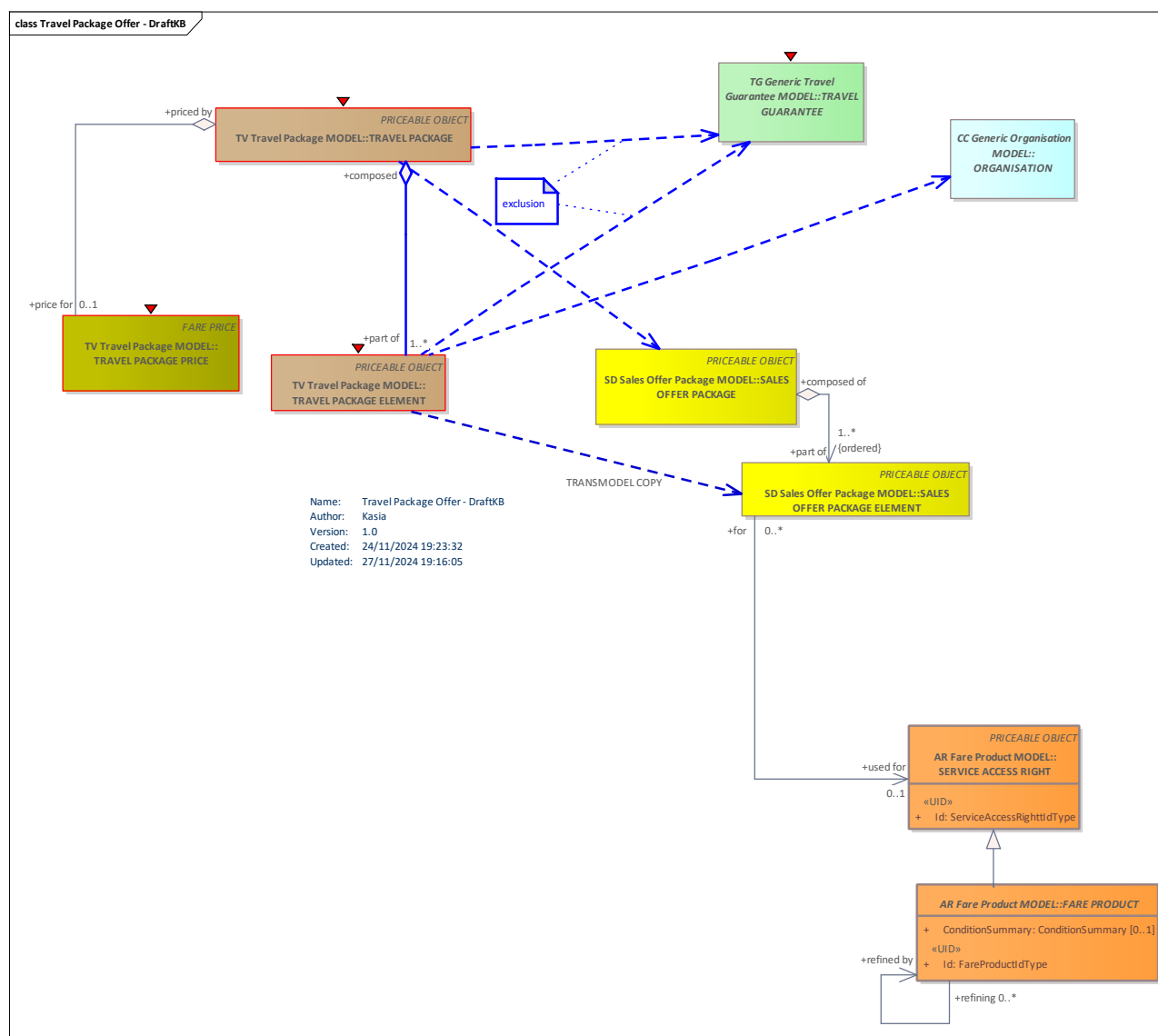


Figure 39 – UML diagram of Travel package offer

8.1.1.3 Travel Party Model

The makeup of the party travelling (single user, group of users, family, companions, pets, etc.) can be described as a TRAVELLING ENTITY. The USER PROFILES and / or GROUP TICKETs of any FARE PRODUCTs making up the TRAVEL PACKAGE should satisfy the TRAVELLING ENTITY as to the number and type of user. for each leg.

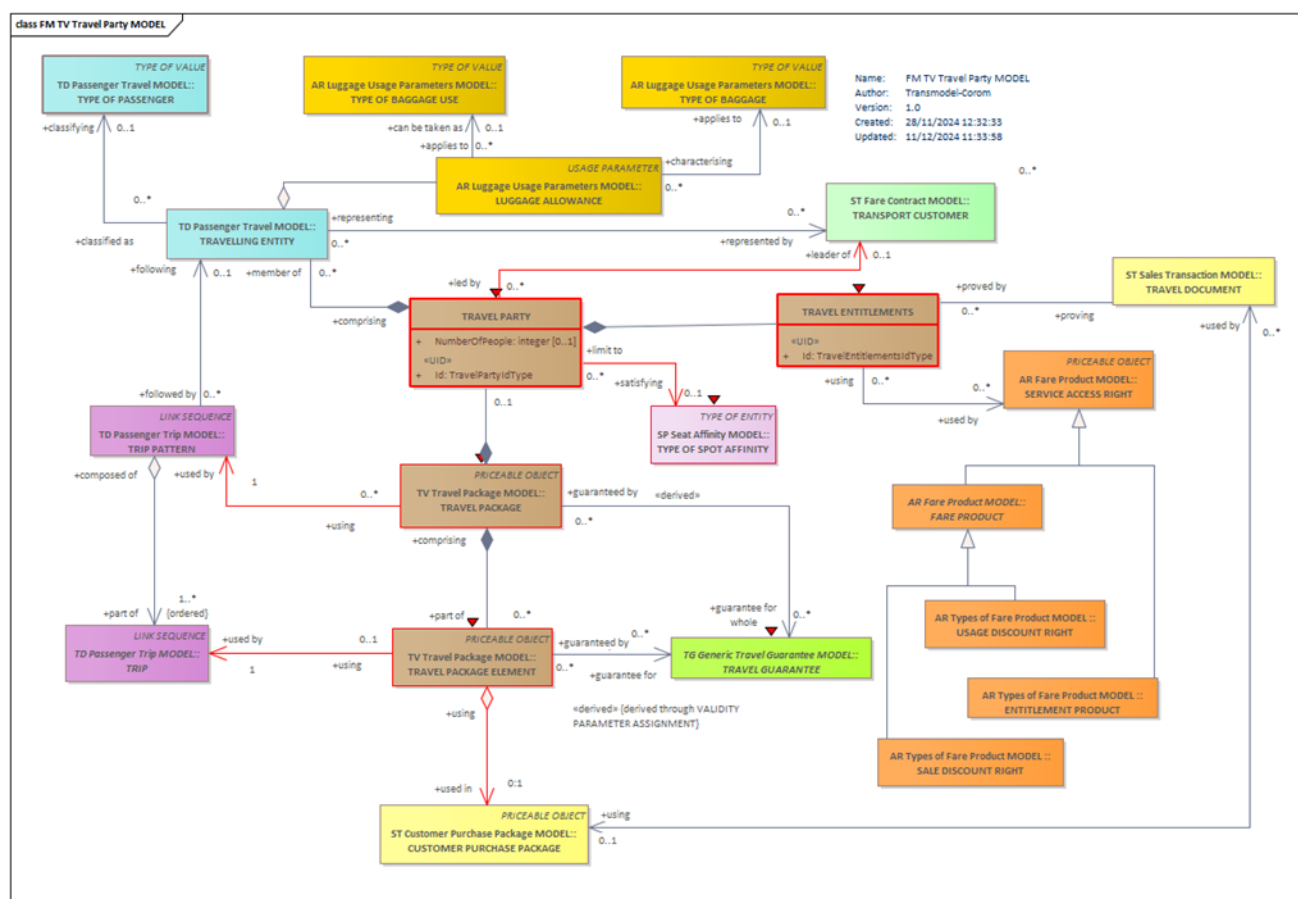


Figure 40 – UML diagram of Travel party

8.1.2 Travel Guarantees & Redresses

One of the requirements identified by the CoRoM Workgroup was the need to explicitly represent the different types of travel guarantees for trips and services that may or may not be provided, as well as the redresses to be offered if the guarantee is not met. Such TRAVEL GUARANTEES may variously apply to a whole TRIP, TRIP LEGs or even several different trips. TRAVEL REDRESSES may apply to specific TRAVEL GUARANTEES.

8.1.3 The different types of Travel Guarantees

Several different types of TRAVEL GUARANTEES have been identified, for example, that the trip will be on time, will complete (i.e. will reach its intended destination), will have certain onboard services, etc.. Each type of TRAVEL GUARANTEE can be regarded as a model element, organised into a hierarchy.

An example hierarchy with some provisional definitions is shown below. There are four main categories of TRAVEL GUARANTEE; firstly, that the travel will meet specified criteria as to timeliness, assistance, etc. (TRAVEL QUALITY GUARANTEES); secondly, that alternative travel will be provided if the planned trip fails (TRIP REPAIR GUARANTEE); thirdly, that there will be adequate information (TRAVEL INFORMATION GUARANTEE); and fourthly, a catchall for miscellaneous other guarantees (OTHER TRAVEL GUARANTEE).

This provisional draft should be further reviewed and modified as part of a formal extension to Transmodel.

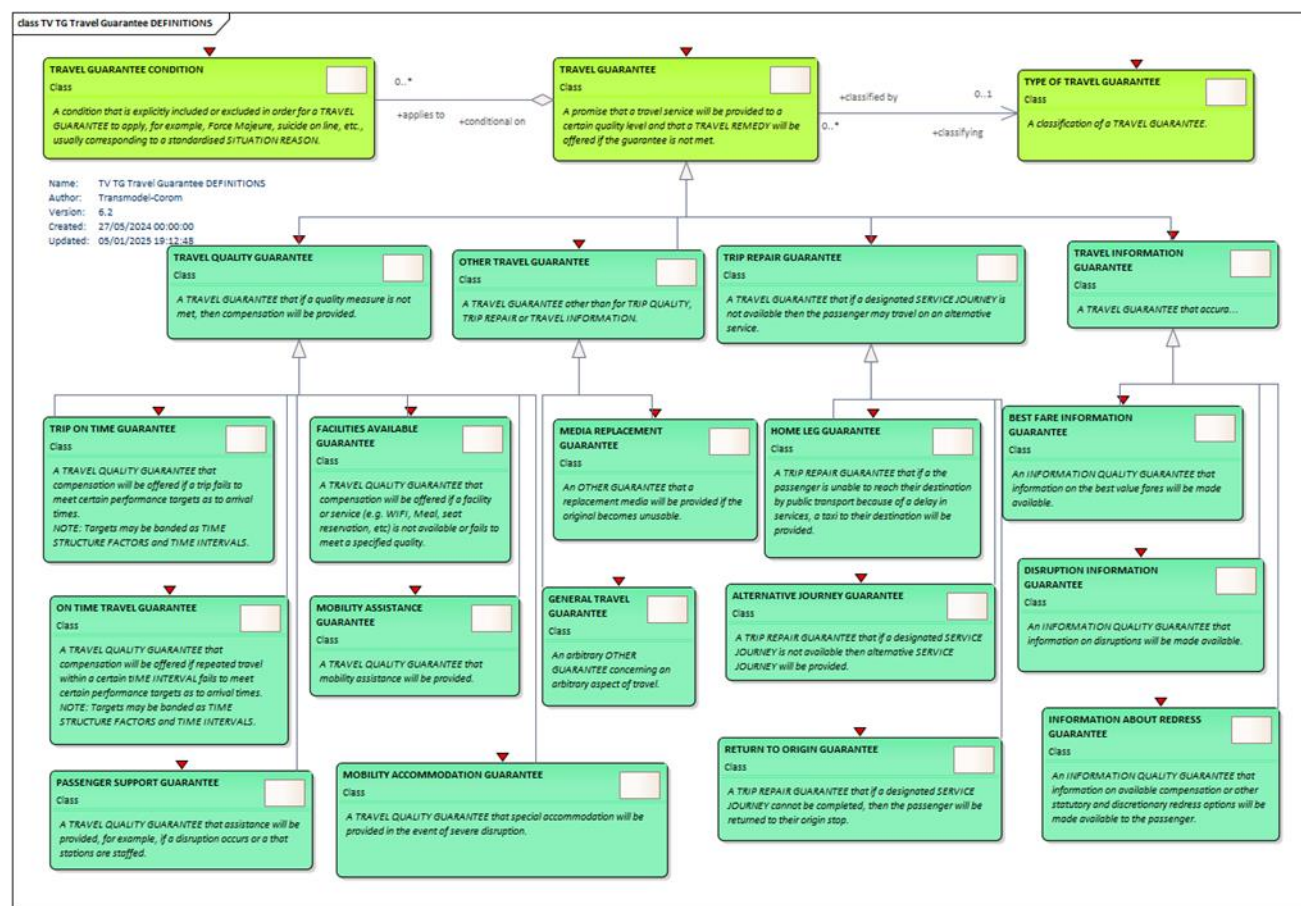


Figure 41 – Definitions of travel guarantees

8.1.4 Redresses

If a TRAVEL GUARANTEE is not met, then a TRAVEL REDRESS may be offered to assist and/or compensate the passenger. Again, each different type of commonly found redress (for example, a new ticket, monetary compensation, free taxi home, free meal, etc.) may be reified as a model element, organised into a hierarchy.

An example hierarchy of TRAVEL REDRESSES with some provisional definitions is shown below. This should be further reviewed and modified as part of a formal extension to Transmodel.

Redresses fall into several categories – TRIP REPAIR REDRESSES which allow alternative travel access rights; REBATE REDRESSES which provide monetary or travel in-kind compensation, and PASSENGER SUPPORT REDRESSES which provide accommodation or meals.

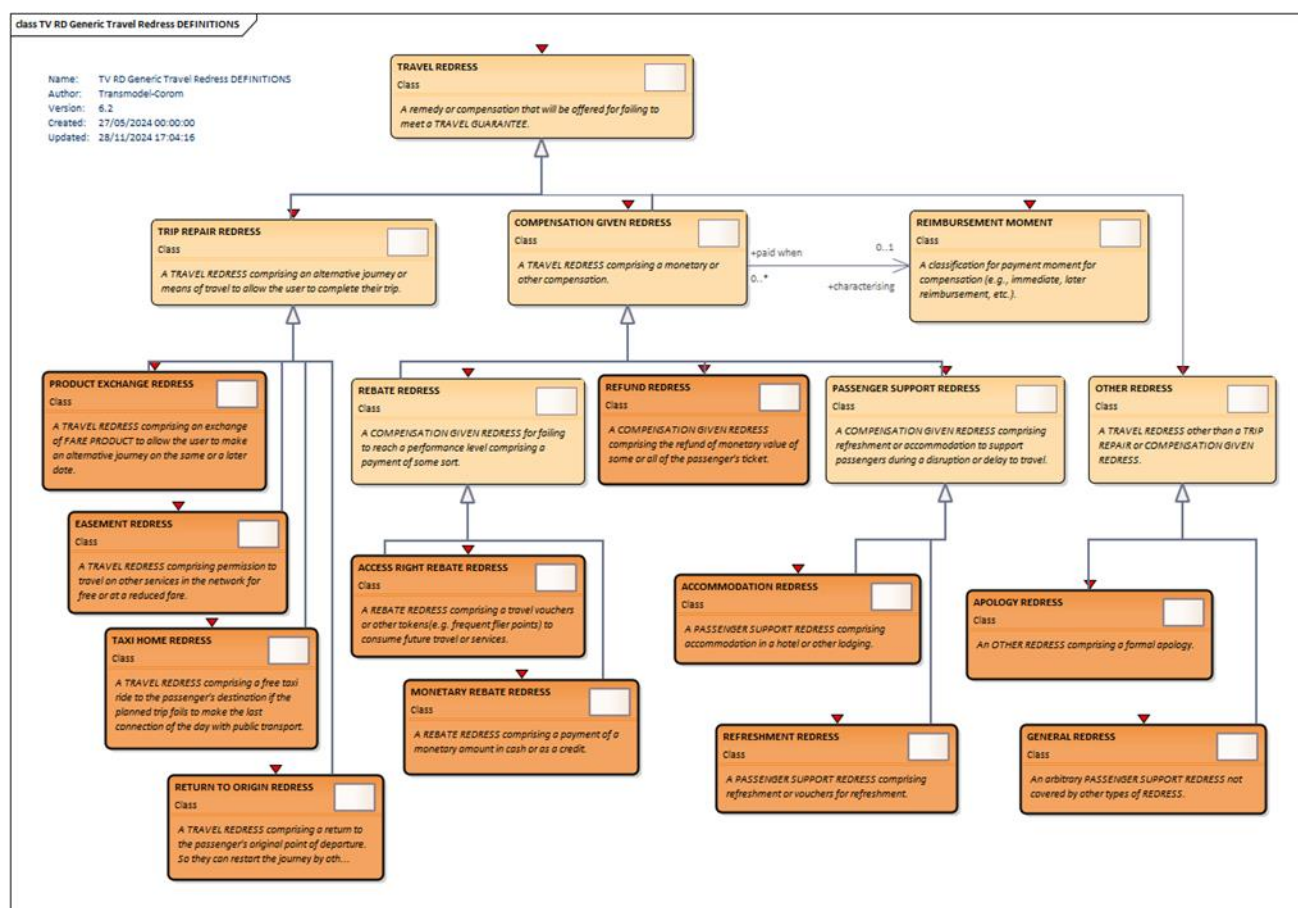


Figure 42 – Definitions of generic travel redress

8.1.5 Associating Travel Guarantees & Redresses with offers

TRAVEL GUARANTEES and TRAVEL REDRESSES are modelled as a separate concern so they can be treated as another type of parameter that may be associated generically at different levels and apply to all or just part of a trip.

A new TRAVEL VALIDITY PARAMETER element could specify which TRAVEL GUARANTEES and TRAVEL REDRESSES apply – see diagram below.

A generic association for example, with a FARE STRUCTURE ELEMENT, FARE PRODUCT, SALES OFFER PACKAGE, SALES OFFER PACKAGE ELEMENT etc, to indicate that the TRAVEL GUARANTEE will apply to the relevant parts of an offer made for the product.

To indicate that the TRAVEL GUARANTEES apply to all or part of a customer's purchase, the TRAVEL VALIDITY PARAMETER can be associated specifically with a CUSTOMER PURCHASE PACKAGE, CUSTOMER PURCHASE PACKAGE ELEMENTs, TRAVEL PACKAGEs, etc., and the TRIP SPECIFICATION can indicate which TRAVEL GUARANTEE applies to each TRIP LEG.

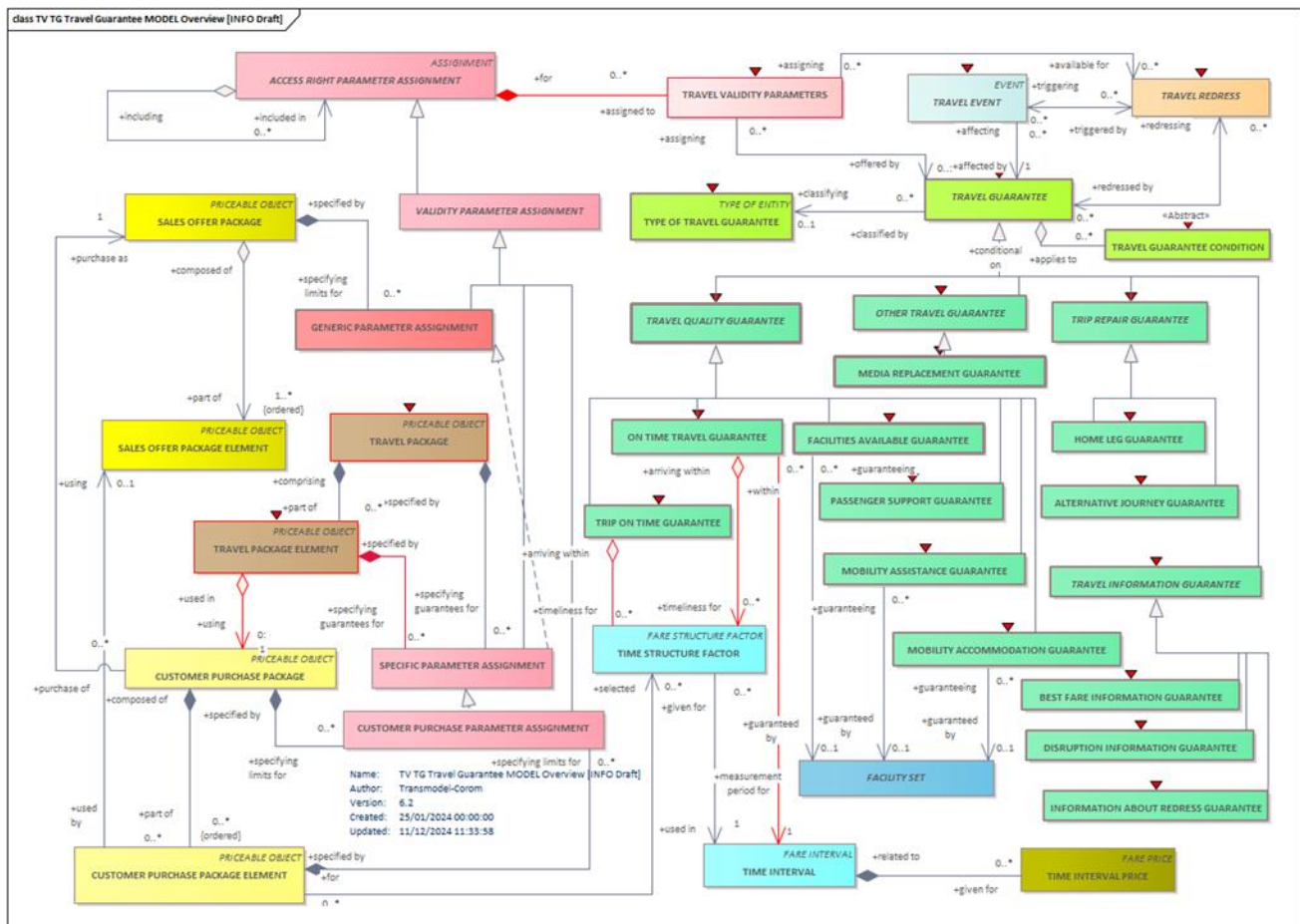


Figure 43 – UML diagram of travel guarantees

8.1.6 Travel Events

A TRAVEL REDRESS may be offered when a TRAVEL GUARANTEE is broken, i.e., a specific disruption event affecting a specific TRIP occurs. In some cases, the triggering is statutory – for example, compensation for long delays to train journeys. In other cases, it is discretionary. In either case it is useful to be able to model the different cases when different events occur and relate them to SITUATIONS and recorded events on the network so that the relevant evidence can be made available to the travel redress process, thus ultimately enabling an automated procedure.

Transmodel already has a generic EVENT model which can be extended with TRAVEL EVENTS to cover the specific types of disruptions to travel that may require redress, for example, SERVICE CANCELLATION EVENT, FACILITIES NOT PROVIDED EVENT, etc. In Transmodel, EVENTS happen in the real world not the system, but relevant data about the event can be recorded in the system as different subtypes of timestamped LOG ENTRIES.

Different types of disruption may trigger different TRAVEL REDRESSES (e.g., a cancelled train vs. a missed connection vs. a missing onboard meal) The different types of disruption can be represented as different TRAVEL EVENTS that will trigger specific redresses.

An example hierarchy of TRAVEL EVENTS with some provisional definitions is shown below. This should be further reviewed and modified as part of a formal extension to Transmodel.

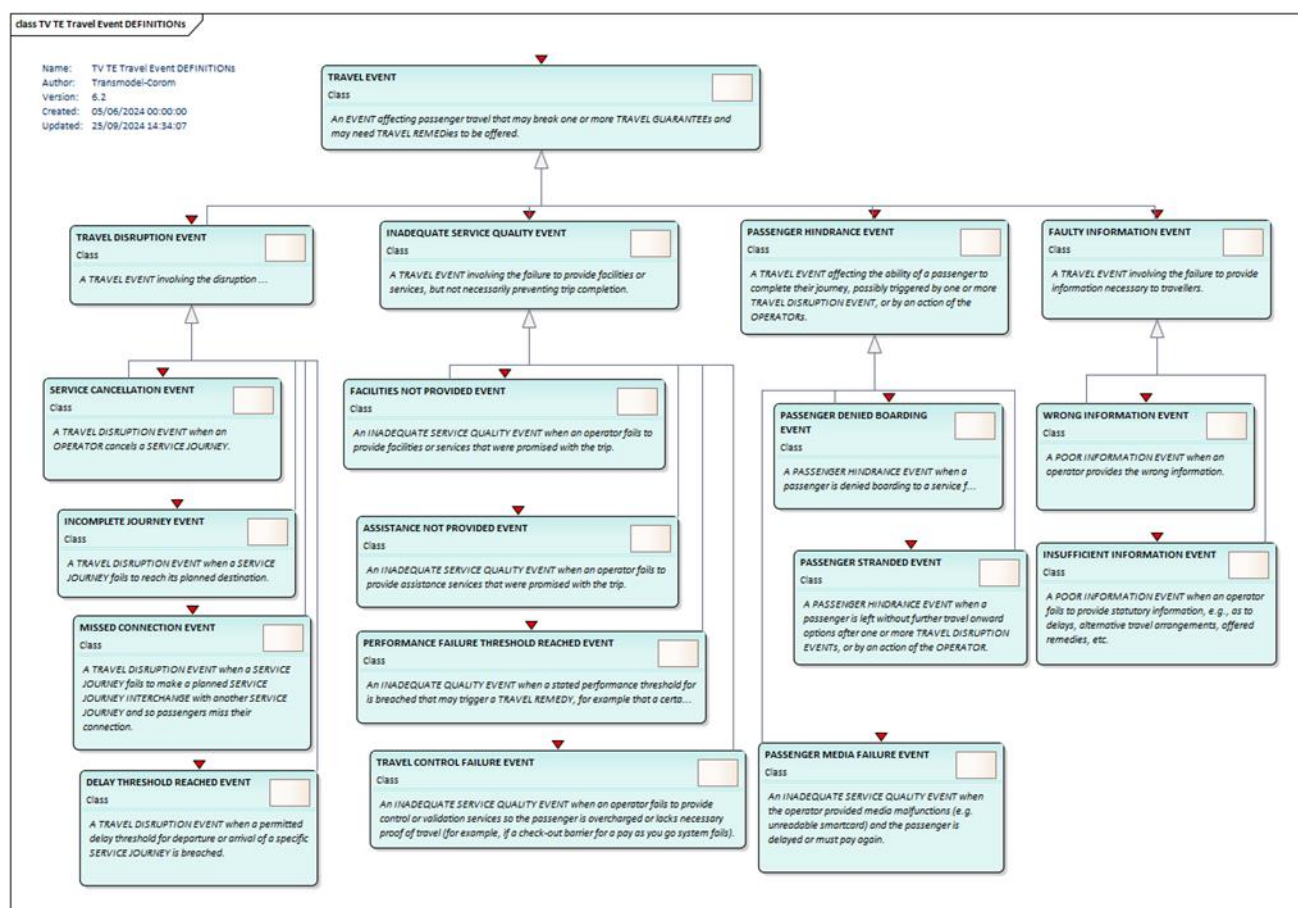


Figure 44 – Definition of travel event

8.1.7 Travel Events & triggering redresses

The TRAVEL EVENT may itself relate to a CONTROL ACTION (i.e., an operational decision by the operator to delay, cancel or otherwise modify a service) and / or to a SITUATION (for example, a crash, bad weather, security incident, planned engineering works, etc. affecting the network) that caused the TRAVEL EVENT. This can usefully be modelled in particular so that statutory TRAVEL REDRESSES can be offered automatically and related to the exact circumstances under which compensation became due.

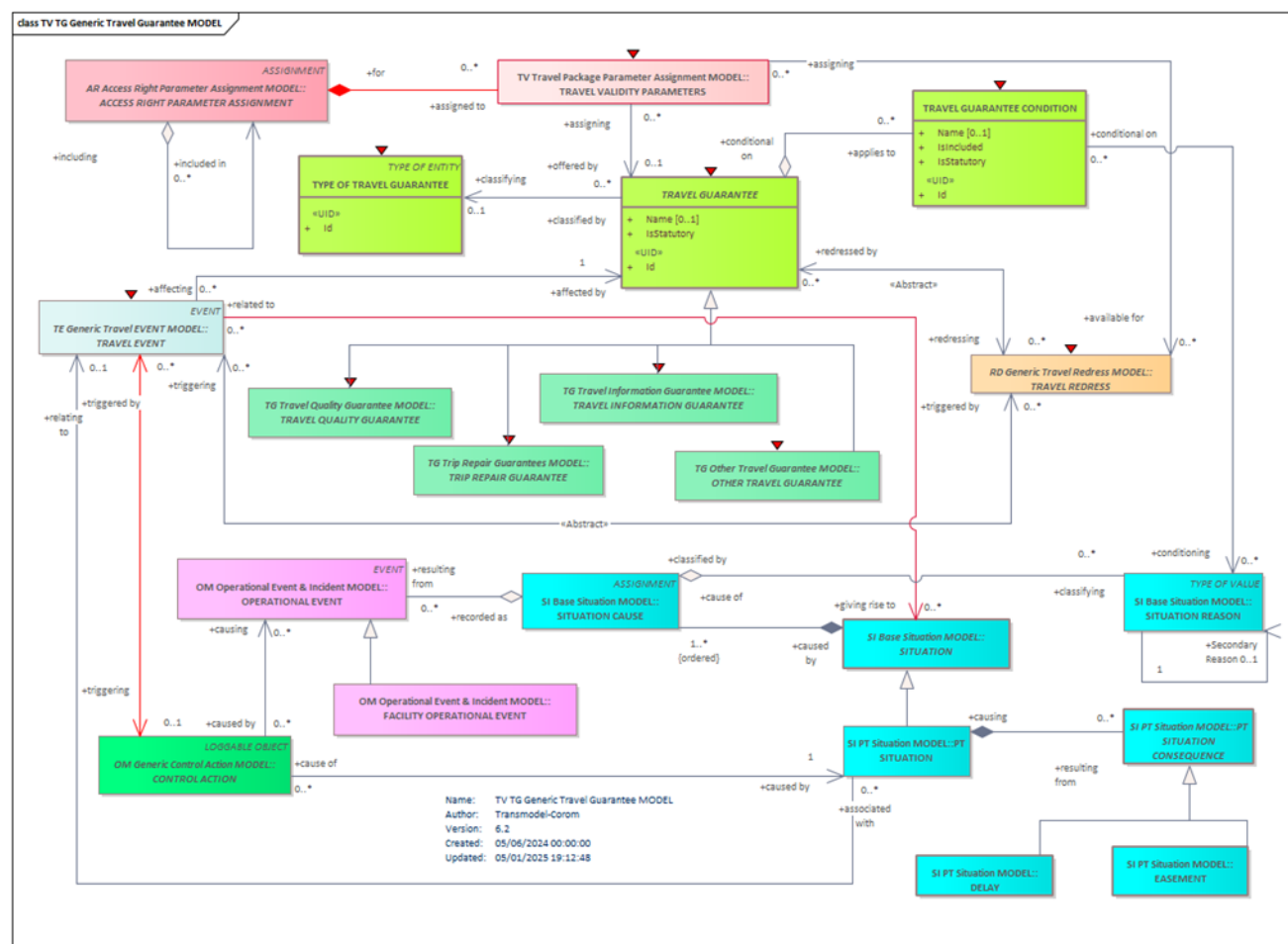


Figure 45 – UML diagram of Generic travel guarantee

8.1.8 Travel Events & Control Actions

Transmodel represents the decisions made to control a Public Transport system as CONTROL ACTIONS. Specific CONTROL ACTIONS may give rise to specific types of TRAVEL EVENT, as summarised below.

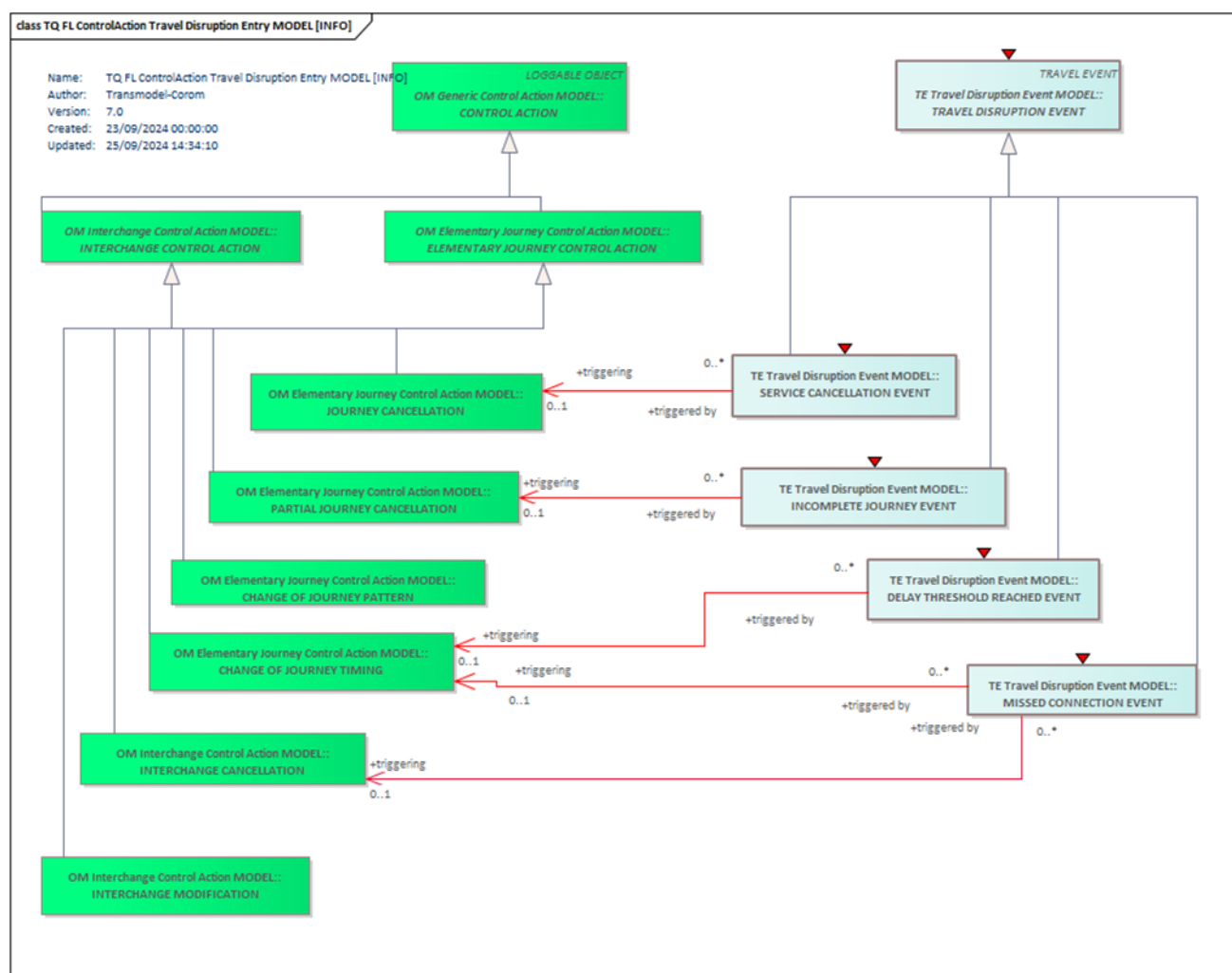


Figure 46 – UML diagram of ControlAction travel disruption entry

8.1.9 Travel Redress Events

TRAVEL EVENTS are generic – that is they describe disruption to the operation of the system, not how the disruption might affect an individual passenger. By comparing the data from a TRAVEL EVENT with each individual passenger's ticket and its access rights (as say modelled by a TRAVEL PACKAGE and/or CUSTOMER PURCHASE PACKAGE), and its associated TRAVEL GUARANTEES and TRAVEL REDRESSES, the system can determine which travel redress to the individual is due, represented by a CUSTOMER ELIGIBLE FOR REDRESS EVENT. The same TRAVEL EVENT may have different consequences for two different passengers travelling the same route. Thus, a delayed train may lead to monetary compensation as a travel redress for a passenger that has a guaranteed arrival time to the next station that is his or her final destination, while a passenger holding a through ticket and who, as a consequence of the delay will miss his or her connection, would need support finding an alternate connection and exchanging tickets.

Further events mark the offer by the operator and acceptance by the customer of a travel redress.

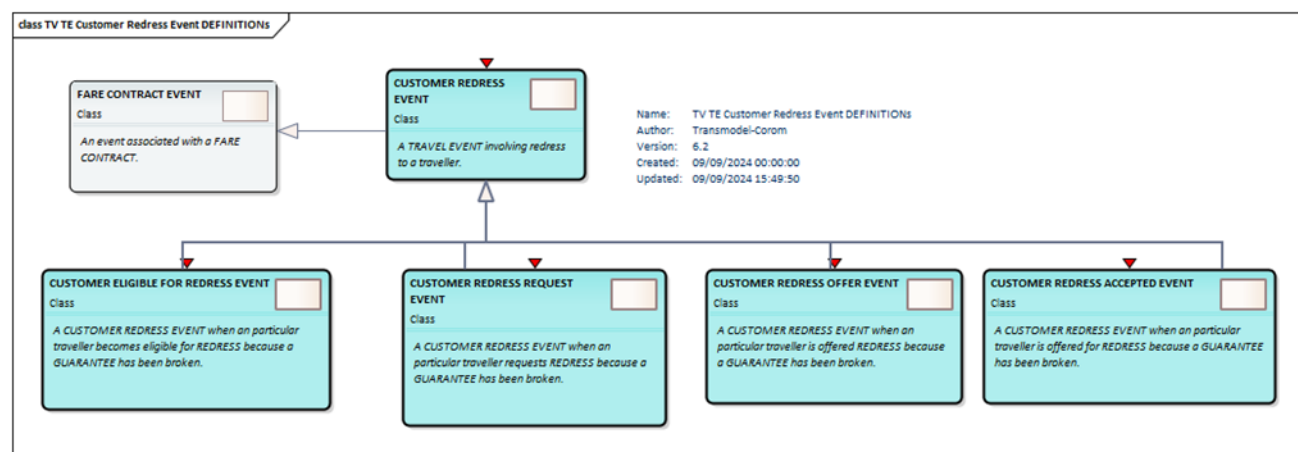


Figure 47 – Definition of customer redress event

8.1.10 Recording Travel Redresses for a Customer

The interactions with passenger arising from the failure to meet a TRAVEL GUARANTEE on a trip, and the offering of a TRAVEL REDRESS can be recorded as transactions applying to the contract for services that exists between the passenger who has bought a ticket and the operator. This FARE CONTRACT may be explicit – as in the case of Account Based Ticketing, or implicit, as in the case of an anonymous walk-up purchase.

In Transmodel, EVENTS happen in the real world, not in computer systems, but relevant data about the event can be recorded in the system as different types of timestamped LOG ENTRIES. CUSTOMER REDRESS LOG ENTRIES record that an individual customer has become eligible for redress, offered it and accepted it. The TRAVEL REDRESS may trigger further SALES TRANSACTIONS to record the payment of compensation, exchange of tickets, etc.

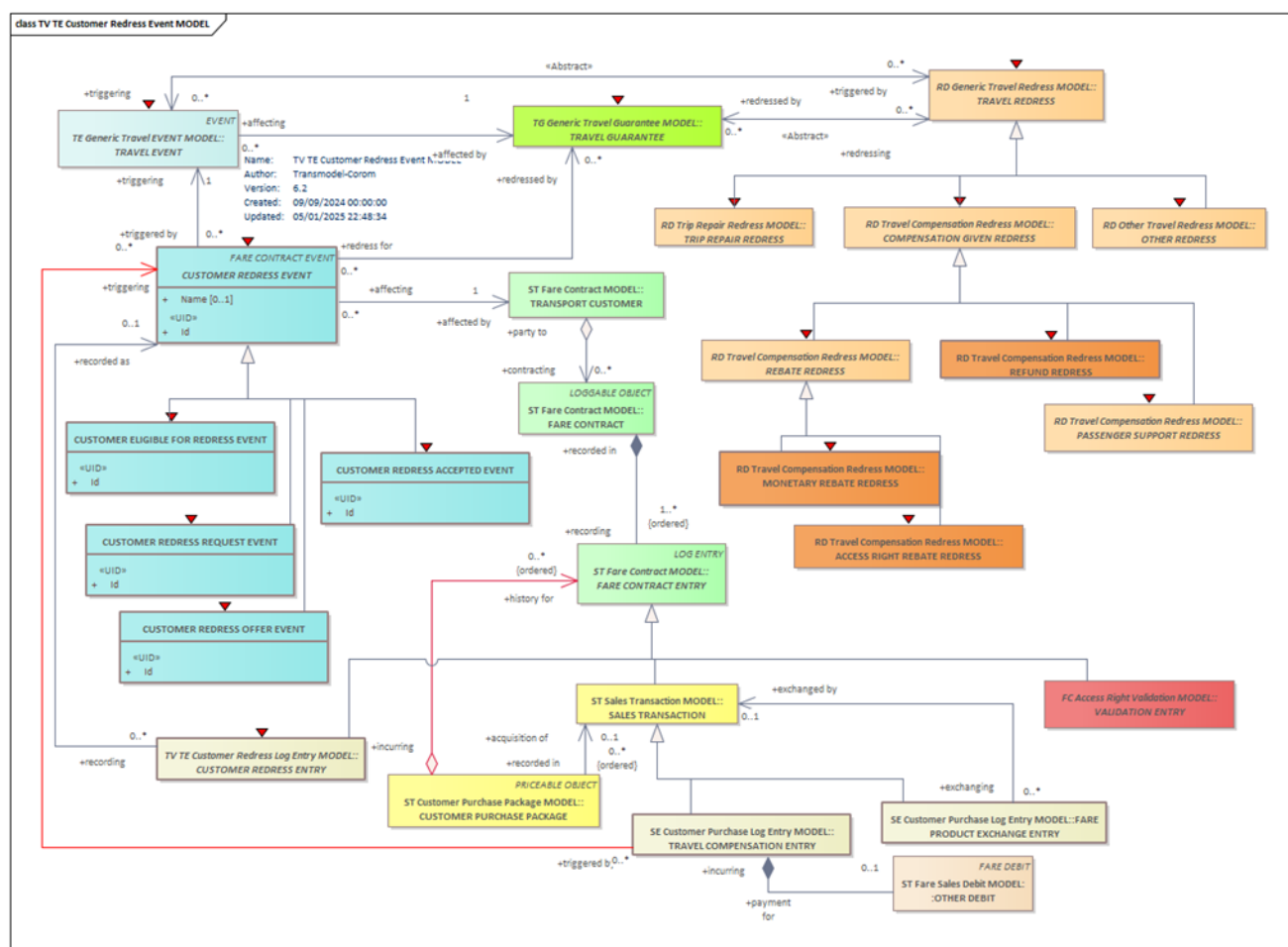


Figure 48 – UML diagram of Customer redress event

8.1.11 Offer Rules

The current Transmodel SALES OFFER PACKAGE is a generic concept, that is to say, it represents the set of all possible options that may be chosen for a given product and ticket media combination, **not** the specific set of choices that represent a purchase of an individual trip. Thus, a SALES OFFER PACKAGE might include multiple alternative options (such as all the origin and destination pairs on the network for a simple trip, or all the different fare classes from which a selection can be made).

In a complex distribution chain, retailers may create new SALES OFFER PACKAGES by combining the products of other suppliers and may choose to place travel guarantees across these or not.

There may however be limitations as to which products they may be combined and as to what travel guarantees they may offer. Transmodel cannot currently specify these rules, but it would be straight forward to model this.

A new concept of an OFFER RULE, specifying constraints on combining any two FARE PRODUCTS and or SALES OFFER PACKAGES should be added to capture this concept. The rule may variously be specific to certain Organisation or apply to all of them.

The open publication of Offer Rules is of particular importance for ensuring transparency and a fair market as without access to them it would be impossible for an external auditing body to determine in some cases of composite sales offers whether distributors are merely acting fairly according to the contracted rules or deliberately biasing certain offers or prices in their favour.

A provisional model for OFFER RULES is shown. This will need revision.

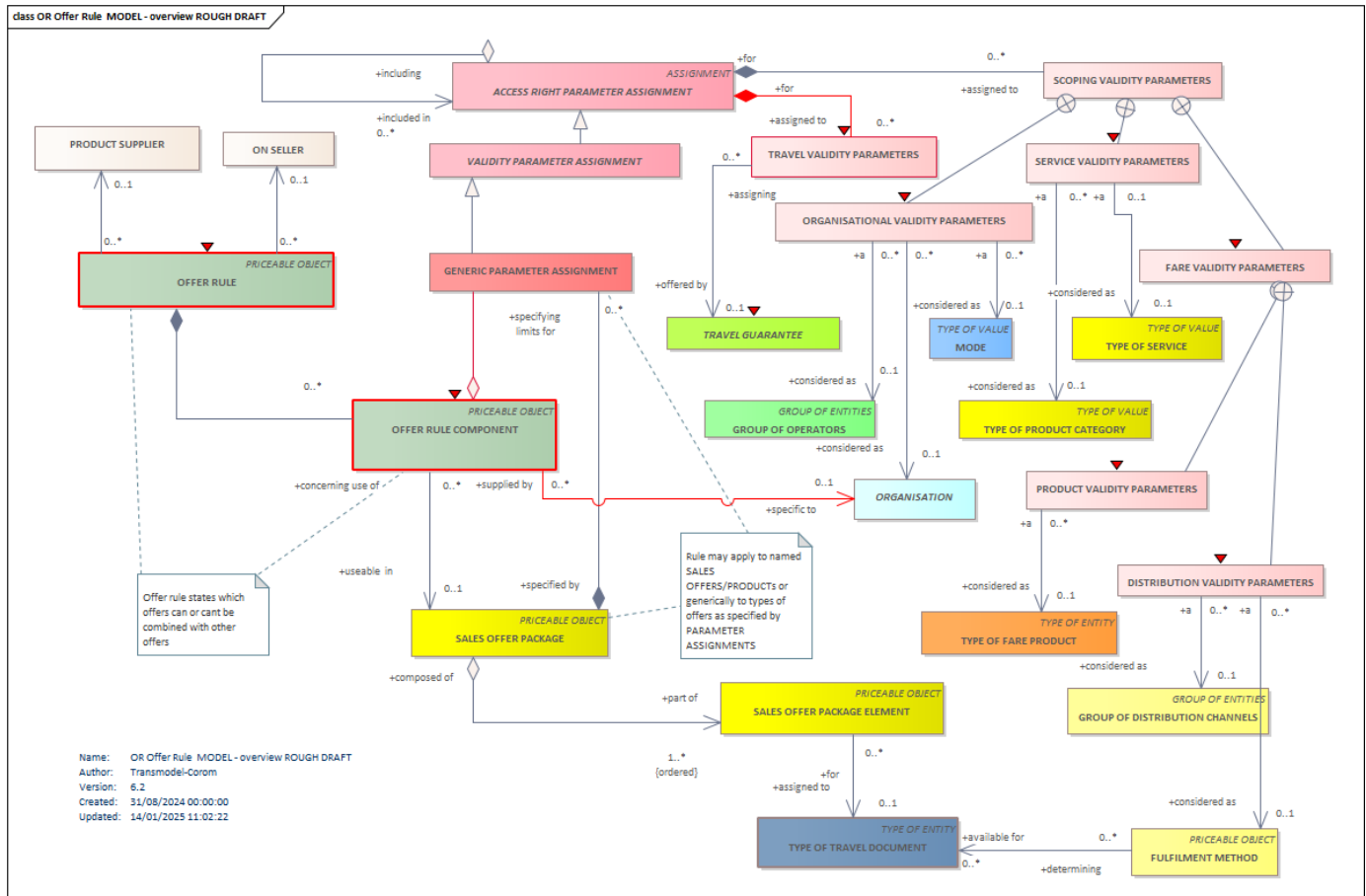


Figure 49 – UML diagram of offer rule

8.1.12 Multimodal Travel Purchase

The current Transmodel can represent the customer information side of a multimodal travel purchase, that is details of what the passenger was offered and has purchased for each leg of the trip.

OFFERED TRIP SPECIFICATION: The detailed specification of the itinerary and other details of travel being offered.

PASSENGER FARE OFFER: The specific choices of the options of a FARE PRODUCT and SALES OFFER PACKAGE to make a specific offer for a trip, with prices.

CUSTOMER PURCHASE PACKAGE the parameters for the specific trip and fare and customer, stored as an electronic representation of the purchase for passenger information and control and validation purposes. This may be made up of several CUSTOMER PURCHASE PACKAGE ELEMENTS, each with specific parameters. Can include a SPOT ALLOCATION to indicate the seat or other reserved space(s).

SALES TRANSACTION recording the creation of the CUSTOMER PURCHASE PACKAGE and its purchase transaction against a FARE CONTRACT and CUSTOMER ACCOUNT.

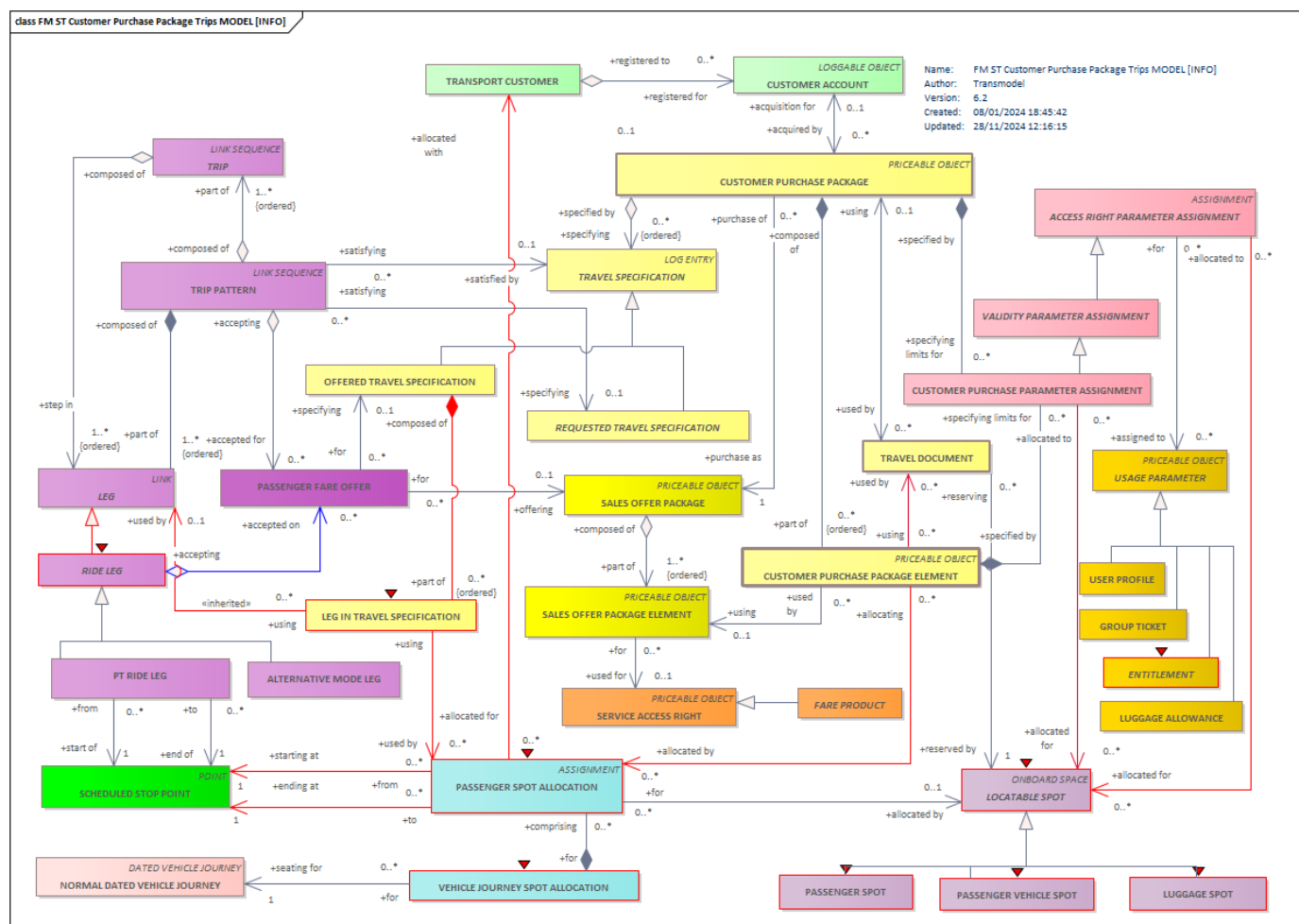


Figure 50 – UML diagram of customer purchase package relaed to trip

8.1.13 Reservation

The reservation of a seat for a specific passenger making a specific trip is represented in Transmodel by a PASSENGER SPOT ALLOCATION; it references a PASSENGER SPOT in a DECK PLAN as well as a customer and their boarding and alighting stops. The DECK PLAN describes the layout of the vehicle and the location, number and facilities of the seats and other bookable spaces (e.g., for vehicles or luggage). For each VEHICLE JOURNEY, there will be a VEHICLE JOURNEY SPOT ALLOCATION holding the set of all PASSENGER SPOT ALLOCATIONS for a given journey running on a specific date (i.e., DATED VEHICLE JOURNEY) with a specified DECK PLAN in effect.

The spot allocation i.e., reservation data above is primarily concerned with the passenger's view. It is likely there is further data relating to the management of reservations for given trip (for example, the time windows for making or changing reservations, the locking periods allowed to support two phase commit etc.). That should also be modelled to support a full reservation model.

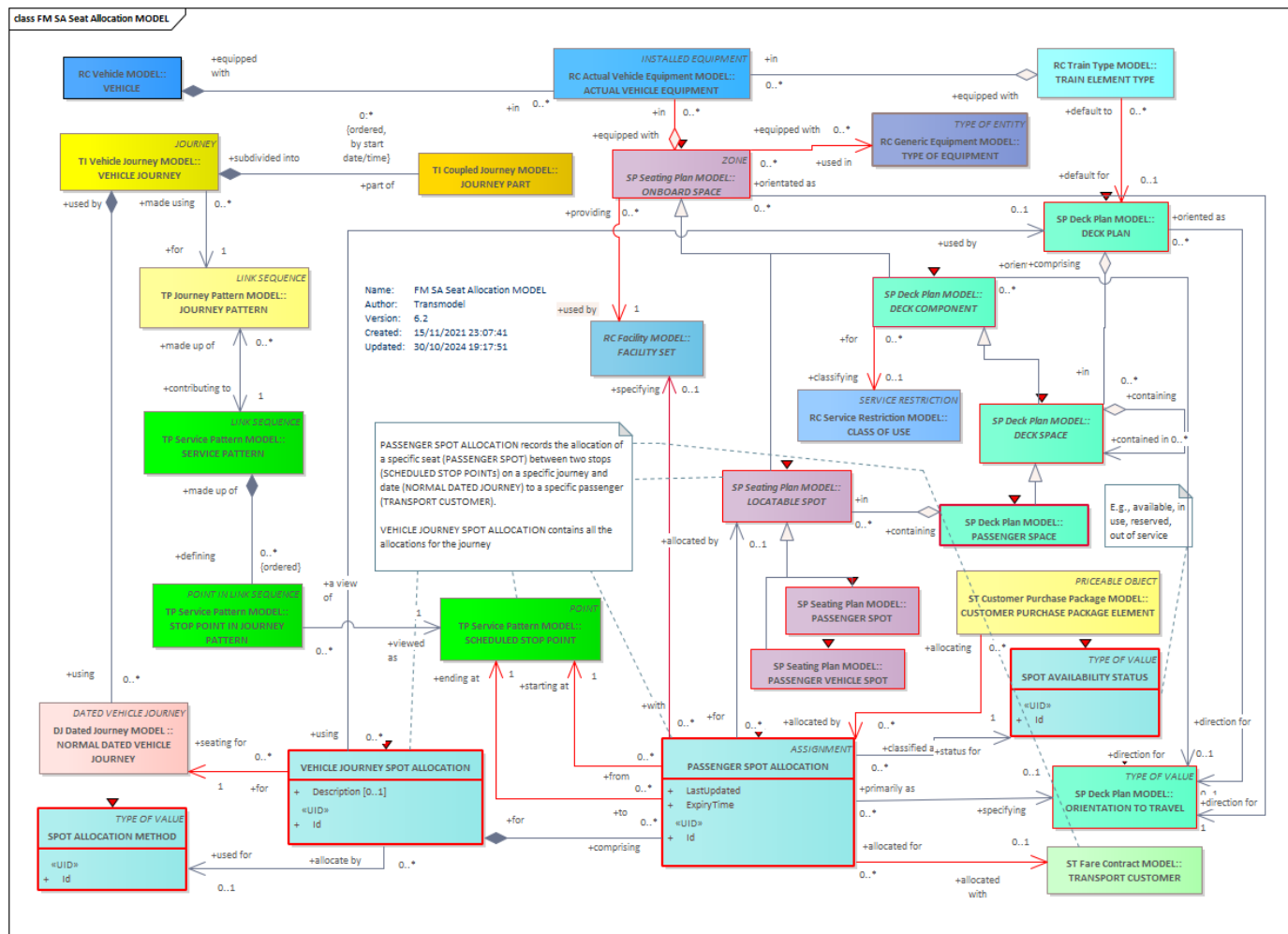


Figure 51 – UML diagram of seat allocation

8.1.14 Identification Card

Transmodel does not explicitly model identification card details as held by a customer, such as citizen ids, passports or driving licences. They could currently be represented as a THIRD PARTY PRODUCT, but it would be clearer to add a new explicit 'IDENTIFICATION DOCUMENT' entity to Transmodel.

8.1.15 Passenger Complaints

Transmodel does not currently have a passenger complaint model.

This could be modelled following the current OSDM model, with individual complaint as entities in their own right that can be raised and progressed through a life cycle.

8.2 Additional Transmodel multimodal travel purchase API

Support for a multimodal travel purchase API will require the addition of several new message pairs that can be used after a Fare Offer has been obtained.

The following are some tentative identifications of new message pairs that will be needed. This is merely a preliminary indication and should be further reviewed and modified as part of a formal extension to Transmodel.

8.2.1 Account Management requests

Messages are needed to create, access and manage customer accounts to support account ticketing. These may follow generic industry practice.

CREATE CUSTOMER ACCOUNT REQUEST/DELIVERY

GET CUSTOMER ACCOUNT REQUEST/DELIVERY (by id or name.. select which details are wanted)

UPDATE CUSTOMER ACCOUNT REQUEST/DELIVERY

CLOSE CUSTOMER ACCOUNT REQUEST/DELIVERY

8.2.2 Purchase & reservation requests

Messages are needed to check availability and make reservations and purchases. It is envisaged these will be REST based operations on a TRAVEL PACKAGE concept.

TRAVEL PACKAGE SPOT AVAILABILITY REQUEST/DELIVERY

TRAVEL PACKAGE PURCHASE REQUEST/DELIVERY i.e., CREATE TRAVEL PACKAGE ?

TRAVEL PACKAGE SPOT RESERVATION REQUEST/DELIVERY

TRAVEL PACKAGE FULFILMENT REQUEST/DELIVERY

8.2.3 After Sales requests

Messages are needed to check handle after sales operations. Some of these may also be represented as operations on a TRAVEL PACKAGE concept.

TRAVEL PACKAGE EXCHANGE REQUEST/DELIVERY

ACCEPT EXCHANGE OFFER REQUEST /DELIVERY

REFUND REQUEST/DELIVERY

ACCEPT REFUND OFFER /DELIVERY

RAISE COMPLAINT REQUEST /DELIVERY

8.3 Example of multi-supplier travel packages and the travel basket concept

Two persons intend to do make trip. After they have consulted a trip planning application, they now turn to an on-line travel agency to understand their ticket options for their intended trip before making a purchase.

In the following the travellers are seen as a transport customer and the on-line travel agency is regarded as a retailer or possibly as an aggregator.

The process begins with the transport customer, based on the result from the trip planning application, handing over the preferred trip pattern option to the on-line travel agency.



Figure 52 Transport customer provides the on-line travel agency with trip details

The on-line travel agency concludes that it needs to use a combination of offers from different suppliers to get tickets that cover the complete trip. Thus, it splits the request into two and sends those two requests to potential suppliers.

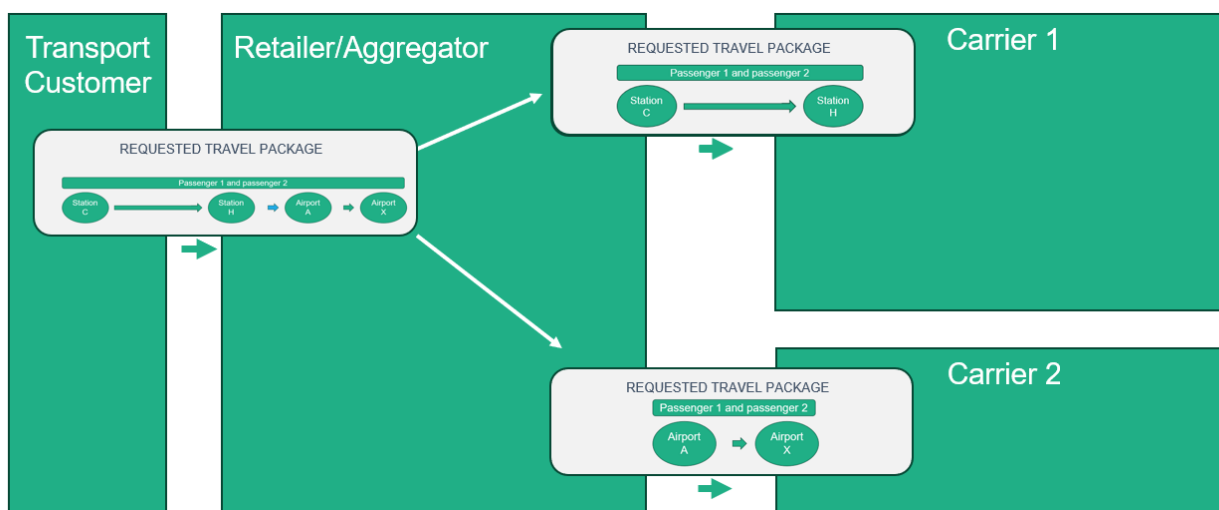


Figure 53 The on-line agency requests offers from potential suppliers

The two suppliers respond with two separate offers and in this first scenario the on-line agency simply provides them to the transport customer as is, possibly after adding a additional services and/or fees. The offer(s) are exposed in a travel basket available to the final transport customer. In this case it is important that the on-line agency clarifies to the transport customer that the two tickets do not constitute a through ticket.

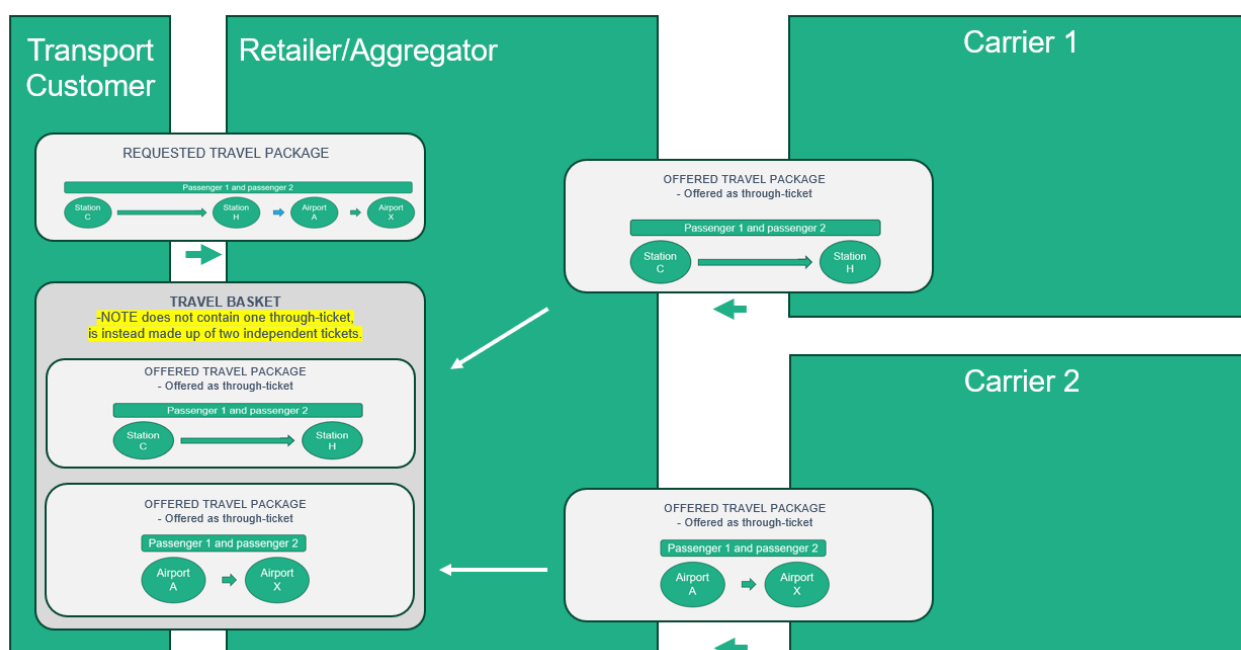


Figure 54 Providing two independent tickets.

Alternatively, the online-agency can act as an aggregator and take the risk of combining the two tickets into one through-ticket. Possibly such a combined through-ticket will be offered to the transport customer at a higher price than the sum of the two separate underlying tickets.

In this case the on-line travel agency thus constructs its own OFFERED TRAVEL PACKAGE where the two OFFERED TRAVEL PACKAGEs are seen as TRAVEL PACKAGE ELEMENTs.

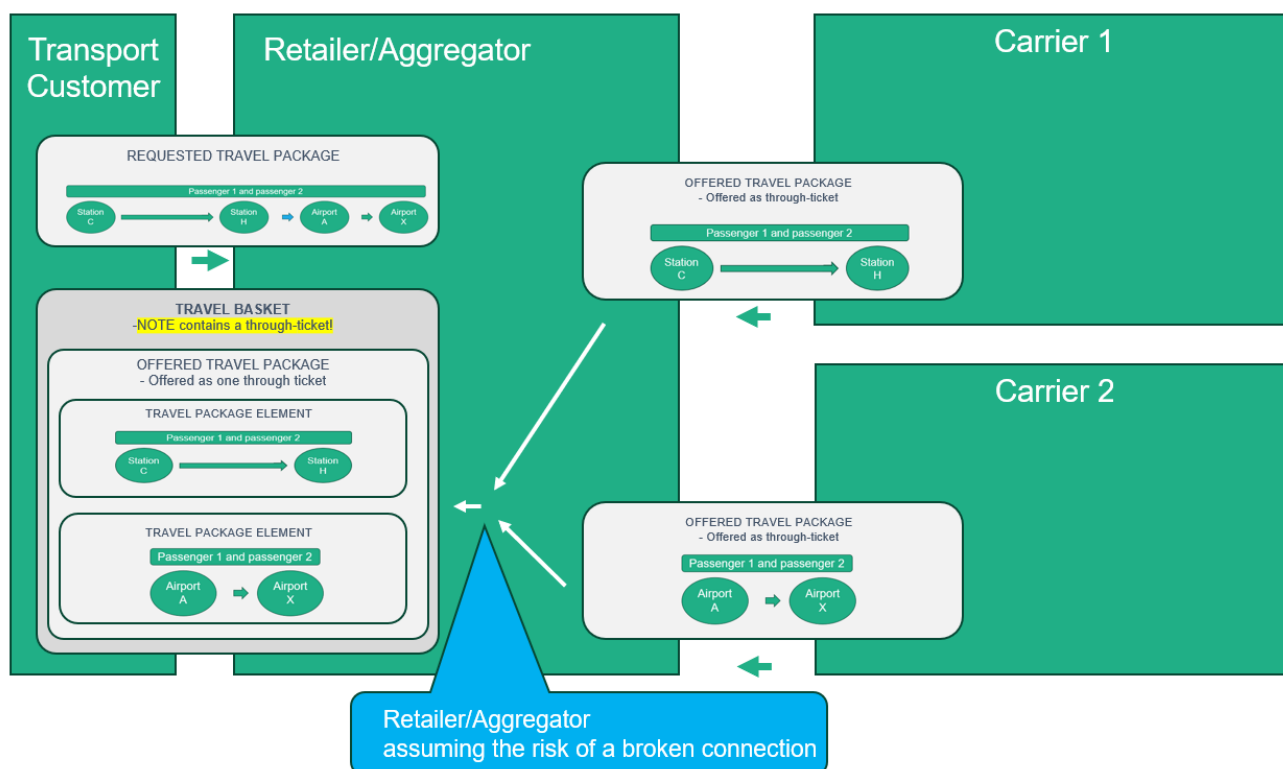


Figure 55 Providing a through-ticket

The above picture is slightly simplified. The travel agency can offer both above variations in parallel. Also, each of the travel packages may be offered in several variations, e.g., having different travel classes, different flexibility, different ancillary products, etc.

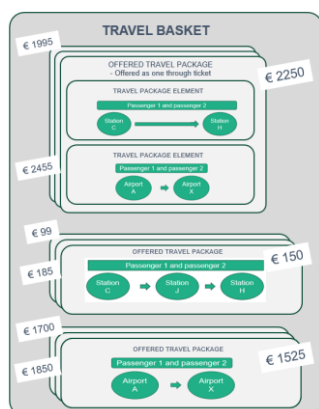


Figure 56 Different prices depending on selected options.

The transport customer can browse the travel basket and select the preferred options before finalizing the purchase.

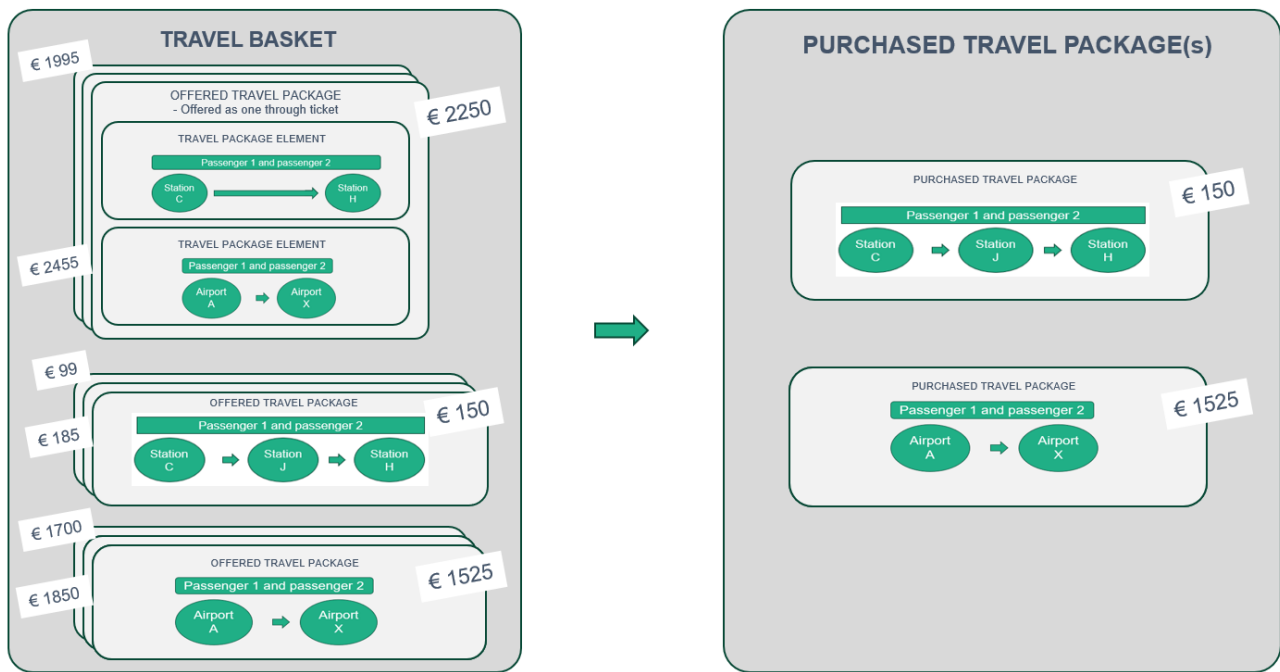


Figure 57 selecting options and finalizing the purchase

Annex A (informative)

BoB

A.1 Overview

- Description: The BoB standard is an open and voluntary standard that defines how public transport tickets can be defined, distributed, combined and validated across a network of independent public transport providers to form a seamless web. The purpose of the BoB standard is to lower the thresholds for travellers to purchase tickets, thereby making public transport more accessible and attractive, while promoting interoperability and innovation within ticketing systems. The BoB standard primarily focuses on regional public transport and provides functionality for reservations/bookings of on-demand travels as an extension.
- Modes it applies to, if relevant: The standard is designed to support regional transport across organisational and geographical borders, with products valid both for single journeys and several journeys, within a defined geographical area and a specific time period. Characteristically, this type of travel is typically not pre-booked. Typical of this mode is Public Transport Authority Operated transport.
- Where it is used: All 21 PTAs (100%) and some regional train systems in Sweden have implemented the BoB standard in their ticketing solutions. The standard is also used by Strætó in Iceland.

A.2 Related queries

Category	Endpoints
Requesting offers	POST /product (a product in BoB is the equivalent of an offer)
Modify offers (add travellers)	N/A (new offer should be requested instead)
Modify offers (add products)	N/A (new offer should be requested instead)
Modify offers (assign seats/assets)	N/A (new offer should be requested instead)
Purchase an offer (2-phase)	POST /manifest POST /ticket
Pre-purchase (pre-book) an offer	POST /manifest
Purchase an offer directly	POST /manifest POST /ticket
Obtain travel documents	POST /ticket GET /ticket POST /ticketbundle GET /ticketbundle
After sales (cancel purchased package)	PUT /ticket/{ticketId}/refundStatus
After sales (reimburse)	PUT /ticket/{ticketId}/refundStatus

A.3 Unique functions

BoB defines four different roles or functions within the infrastructure; the ticket vendor, the price & product service, the ticketing service and the validation service.

The infrastructure is centred around the ticket vendors, who need to maintain contracts (though bilateral agreements) with all public transport providers whose products are going to be offered through the vendor. The ticket vendor also maintains the customer relationship and provides the tools required for planning the trip and paying for tickets. Hence, being abstracted behind the vendor, travel accounts, trip planning and payments are outside of the scope of the infrastructure.

Unique functions for the BoB infrastructure, include:

- Universal Ticket Bearers
- A common ticket bearer called the MTB, that defines a data structure where each public transport provider can use its specific “slot” to encode the ticket information in a compact and standardised way. The specifications also include security functions for authenticating ticket information, copy protection and optical encoding.
- An id-based token bearer that can be used with NFC-based smart cards as well as mobile phones, incorporating strong public-key cryptography for authentication.
- Product definitions.
- The Product API is used by the ticket vendor for retrieving information on available products, including pricing, based on certain criteria such as route retrieved from trip planner.
- The Product API is also used to issue so called Manifests, which are fixed offers valid for a certain amount of time. Based on these Manifests, the Ticketing service will issue tickets.
- Ticket issuance and distribution.
- The Ticket API is used to issue tickets based on Manifests.
- It is also used to retrieve travel documents and activate dormant tickets, alongside other ticket life-cycle events (such as revocations, refunds, etc.).
- Ticket validation and inspection.
- A central component of the standard is the domain-specific language to express ticket validity, called TICKLE. It is used to express both geographical (spatial) and time-based (temporal) validity and can also set other related pre-conditions such as a specific mode of transport or type of service. The TICKLE language is universal and can be used to express any type of travel entitlement. It can be used both internally and to facilitate interoperability of the same ticket (issued by one party) over multiple public transport providers.
- The Validation API supports submitting ticket validation and inspection events and may also be used for on-line validation.
- For id-based travelling, the concept of white-lists is used to distribute ticket information beforehand to the validation devices, to work independent of communication networks (in offline scenarios) as well as improving validation speed. This is also supported by the Validation API.

- On-demand responsive transport and pre-booked tickets.
- For on-demand responsive transport and other types of tickets requiring pre-booking, the Booking API may be used.
- For tickets requiring booking, an additional loop is added for the issuance of tickets. Preliminary bookings are made before the purchase, which is then finalised when the purchase of the ticket has been completed. It adds the possibility for an interactive user-confirmation to accept the specific conditions that may apply for a booking (such as final departure time, seating or other types of conditions that may be dynamic).
- Traveller support.
- The Traveller API is used to manage traveller-related functions, such as renewing tickets (i.e., purchasing the same product again), looking up products tied to id-based tokens, sending notifications to the traveller and providing traveller support during the journey.

In addition to the functions and services outlined above, it also incorporates a few additional services to support the infrastructure, for example, the functions that are required for secure communication between the participants.

Annex B (informative)

OSDM

B.1 Overview

- OSDM (Open Sales and Distribution Model) defines an API to enable and simplify the sale of transport products. The API allows Retailers to access transport products provided by distributors. It also allows distributors to access transport product bricks provided by carriers or fare providers to build combined transport products. The aim of OSDM is to provide a simple API to access required information online, however OSDM also provides an offline data exchange of fares.

The API covers the full sales process including

- timetable
- offer search,
For the offer search, OSDM allows to find offers providing in the request
 - Origin & Destination: In this case the provider will use its own journey planner to find the different itineraries for this O&D and will provides the different offers for those different trips.
 - Trip details: The distributor is using its own journey planned to find all the routes to go from the Origin to the destination and will send in the OSDM offer request detailed information on the trip for which offers are required. The Trip data model is OPJ compliant to secure OSDM interoperability with a Transmodel journey planner
- prebooking and booking,
- Travel tickets delivering
- Refund including a full refund capability in case of re accommodation (equivalent to the COROM Redress)
- exchange including special processes to handle delays, change of material in case of re accommodation (equivalent to the COROM Redress)
- compensation.

OSDM covers scheduled multimodal transport services (trains, busses, trams, ...). On Demand services are currently covered with some functional limitations.

The OSDM API is rich and can be overwhelming at the start. The reason for its feature of richness (leading to its complexity) is the fact that it aims to cover all possible public transportation products and distribution processes within Europe and beyond.

The products can be split roughly into three categories:

- *admissions*: the right to ride on a train, resulting in a ticket.
- *reservations*: seat or couchette or bed in case of night trains
- *Ancillary*: Ancillary services such WIFI, 3-Menu plate

Between these categories, relationships are modelled to express whether it is optional or mandatory to have a certain seat reservation or ancillary on a given vehicle. In an open system (e.g. Switzerland) having a reservation is optional. In contrast on a closed system (e.g. France) a seat reservation is a mandatory.

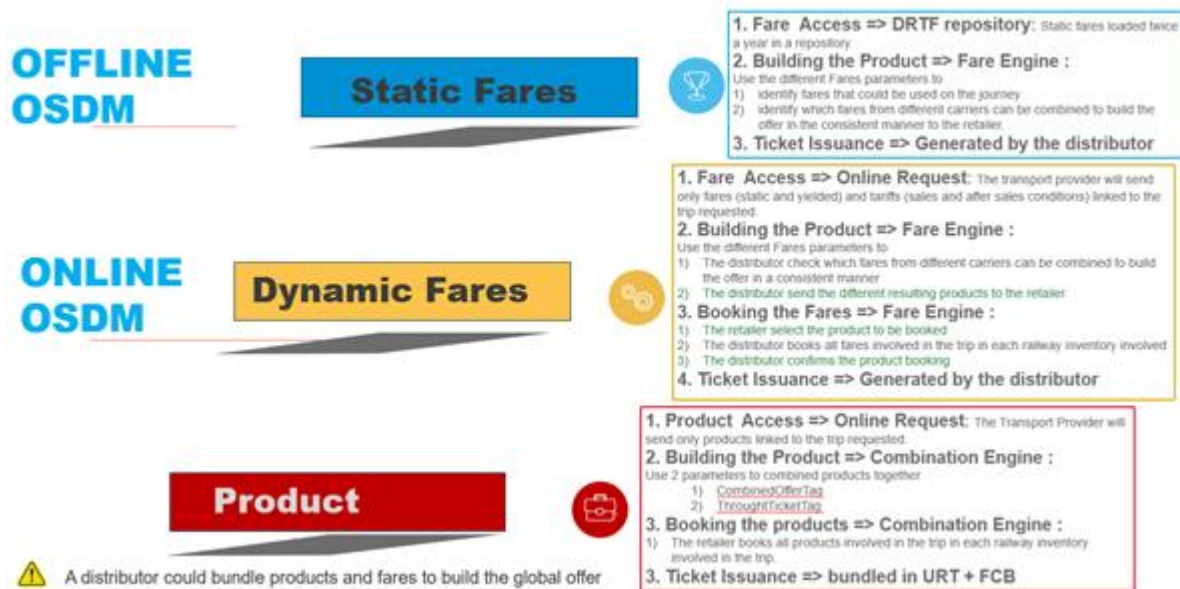
More detailed information can be found in OSDM specifications (URL link: <https://osdm.io/spec/>)

B.2 Related queries

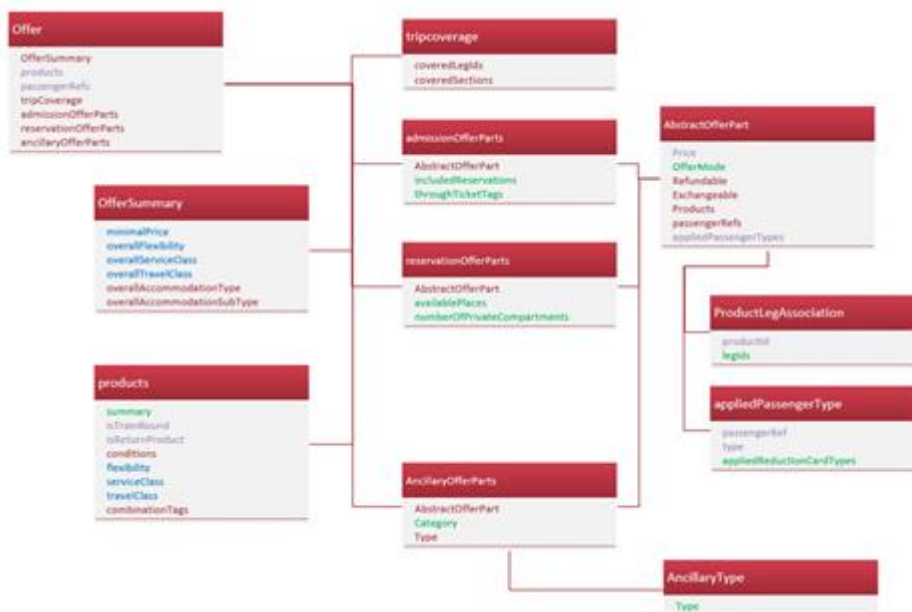
Category	Endpoints
Requesting offers	POST/offers
Modify offers (add travellers)	POST//bookings/{bookingId}/booked-offers/{bookedOfferId}/offer-parts
Modify offers (add products)	POST//bookings/{bookingId}/booked-offers/{bookedOfferId}/offer-parts
Modify offers (assign seats/assets)	POST/places
Purchase an offer (2-phase)	POST/ /bookings/{bookingId}/fulfillments
Pre-purchase (pre-book) an offer	POST/ /bookings
Purchase an offer directly	N/A as always 3 steps process at API level.
Obtain travel documents	GET/ /fulfillments/{fulfillmentId}
After sales (Refund offers request)	POST/ /bookings/{bookingId}/refund-offers
After sales (Refund confirmation)	PATCH/ /bookings/{bookingId}/refund-offers/{refundOfferId}
After Sales (Exchange offer request)	POST/ /bookings/{bookingId}/exchange-offers
After Sales (Exchange Confirmation)	PATCH/ /bookings/{bookingId}/exchange-operations/{exchangeOperationId}

B.3 Unique functions

OSDM supports in a consistent and standardized manner the different levels of interactions between the actors from the static fares filed to the most advanced interactive sales at product level. The following diagram provides a summary of OSDM capabilities



An abstract of the offer data structure is helping to understand the way OSDM is linking the products, trips, passengers in an offers that can be composed by an admission allowing to travel on a defined trip, a reservation allowing you to use a dedicated seat, berth, couchette in a train and ancillaries that could be used.



More details can be found on the following link <https://osdm.io/spec/models/>

B.4 Notes on terminology

The term *place* is used for two different concepts in OSDM (firstly, physical locations between which travel may be desired; and secondly available seating locations on vehicles). A suggestion for a new API

would be to keep the current usage (that matches well with Transmodel) for the former, i.e. travel location sense:

- startPlace
- endPlace

, But to replace the OSDM usage for seating locations with 'spot' as shown in the examples below:

- placeSelections -> spotSelections
- referencePlace -> referenceSpot
- placeNumber -> spotNumber
- placeProperties -> spotProperties

and additionally, to align with above new terms and clarify the scope of the term *reservation* also replace as below:

- reservationId -> spotReservationId
- reservationOfferId -> spotReservationOfferId

OSDM- current terminology	OSDM - possible CoRoM inspired renaming?
/bookings/{bookingId}/booked-offers/{bookedOfferId}	/travelPackages/{travelPackageId}/travelPackageElements/{travelPackageElementId}
/bookings/{bookingId}/booked-offers/{bookedOfferId}/reservations/{reservationId}	/travelPackages/{travelPackageId}/travelPackageElements/travelPackageElementId/spotReservations/{spotReservationId}
GET /availabilities/nearby?coachNumber=5;placeNumber=3	GET /spotAvailabilities/ nearby?coachNumber=5;spotNumber=3
.. or possibly, to separate passengerSpot passengerVehicleSpot passengerLuggageSpot passengerPetSpot	GET /passenger-spot-availabilities/ nearby?coachNumber=5;passengerSpotNumber=3
/bookings/{bookingId}/booked-offers/{bookedOfferId}/additional-offers	/travelPackages/{travelPackageId}/travelPackageElements/{travelPackageElementId}/ancillaryOffers
/bookings/{bookingId}/booked-offers/{bookedOfferId}/ancillaries/{ancillaryId}	/travelPackages/{travelPackageId}/travelPackageElements/{travelPackageElementId}/ ancillaries/{ancillaryId}

/bookings/{bookingId}/booked-offers/{bookedOfferId}/admissions/{admissionId}	/travelPackages/{travelPackageId}/travelPackageElements/{travelPackageElementId}/admissions/{admissionId}
GET /bookings/{bookingId}/fulfillment-check <i>(step 1 in a two phase commit process. Provider do all checks to "guarantee" that fulfillment will work.)</i>	GET /travelPackages/{travelPackageId}/purchase-lock <i>(step 1 in a two phase commit process. Returns OK if the purchase is guaranteed)</i>
POST /bookings/{bookingId}/fulfillments <i>(in OSDM this represents both the purchase and may optionally also include a direct fulfilment)</i>	POST /travelPackages/{travelPackageId}/purchases <i>(used for direct purchase or as stage2 in a two phase commit after successful purchase-lock. Response contains fulfillmentId)</i>
GET /fulfillments/{fulfillmentId} <i>(returns the fulfilment, aka the ticket)</i>	GET /fulfillments/{fulfillmentId} <i>(returns the fulfilment, aka the ticket for specified fulfillmentId)</i>

Annex C (informative)

TOMP-API

C.1 Overview

- **Description:** the TOMP-API is the API between Transport Operator (TO) and MaaS Service Provider (MP) but is also used by resellers and brokers. It serves the digital communication about planning (GBFS-like), booking, trip execution, support and payments.

Vision: Effortless mobility for everyone

Mission: The TOMP working group aims to develop and sustain an internationally governed interoperable open standard for technical communication between Transport Operators and MaaS Providers, by means of definition, improvement, alignment and dissemination.

- **Modes it applies to, if relevant:** The TOMP-API serves all modes as described in NeTEx, GBFS, GTFS (also GTFS-on demand), APDS, CDS, OCPI, DATEX-II so it was extended with 'PARKING', 'MOPED', and 'STEP'. The TOMP-API working group strongly believes in working together and use as much as possible what is already out there, to reduce implementation costs. It can be applied on all assets (or asset types) in the mobility arena; the TOMP-API is agnostic.
- **Where it is used:** geographical coverage: global. A few of the first implementations were in Australia, some are in South Afrika, Columbia and the USA, but most of the implementations are made in Europe. There is no official registration of implementations, and therefore we have to get this indication based on given expert's advice in these countries. Most of the implementations are made for the modes bike, scooter, taxi and on-demand services. Some in the conventional public transport, and some prototyping has been done in combination with APDS in the domain of parking.

C.2 Related queries

C.2.1 Version 1.x

Category	Endpoints
Requesting offers	POST /planning/offers
Modify offers (add travellers)	PUT /bookings/{id}
Modify offers (add products)	PUT /bookings/{id}
Modify offers (assign seats/assets)	POST /bookings/{id}/events - ASSIGN_ASSET
Purchase an offer (2-phase)	POST /bookings/{id}/events - COMMIT
Pre-purchase (pre-book) an offer	POST /bookings
Purchase an offer directly	POST /bookings/one-stop
Obtain travel documents	GET /bookings/{id}
After sales (cancel purchased package)	POST /bookings/{id}/events - CANCEL

After sales (reimburse)	
-------------------------	--

C.2.2 Version 2.0 (draft)

Category	Endpoints
Requesting offers	POST /processes/search-offers/execution
Modify offers (add travellers)	POST /processes/update-traveller/execution
Modify offers (add products)	POST /processes/assign-ancillary/execution
Modify offers (assign seats/assets)	POST /processes/assign-asset/execution
Purchase an offer (2-phase)	POST /processes/purchase-package/execution AND POST /processes/confirm-purchase/execution
Pre-purchase (pre-book) an offer	POST /processes/select-offers/execution
Purchase a package directly (not based on offers)	POST /processes/purchase-package/execution POST /processes/purchase-product/execution
Obtain travel documents	GET /collections/travel-documents/items/
After sales (cancel purchased package)	POST /processes/cancel-package/execution
After sales (reimburse)	GET /collections/redress-options/items POST /processes/claim-redress-option/execution POST /processes/confirm-redress-option/execution
Purchase an offer directly, not reserving nor ending up in a 2-phase-purchase	POST /processes/purchase-offers/execution

C.3 Unique functions

The TOMP-API 1.x delivers functionality for planning ('level 1 MaaS Maturity Level' – RI.SE), and a lot of functions for executing trips (starting legs, modifying legs, reporting progress, support dialogs between TO and MP, but also between travellers and drivers, operating assets), support (request support, support information, communication between end users and TOs), and payment (journals of orders to settle between TO and MP, request non-fare payments, like fees, fines, damages, additional payments (form of yield payments), etc).

Another unique feature is that we completely rely on external data sources, like NeTEx files (since version 2.0). The 'planning information' is removed from the TOMP-API, so any source can be used, although the TOMP-API is expressed in Transmodel concepts.

TOMP-API v2.0 also supports extensions, that can be added to the 'core': offering, purchasing and executing. Functional extensions are "pay", "support", "notify" (the reseller extension), and more

technical oriented extensions are “contract” (digitally discover & sign contracts on the fly), “discovery” (digital discovery of the API using the meta-standards of OGC), and “data spaces” (connect to data spaces like Gaia-X, Eona-X or MDS etc).

With version 2.0 the TOMP-API includes better support for public transport and the support for integration in the environment/ecosystem (administration of customers, contracting, connecting to data spaces, and full discovery, compliant with OGC standards and IDS).

The TOMP covers all the process handling, except for data exchange for planning purposes and journey planning itself. Customer registration, Contracts, Booking (=purchase), Payments, Trip execution, Support, and Notifications (to/from end users, drivers, resellers).

C.4 Rough equivalences

Requests/deliveries

Only relevant endpoints are enlisted.

Offers

- POST /processes/search-offers/execution
 - o TRIP REQUEST & DELIVERY
 - o (REPEATED, SINGLE) TRIP FARE REQUEST & DELIVERY

Purchase

- POST /processes/select-offers/execution
 - o TRAVEL PACKAGE PURCHASE REQUEST & DELIVERY
- POST /processes/purchase-package/execution
 - o TRAVEL PACKAGE PURCHASE REQUEST & DELIVERY
- POST /processes/purchase-package/execution
 - o TRAVEL PACKAGE PURCHASE REQUEST & DELIVERY
- POST /processes/purchase-offers/execution
 - o TRAVEL PACKAGE PURCHASE REQUEST & DELIVERY
- POST /processes/purchase-product/execution
 - o TRAVEL PACKAGE PURCHASE REQUEST & DELIVERY
- POST /processes/confirm-purchase/execution
- POST /processes/release-purchase/execution
- POST /processes/rollback-purchase/execution
- POST /processes/extend-expiry-time/execution

Packages

- GET /collections/travel-packages/items
 - o FULFILLMENT REQUEST & DELIVERY
 - o TRAVEL PACKAGE DOCUMENT FULFILMENT REQUEST & DELIVERY
- GET /collections/assets/items

- TRAVEL PACKAGE SPOT AVAILABILITY REQUEST & DELIVERY
- POST /processes/assign-asset/execution
 - SPOT REQUEST & DELIVERY
 - SPOT RESERVATION REQUEST & DELIVERY
 - TRAVEL PACKAGE SPOT RESERVATION REQUEST & DELIVERY
 - TRAVEL PACKAGE SPOT CANCELLATION REQUEST & DELIVERY
- POST /processes/assign-traveller/execution
 - TRAVEL PACKAGE ALTERATION REQUEST & DELIVERY
- POST /processes/assign-ancillary/execution
 - TRAVEL PACKAGE ALTERATION REQUEST & DELIVERY
- POST /processes/update-travel-specification/execution
 - TRAVEL PACKAGE ALTERATION REQUEST & DELIVERY

After sales

- POST /processes/cancel-package/execution
 - TRAVEL PACKAGE CANCELLATION REQUEST & DELIVERY
- GET /collections/redress-options/itemion
 - PARTIAL JOURNEY CANCELLATION REQUEST & DELIVERY
- POST /processes/claim-redress-option/execution
 - TRAVEL PACKAGE REFUND REQUEST & DELIVERY
- POST /processes/confirm-redress-option/execution
 - TRAVEL PACKAGE REFUND REQUEST & DELIVERY
- GET /collections/redress-options/items
- POST /processes/claim-redress-options/execution
- POST /processes/confirm-redress-option/execution
 - CUSTOMER REDRESS REQUEST & DELIVERY

Customer registration

- POST /collections/customers/items
 - CREATE CUSTOMER ACCOUNT REQUEST/DELIVERY
- GET /collections/customers/items
 - GET CUSTOMER ACCOUNT REQUEST/DELIVERY
- PATCH /collections/customers/items
 - UPDATE CUSTOMER ACCOUNT REQUEST/DELIVERY

Concepts

Only relevant concepts are enlisted.

TOMP-API v2	Transmodel	Comments
Package	SALES OFFER PACKAGE CUSTOMER PURCHASE PACKAGE TRAVEL PACKAGE	Dependent of the state of the package & whether it is specifically constructed for the traveller (seat assigned, etc).
Leg	LEG	
Product	FARE PRODUCT	
PackageStatus	CUSTOMER PURCHASE STATUS	only when purchased
TravelSpecification	TRAVEL SPECIFICATION	
TravellerRequirements	ACCESS RIGHT PARAMETER ASSIGNMENT	the parameters are used to describe the traveller's demands.
Requirements	SCOPING VALIDITY PARAMETERS	
Operators	ORGANISATIONAL VALIDITY PARAMETER	
Equipment	EQUIPMENT VALIDITY PARAMETERS	
Temporal	TEMPORAL VALIDITY PARAMETERS	
Asset	SEATING VALIDITY PARAMETERS	
Network	NETWORK VALIDITY PARAMETERS	
Routing	ROUTING VALIDITY PARAMETERS	
Service	SERVICE VALIDITY PARAMETERS	
Distribution	DISTRIBUTION VALIDITY PARAMETER	
Traveller	INDIVIDUAL TRAVELLER	
userProfile	USER PROFILE	
groupProfile	GROUP PROFILE	
companionProfile	COMPANION PROFILE	
entitlement	ENTITLEMENT	
commercialProfile	COMMERCIAL PROFILE	
traveller	INDIVIDUAL TRAVELLER	

Asset	VEHICLE PARKING BAY CYCLE STORAGE EQUIPMENT RECHARGING STATION	Not really compatible due to missing generic concept in TM.
Mode	MODE	With some extensions
Customer	CUSTOMER ACCOUNT	
FareStructure	TARIFF	
FareStructureElement	TIME INTERVAL PRICE TIME UNIT PRICE GEOGRAPHICAL UNIT PRICE GEOGRAPHICAL INTERVAL PRICE	
TravelDocument	TRAVEL DOCUMENT	

Annex D (informative)

Entur API

D.1 Overview

- Description
- Modes it applies to, if relevant
- Where it is used

D.2 Related queries

Category	Endpoints
Requesting offers	POST /offers/v2/search/stop-places POST /offers/v2/search/zones POST /offers/v2/search/trip-pattern POST /offers/v2/search/authority
Modify offers (add travellers)	POST /v1/update-order POST /v1/orders/{orderId}/travellers (also with UPDATE and DELETE)
Modify offers (add products)	POST /v1/update-order
Modify offers (assign seats/assets)	POST /v1/update-order/supplement-products POST /v1/orders/{orderId}/reservations (also with UPDATE AND DELETE)
Purchase an offer (2-phase)	?? POST /v1/orders/{orderId}/confirm OR when obtained a travel document ??
Pre-purchase (pre-book) an offer	POST /v1/reserve-offers
Purchase an offer directly	
Obtain travel documents	POST /v1/pdf-tickets/{orderId}/send POST /v1/pdf-tickets/{orderId}/createOrGet
After sales (cancel purchased package)	GET /v1/refunds/options/{orderId}
After sales (reimburse)	

D.3 Unique queries

Apply discount on an offer, account-based ticketing, validate tickets (Norwegian Database), recurring payments, payment, receipts

Annex E (informative)

FerryGateway

E.1 Overview

Description: FerryGateway is an open source and free-to-use worldwide standard for communication between ferry operators and agent reservation systems. It provides a standardised set of messages that enables bookings, amendments and cancellations of ferry trips including all products and services on-board. It is stateless, SOAP-based and is defined using XML-schemas.

Modes it applies to: The scope of the FerryGateway API is passengers travelling with ferries. This includes managing things that passengers bring along on a trip, e.g., vehicles and pets, as well as managing on-board services such as accommodation, meals and events and some land services such as bus-transfers and events. The API caters for up to three fare classes, e.g., Economy, Flex and Premium and is capable of handling single leg, return and cruise pricing. Rich information regarding service offerings can be presented to the customer through URL's that points to information and can include media such as text, pictures and video.

Where it is used: FerryGateway is an initiative that was started by 8 major ferry companies in Northwest Europe in 2014. This is still the area with the highest coverage, but there is no geographical limit and the number of ferry operators using the standard is steadily increasing.

E.2 Related queries

This API is mainly stateless, meaning that, for most interactions, a complete set of information is passed back and forth between the parties involved in each interaction and amended and adjusted as needed in each step.

Category	Endpoints
Requesting offers	GetSailingsRequest(anonymous Passengers, Pets, Vehicles, SearchCriterion(list of requested Sailings (i.e., legs/trips) with detail of their respective DeparturePlace, DestinationPlace and DepartureDateTime-window)) resulting in a GetSailingsResponse containing a basket of all Sailing-options (i.e trip/leg options) that fulfil any of the listed the search-criteria. Each Sailing-option is described in detail, including availability and pricing. At this stage optional services (i.e., ancillaries) are not included. The customer adjusts the returned information-set by removing all but the preferred Sailing-option for each required departure-destination-pair. The next step is to investigate what possible optional services (i.e., ancillaries) are available for each selected Sailing (i.e., trip/leg) option. This is done by posting a

	GetServicesRequest resulting in a GetServicesResponse. The customer then removes unwanted optional Services (ancillaries) from the returned information-set. Now the information-set should reflect the complete package to be bought. Next step is to post a GetPriceRequest resulting in a GetPriceResponse containing the complete information-set including total cost-details and possibly also a unique Token with an expiryTime showing for how long the price-info is valid.
Modify offers (add travellers)	If additional anonymous Passengers are to be added: Redo the complete sequence above starting with a GetSailingsRequest query.
Modify offers (add products)	Redo the complete sequence above starting with a GetSailingsRequest query.
Modify offers (assign seats/assets)	Redo parts of the sequence above starting with a GetServicesRequest query.
Purchase an offer (2-phase)	Use the returned information-set from the GetPriceResponse possibly also including its Token for reference. Amend the Passengers-list with needed details (Name, ID-card etc) and then (phase 1) post a ReservationRequest resulting in a ReservationResponse That assures that the purchase is possible and the complete package is locked exclusively for a certain time span. (phase 2) post a ConfirmReservationRequest resulting in a ConfirmReservationResponse finalizing the purchase.
Pre-purchase (pre-book) an offer	Use the returned information-set from the GetPriceResponse possibly also including its Token as the basis. Amend the Passengers-list with needed details (Name, ID-card etc) and then (phase 1) post a ReservationRequest resulting in a ReservationResponse
Purchase an offer directly	Use the returned information-set from the GetPriceResponse possibly also including its Token as the basis. Amend the Passengers-list with needed details (Name, ID-card etc) and then post a BookRequest resulting in a BookResponse finalizing the purchase.

Obtain travel documents	Use the BookingReference supplied in the BookResponse/ConfirmReservationResponse and post a RecallBookingRequest(BookingReference) resulting in a RecallBookingResponse providing the expanded information-set. This information-set is then used as a basis for the ticketing provider to produce the travel documents.
After sales (cancel purchased package)	Use the BookingReference supplied in the BookResponse/ConfirmReservationResponse and post a GetCancelChargeRequest(BookingReference) resulting in a GetCancelChargeResponse showing the cost of cancelling. To complete the cancellation post CancelBookingRequest(BookingReference) resulting in a CancelBookingResponse
After sales (reimburse)	Use the BookingReference supplied in the BookResponse/ConfirmReservationResponse and post a GetCancelChargeRequest(BookingReference) resulting in a GetCancelChargeResponse showing the cost of cancelling. To complete the cancellation post CancelBookingRequest(BookingReference) resulting in a CancelBookingResponse

E.3 Unique queries/ information included in the API that should be highlighted:

The FerryGateway API provides a two-step approach where you first select from a basket of all possible available leg-offers and then in a second step, after deciding the preferred leg-offer for each leg, adding optional ancillary accommodation, meals and special events to the respective selected leg-offer.

The FerryGateway API provide a detailed set of attributes and constructions to handle Passengers, Pets, Vehicles with or without trailers, Accommodation including cabins, Meals with different diet-preferences, as well as support for vouchers covering events onboard or on land.

E.4 Rough equivalences:

Below a proposal of how to interpret (or align) the naming of the FerryGateway APIs in relation to other APIs in this document:

=====	
=== GetSailingsRequest === Get pricing info for all available basic PackageElement offers (without optional ancillaries) that matches the search criteria ==	
=====	
OfferedPackageElementsRequest	ann:Ask for multiple candidate-PackageElements to select among, possibly also for different Trip Patterns in parallel.
OfferedPackageElementsDelivery	ann:Get basic availability and pricing info for possible different PackageElements with the mandatory included ancillaries
=====	
=== GetServiceRequest === Ask to see which optional ancillaries you can add to a selected set of PackageElements ===	
=====	
OptionalAncillaryOffersRequest	ann:Ask to see which optional ancillaries you can add to a selected set of PackageElements
OptionalAncillaryOffersDelivery	ann:Get a list of optional ancillaries that can be added with pricing info
=====	
=== GetPriceRequest === Ask for an Package offer based on a detailed list of selected PackageElement offers and selected optional ancillaries ===	
=====	
OfferedPackageRequest	ann:Ask for an offer based on a detailed list of the PackageElements and the optional ancillaries you have added
OfferedPackageDelivery	ann:Get a single offer based on the requested combined selection
=====	
=== BookRequest === Purchase the offered Package ===	
=====	
PurchaseOfferedPackageRequest	ann:Request to Purchase an offered Package according to supplied requested TripPaclakes and ancillaries and Passenger-details. This constitutes a commitment to pay.
PurchaseOfferedPackageDelivery	ann:Confirms or rejects purchase. A confirmation constitutes a commitment to deliver.
=====	
=== ReservationRequest === Reserve the offered Package and request that a lock be held for a specified duration ===	
=====	
LockPackageOfferRequest	ann:Request to lock, for a short duration, all parts of an offered Package according to supplied requested TripPaclakes and ancillaries and Passenger-details.
LockPackageOfferDelivery	ann:Confirms or rejects that a purchase is now possible and that all parts of the offer will remain exclusively reserved until a certain time. This is a time-limited commitment to deliver.
=====	
=== ConfirmReservationRequest === Purchase the reserved Package ===	
=====	
PurchaseLockedPackageOfferRequest	ann:Request to finalize Purchase. This constitutes a commitment to pay.
PurchaseLockedPackageOfferDelivery	ann:Confirmation. At this stage a negative response is not acceptable and would be a breach of contract.
=====	
=== RecallBookingRequest === Extract all details of a purchased Package, for ticketing purposes or to alter the date of a flexible booking ===	
=====	
DisplayPurchasedPackageRequest	ann:Request full details of an already purchased Package as a basis for ticketing or as a basis for an alteration such as changing date of a trip in a flexible Package.
DisplayPurchasedPackageDelivery	ann:Provide full details of purchased Package.
=====	
=== CancelBookingRequest / GetCancelChargeRequest === Cancel all or parts of a purchased Package/ Get cancellation cost ===	
=====	
RefundPurchasedPackageRequest	ann:
RefundPurchasedPackageDelivery	ann:
GetRefundChargeRequest	ann:
GetRefundChargeResponse	ann:
=====	

Annex F (informative)

TAP TSI

F.1 Overview

Description: The TAP TSI is a European regulation for the data exchange between railway undertakings, Infrastructure managers and ticket vendors. The Regulation is legal binding within EU.

The TAP TSI-API is the API between the “Attributor¹¹” – usually a railway undertaking – and a “Distributor¹²” – usually another railway undertaking or ticket vendor. It serves as an interface to retrieve the availability of products including the available fares, the booking and the modification of a booking of such a specific product. Booking covers spot allocations (e.g., seats, berths, bicycle places, accompanied car transport by train) and/or products subject to an availability check.

The precondition to use this booking interface is the availability of the product catalogue, including the sales, usage and after sales conditions of those products; this catalogue must be available. The TSI therefore requires the provision of timetable and fare data to all participant systems.

Modes it applies to, if relevant: The standard applies to all rail transport modes, covered by the Regulation (EU) 2016/797 (Interoperability Directive), further specified in the regulation (EU) 454/2011 (TAP TSI). The scope is the single European Union rail system, as defined in the Rail Interoperability directive (EU) 2026/797. Some exemptions for rail services are allowed according to this directive: Metros, Trams and fully separated networks are not included in the scope of this European regulation. Other exemptions might be applied by the member states, for example, for lines for strictly local, touristic, or historical use.

Where it is used: This standard is used for the reservation of accommodation (e.g., seats, couchettes, berths, bike places, car transport) and fares subject to the management of availability of rail services.

F.2 Related queries

Category	Endpoints
Requesting offers	AvailabilityRequest / AvailabilityReply
Modify offers (add travellers)	AddPersonalDataRequest / AddPersonalDataReply
Modify offers (add products)	Separate call of ReservationRequest / ReservationReply
Modify offers (assign seats/assets)	Separate call of ReservationRequest / ReservationReply
Purchase an offer (2-phase)	-
Pre-purchase (pre-book) an offer	-
Purchase an offer directly	ReservationRequest / ReservationReply

¹¹ Means a company managing an attributing system. May be a carrier.

¹² Means an undertaking providing legal and technical capacity to issuers to sell rail products or to provide on line-facilities to customers to buy rail products. Besides, the distributor can offer services to issuers by assembling O-Ds carried out by different carriers into complete journeys as required by the traveller. The distributor may be a carrier

Obtain travel documents	Not supported as function call, travel documents have to follow the technical documents for rail tickets B.11 / B.12 ¹³
After sales (cancel purchased package)	CancelRequest / CancelReply PartialCancelRequest / PartialCancelReply
After sales (reimburse)	No specific request/reply. To be implemented with CancelRequest / PartialCancelRequest-

F.3 Unique queries

RollbackRequest / RollbackReply

If, for any reason, the issuing system¹⁴ cannot send confirmation of an original request or confirmation of a partial cancellation to the requesting terminal, the issuing system sends a correction request to the allocating system.

SynchroRequest / SynchroReply The synchronisation message is used in case of lost reservation dialogs to reestablish the state of the booking on the attributor and issuers system.

The full description of the messages is defined in the TAP TSI technical document B.5, available at the website of the European Union Agency for railways¹⁵.

¹³ https://www.era.europa.eu/domains/technical-specifications-interoperability/telematics-applications-passenger-service-tsi_en

¹⁴ Means an undertaking selling the ticket and receiving payment. May be a carrier and/or a distributor.

¹⁵ https://www.era.europa.eu/domains/technical-specifications-interoperability/telematics-applications-passenger-service-tsi_en

Annex G (informative)

Comparison between OSDM & Transmodel elements

The establishment of a common Multimodal Travel Purchase API will require the harmonisation of Transmodel, OSDM and the other APIs, at least as far as a mapping of the essential data elements. Close equivalences can already be established for most OSDM and Transmodel elements. The following tables summarise suggested equivalences between current OSDM online elements and their Transmodel equivalents.

The current OSDM API message set can be viewed at <https://editor-next.swagger.io/?url=https://raw.githubusercontent.com/UnionInternationalCheminsdeFer/OSDM/master/specification/v3.5/OSDM-online-api-v3.5.0.yml>.

This is a preliminary assessment intended to show that common semantics can readily be established for many elements. Further work is required to complete a full mapping. The main gap in the current Transmodel is a representation of travel purchases and reservations on the distributor system or systems (there is already a representation of the customer's individual travel purchase and reservation as passenger information).

G.1 OSDM vs Transmodel: General Purpose Elements

Framework	OSDM	Transmodel	Comment
General	Geoposition	LOCATION	Partial coordinates
	Link	INFO LINK	Associated URLs.
Mode	Mode	MODE, (SUBMODE & MODE OF OPERATION)	NeTEx & OSDM mostly as enums
	IndividualMode, ContinuousMode, PTMode	PERSONAL MODE, PUBLIC TRANSPORT MODE, ACCESS MODE	
Organisation	NamedCompany / Company	ORGANISATION	
	Carrier	TRANSPORT ORGANISATION, OPERATOR	
	ServiceBrandCode	OPERATOR	Transmodel also has the concept of an arbitrary BRAND which may be applied to elements.

G.2 OSDM vs Transmodel: Vehicle & Coach layout/Deck plan Elements

Reusable Components	OSDM	Transmodel	Comment
---------------------	------	------------	---------

Reusable vehicle	Vehicle	VEHICLE TYPE, TRAIN	Clarify Instance versus type in OSDM?
Vehicle layout	Coach	TRAIN ELEMENT TYPE, TRAIN COMPONENT, TRAILING ELEMENT TYPE	
	Coach Layout	DECK PLAN	
	Compartment	PASSENGER SPACE	
Seat	Place Position (Seat)	ONBOARD SPACE / LOCATABLE SPOT / PASSENGER SPOT	In OSDM Generally just referenced by number
	Place Properties	FACILITY SET / ACCOMMODATION, FACILITY, TYPE OF EQUIPMENT	In OSDM Generally just referenced by number
Seating Map	CoachLayoutGridSize	SCHEMATIC MAP +	
	CoachLayoutInternal	SCHEMATIC MAP MEMBER + SIGN EQUIPMENT	
	Coach LayoutSign	SCHEMATIC MAP MEMBER	
	CoachLayoutPlace	ONBOARD SPACE / LOCATABLE SPOT / PASSENGER SPOT	In OSDM Generally just referenced by number Clarify Instance versus type in OSDM?

G.3 OSDM vs Transmodel: Network Elements

Network	OSDM	Transmodel	
Place	Place (Site)	PLACE; SITE	
Stop	Stop Place	STOP PLACE; SCHEDULED STOP POINT, STOP PLACE ASSIGNMENT	OSDM does not separate physical (interchange layout from timetable (logical) stop concepts
	StationSet	GROUP OF STOP PLACES	
	FareConnectionPoint	FARE POINT IN JOURNEY PATTERN, BORDER POINT	
Point Of Interest	PointOfInterest	POINT OF INTEREST	
	PointOfInterestCategory	POINT OF INTEREST CLASSIFICATION	
Tariff Zone	Zone	TARIFF ZONE / FARE ZONE	OSDM zone is used for other purposes as well? Transmodel has additional zone types.

Routing	<i>Not used??</i>	SERVICE LINK	OSDM does not define allowed links / routes?
	<i>Not used??</i>	SERIES CONSTRAINT	Transmodel can specify which routings are allowed for a particular fare.

G.4 OSDM vs Transmodel: Service Elements

	OSDM	Transmodel	
Calendar	Calendar	SERVICE CALENDAR	
	Operating Day	OPERATING DAY, OPERATING PERIOD	
	<i>Not used??</i>	DAY TYPE	
Timetable	<i>Not used??</i>	SERVICE JOURNEY	
	Continuous Service	TEMPLATE VEHICLE JOURNEY	Frequency based services every x minutes
	Dated Journey	DATED VEHICLE JOURNEY	OSDM does not detail No JOURNEY PATTERN, POINT IN PATTERN, INTERCHANGE etc

G.5 OSDM vs Transmodel: Trip Elements

	OSDM	Transmodel	Comment
Trip	Trip	TRIP ; TRIP PATTERN	
	Trip Leg	TRIP LEG / PT RIDE LEG	
LegStop	(LegStop)	Derived from TRIP LEG → SERVICE JOURNEY → CALL POINT IN JOURNEY PATTERN	Trip Stop arrival and departure details are not restated in current Transmodel TRIP LEG but can be derived from the POINT IN JOURNEY PATTERN/ CALL . Transmodel should add explicit elements so that an itinerary can be specified on individual passenger

			information without recourse to the timetable
	Board	Can be derived from → CALL / ARRIVAL or POINT IN JOURNEY PATTERN / PASSING TIMES, etc.	See above.
	Alight	Can be derived from → CALL / DEPARTURE or POINT IN JOURNEY PATTERN / PASSING TIMES etc	See above.
	TripVia	Can be derived from TRIP LEG → SERVICE JOURNEY → CALL / VIA	See above.
	NotVia	Can be derived from CALL / VIA	See above.

G.6 OSDM vs Transmodel: Pricing Elements

	OSDM	Transmodel	
Amount	Price	FARE PRICE	
	Fee	PRICING RULE FARE PRICE (also use RESULT STEP o show calculation)	Not quite the same but can be used to specify Tax, VAT , commission, etc.
	Currency	CURRENCY	

G.7 OSDM vs Transmodel: Product Elements

OSDM can be regarded as a set of

	OSDM	Transmodel	
Product	Product	FARE PRODUCT / PREASSIGNED FARE PRODUCT, etc	
	TravelClass	CLASS OF USE	
Travel	CombinationTag		Add COMBINATION RULE?
	TripInterruptionConstraint	USAGE PARAMETER / INTERCHANGING	NeTEx Gap – interruption process
Promotional Codes	PromotionalCode	QUALITY STRUCTUREFACTOR + PRICING RULE	Arbitrary values associated with product or product element
Travel Cards	Reduction CardType	FARE PRODUCT / SALES DISCOUNT RIGHT	Associated with offered product as

			REQUIRED ENTITLEMENT parameter Associated with user via CUSTOMER ACCOUNT / FARE CONTRACT
--	--	--	--

G.8 OSDM vs Transmodel: Fare Structure Validity Conditions Elements

OSDM TravelValidity can be regarded as a set of USAGE PARAMETERS concerned with when and how a journey can be made.

	OSDM	Transmodel	
Usage Conditions	TravelValidity	VALIDABLE ELEMENT + FARE STRUCTURE ELEMENT + ACCESS RIGHT PARAMETER ASSIGNMENT +	Groups temporal and other parameters.
Travel	ValidityType (Enum)	USAGE PARAMETER / ROUND TRIP	
	TripInterruptionConstraint	USAGE PARAMETER / INTERCHANGING	Transmodel Gap – interruption process
Temporal	ValidityRange	FARE STRUCTURE ELEMENT → TIME INTERVAL TIME STRUCTURE FACTOR	Days, etc for passes
	ExcludedTimeRange	FARE STRUCTURE ELEMENT → FARE DEMAND FACTOR + VALIDITY CONDITION, DAY TYPE, TIME BAND	Use of off peak
	TrainValidity	FARE STRUCTURE ELEMENT → VALIDITY CONDITION TIME BAND	OSDM has absolute dates . Does not specify specific train number.
	TripAllocationConstraint	USAGE PARAMETER / USAGE VALIDITY PERIOD	How long valid NeTEx has detailed params
	ReturnConstraint	USAGE PARAMETER / MINIMUM STAY	TM gap add excluded days of week
Carrier	CarrierConstraint	ACCESS RIGHT PARAMETER ASSIGNMENT → OPERATOR, BRAND	

PromotionalCode	PromotionalCode	QUALITY STRUCTUREFACTOR + PRICING RULE	Arbitrary values associated with product?
Reduction CardType	FARE PRODUCT / SALES DISCOUNT RIGHT	Associated with offered product as REQUIRED ENTITLEMENT parameter Associated with usere via CUSTOMER ACCOUNT / FARE CONTRACT	Reduction CardType

G.9 OSDM vs Transmodel: Fulfillment option elements

Transmodel specifies the possible allowed means of fulfillment as properties of the SALES PACKAGE. - in particular through the DISTRIBUTION ASSIGNMENT, DISTRIBUTION CHANNEL and FULFILLMENT METHOD. The actual details of how a specific fulfillment was done can be recorded on the CUSTOMER PURCHASE PACKAGE.

	OSDM	Transmodel	
Fulfillment options	FulfillmentOption	TYPE OF TRAVEL DOCUMENT, TYPE OF MEDIUM ACCESS DEVICE NB associated with product by SALES OFFER PACKAGE + SALES OFFER PACKAGE ELEMENT DISTRIBUTION ASSIGNMENT	Ticket media types Some Gaps as to detailed attributes – add to TM/NeTex
	FulfillmentType (enum)	FULFILLMENT METHOD,	TM also has DISTRIBUTION CHANNEL etc.
	FulfillmentConstraint	some missing (GROUP TICKET usage parameter has one parameter)	GAP – add to TM

G.10 OSDM vs Transmodel: Passenger elements

OSDM has several different types of concrete CUSTOMER ACCOUNTS

	OSDM	Transmodel	
Passenger	PassengerType	USER PROFILE	Allowed types of passenger. Type OSDM is enum
	Passenger	TRANSPORT CUSTOMER	Specific instance Not linked to type
	PersonDetail	TRANSPORT CUSTOMER	
Account	AbstractTravelAccount	CUSTOMER ACCOUNT	

	TravelPassAccount	CUSTOMER ACCOUNT + TYPE OF CUSTOMER ACCOUNT + SALES TRANSACTION /CUSTOMER PURCHASE PACKAGE	
	MultirideAccount	CUSTOMER ACCOUNT + TYPE OF CUSTOMER ACCOUNT + SALES TRANSACTION /CUSTOMER PURCHASE PACKAGE	
	MultirideBalance	CUSTOMER PURCHASE PACKAGE / CUSTOMER PURCHASE PACKAGE / parameters	
	TravelAccountConsumption	CUSTOMER PURCHASE PACKAGE / CUSTOMER PURCHASE PACKAGE / parameters	Check Can be derived . View might help
	ReductionCardAccount	CUSTOMER ACCOUNT / SALES TRANSACTION	Can be derived View might help
	AbstractAccountUnit	TODO – TIME UNIT, GEOGRAPHICAL UNIT?	

G.11 OSDM vs Transmodel: Sales elements

An OSDM FARE corresponds broadly to a Transmodel PASSENGER FARE OFFER. OSDM has a fixed set of different named constraints (e.g., Carrier Constraint, Regional Constraint, etc) describing a specific trip for a specific purchase by a specific customer,

Transmodel can specify both (i) a generic set of constraints as a FARE STRUCTURE made up of general purpose FARE STRUCTURE ELEMENTs (using GENERIC ACCESS RIGHT PARAMETER ASSIGNMENTs to assign USAGE PARAMETERS and other properties), and (ii) the specific selection of individual constraints parameters for a specific trip as SPECIFIC ACCESS RIGHT PARAMETER associated with a PASSENGER FARE OFFER and /or CUSTOMER PURCHASE PACKAGE. Semantically equivalent constraints to OSDM can be specified using the Transmodel elements.

The OSDM constraints can be regarded as a set of fixed combinations of scoping and usage parameters that cover certain commonly found conditions for basic rail trips and simple zonal passes, i.e., a specialised subset of the Transmodel constraints. As such they are easy to understand and efficient to use. Equivalents can be found in Transmodel by combining one or more Transmodel fare components.

	OSDM	Transmodel	
Fare Structure	Fare	PASSENGER FARE OFFER	Groups eligibility constraints and other parameters.
	TravelClass (enum)	CLASS OF USE	
Usage Conditions	PassengerConstraints	USAGE PARAMETER / USER PROFILE+	

	LuggageConstraint	USAGE PARAMETER / LUGGAGE ALLOWANCE +	
	AfterSalesCondition	USAGE PARAMETER / EXCHANGING, REFUNDING,	Specifies whether ticket can be exchanged, etc. Place change?
Travel	RegionalConstraint	GENERIC PARAMETER ASSIGNMENT / TOPOGRAPHICAL PLACE	Can restrict to TARIFF ZONE, or even TOPGRAPHICAL PLACE (i.e., area)
Combining	FareCombinationModel	Gap	Not currently supported
Carrier	CarrierConstraint	ACCESS RIGHT PARAMETER ASSIGNMENT → OPERATOR, BRAND	
	RegulatoryCondition	Could use QUALITY STRUCTURE FACTOR	Gap

Bibliography

- [1] EN 12896-2, *Public transport - Reference data model – Part 2: Public transport network topology*
- [2] EN 12896-7, *Public transport - Reference data model – Part 3: Timing Information and Vehicle Scheduling*
- [3] EN 12896-4, *Public transport - Reference data model – Part 4: Operations monitoring and control*
- [4] EN 12896-7, *Public transport - Reference data model – Part 7: Driver management*
- [5] CEN/TS 16614-1, *Public transport - Network and Timetable Exchange (NeTEx) - Part 1: Public transport network topology exchange format*
- [6] CEN/TS 16614-2, *Public transport - Network and Timetable Exchange (NeTEx) - Part 2: Public transport scheduled timetables exchange format*
- [7] CEN/TS 16614-4, *Public transport - Network and Timetable Exchange (NeTEx) - Part 4: European Passenger Information Profile*
- [8] CEN/TS 16614-5, *Public transport - Network and Timetable Exchange (NeTEx) - Part 5: Alternative modes exchange format*
- [9] CEN/TS 16614-6, *Public transport - Network and Timetable Exchange (NeTEx) - Part 6: European Passenger Information Accessibility Profile*
- [10] CEN/TR 00278582:2022, *Public transport – Distribution APIs for MaaS*
- [11] *The Fares and Timetable, Use-Cases description and mapping for PT and Rail, CEN and UIC*
- [12] ISO 24014-1:2021, *Public transport — Interoperable fare management system — Part 1: Architecture*